

Battery Impedance Analyzer(*BZA series*)
Impedance Manager(ZM)

Operation Manual

Ver. 1.08*(English)

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This manual has to be read carefully
before switching on the instrument.
The instructions must be followed strictly.

Non-observance of these instructions can
lead to the loss of right to claim for damages
or guarantee.



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Safety

- Do not open the device. No user serviceable parts insides. Refer servicing to qualified service personnel.



If the battery or battery pack is under charging, do not use this device to measure EIS of the battery or battery pack.



The battery or battery pack's voltage might be high so it is dangerous to take care of the device. Do not touch the terminal of the battery or battery pack.



If the system is under stuck and if it is not released, you need reboot the system.



If the battery or battery pack is under high current flow, the heating will happen. Proper cooling is required.

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Notice

1. This manual

This manual is for Battery Impedance Analyzer with ZM (Impedance Manager) software operation guide.

Electronic version of this manual is on c:\program files(x86)\ZIVELAB\ZM\manual folder

2. LAN connection fail/PC failure

WonATech SIF control board has internal memory and independent CPU, so it works independently nevertheless connection with PC or not. You can reconnect working system anytime.

If PC failure happens, just reboot the PC and run ZM.exe then you can see running experiment normally without any loss of data if data points are less than internal memory.

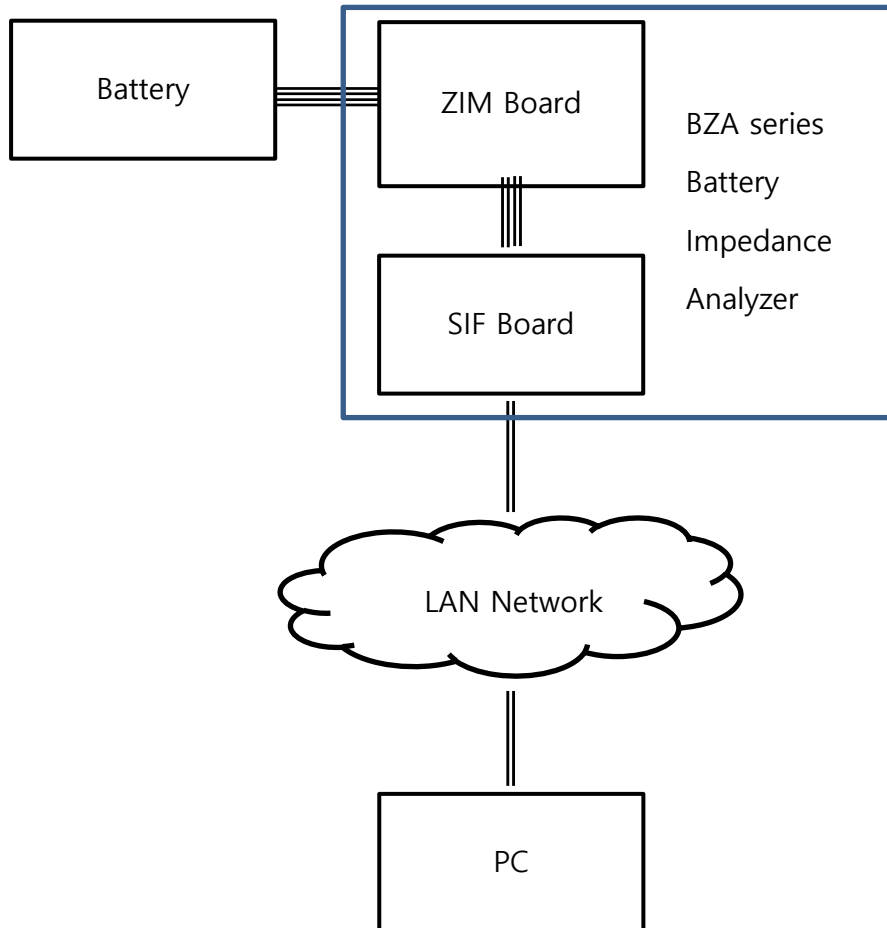
- If the problem happens, don't turn off system power. It will make all experiment data disappeared.
- Please note that if LAN connection was lost, you must try to connect it after 10 seconds

3. Caution

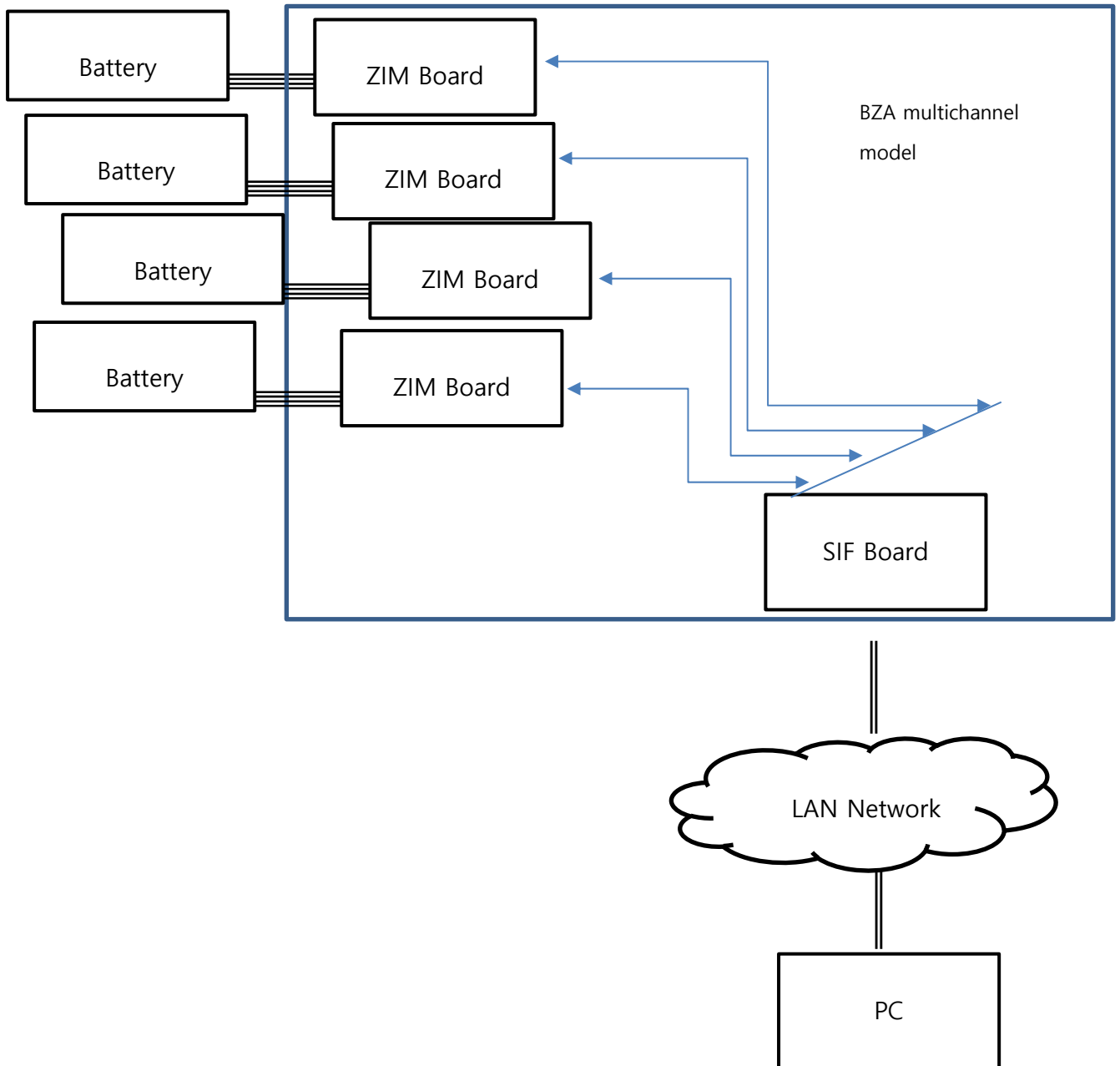
- **You must set power saving option as disable in Windows OS.**
- **There is 70C degree thermostat at heat sink. This will stop the BZA operation to prevent overheat of instrument. Please try to use BZA after 5 minutes**
- Don't place EMI producing unit nearby this instrument (such as radio, vibration etc.).
- The cell cable itself can work as antenna. (It can receive EMI noise). Especially don't place cell cable on power line cable.
- The cell cables which WonATech supply are different by model. The cell cable characteristic give the different EIS result. You must use the cell cable only supplied by WonATech with BZA
- If the result data is so noisy, make the cell cable as short as possible. (It reduces noise)
- Maximum data point number is 100,000
- Do not touch battery terminal especially for high voltage battery.
- If the problem still remains, please contact our service department via [service@ WonATech.com](mailto:service@WonATech.com).

4. Block diagram

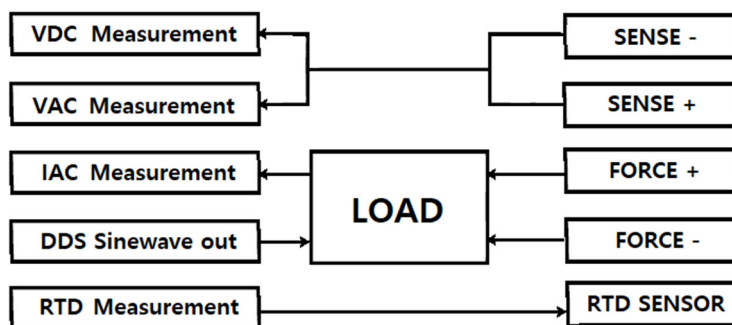
A. BZA Hardware Block Diagram.



B. BZA60M, BZA500M 4channel hardware block diagram.

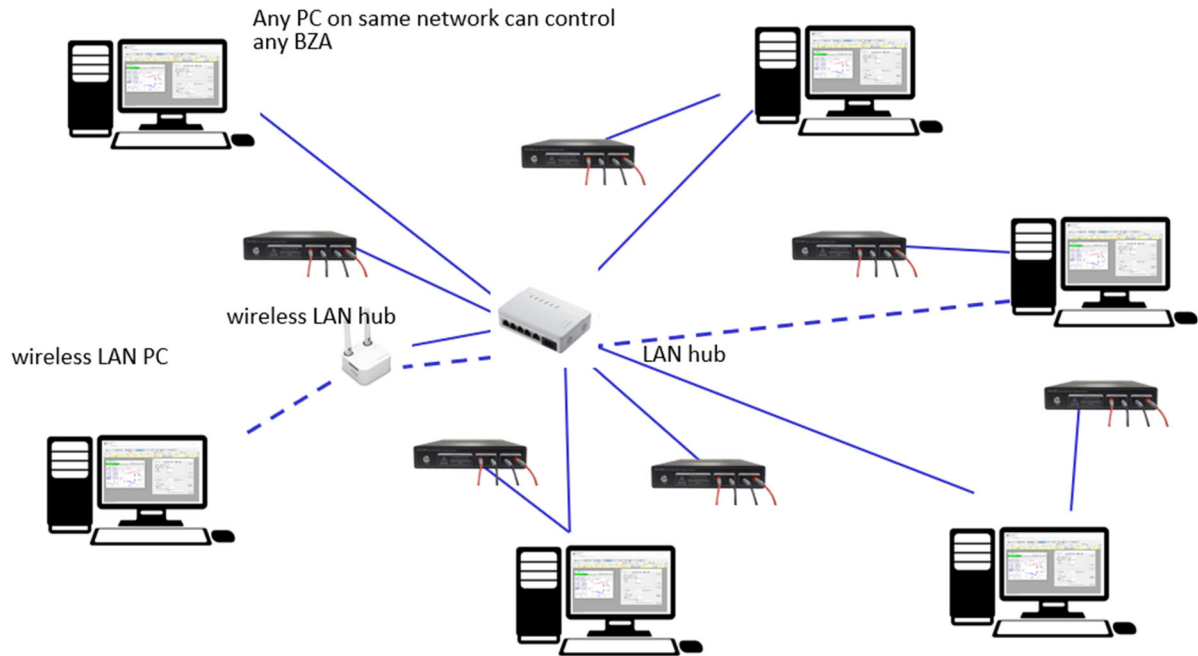


C. Analog part.

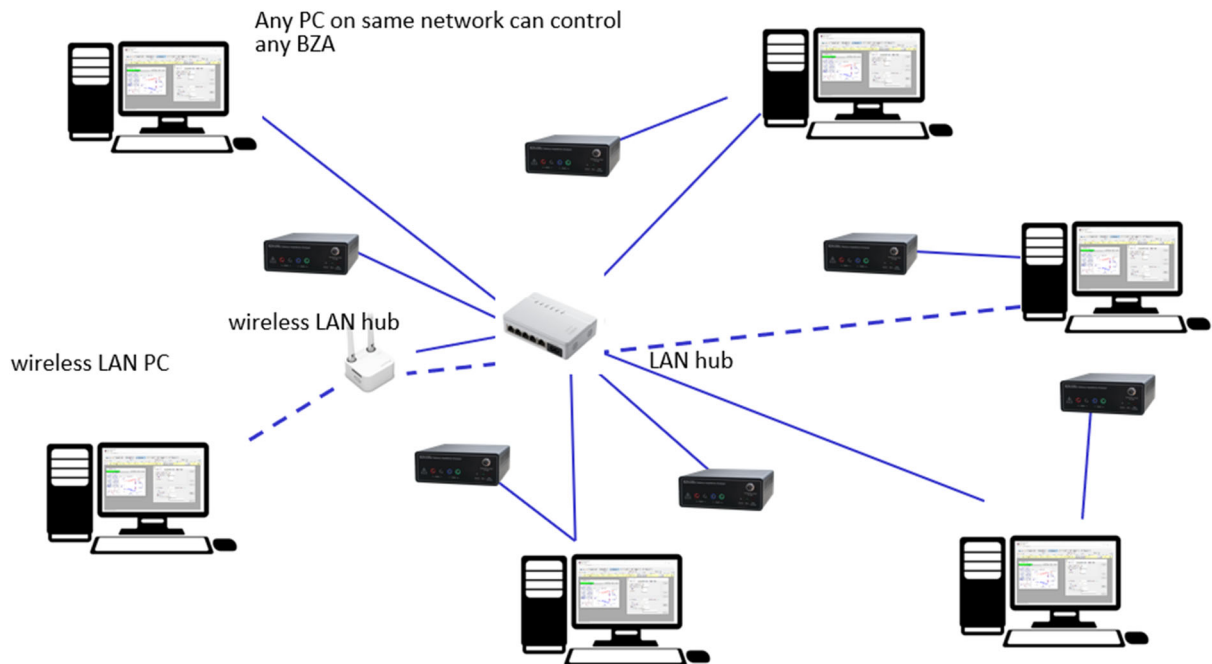


5. Connection diagram

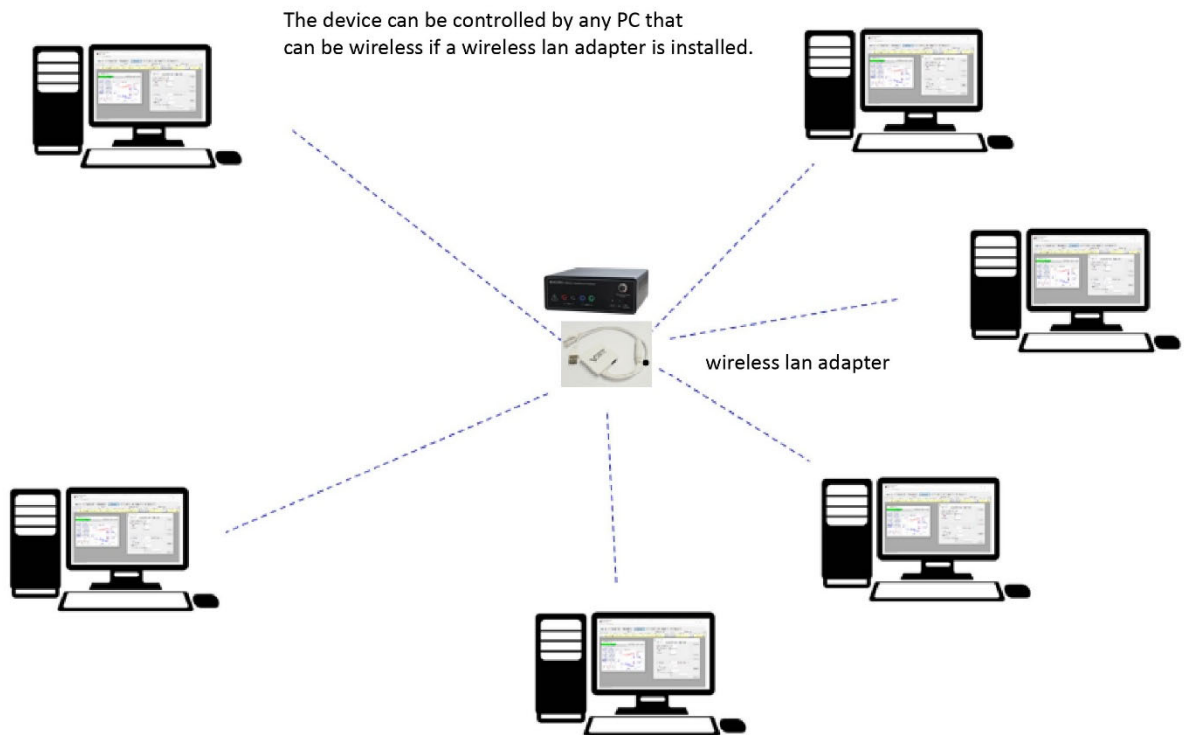
(1) BZA1000 (BZA500M)



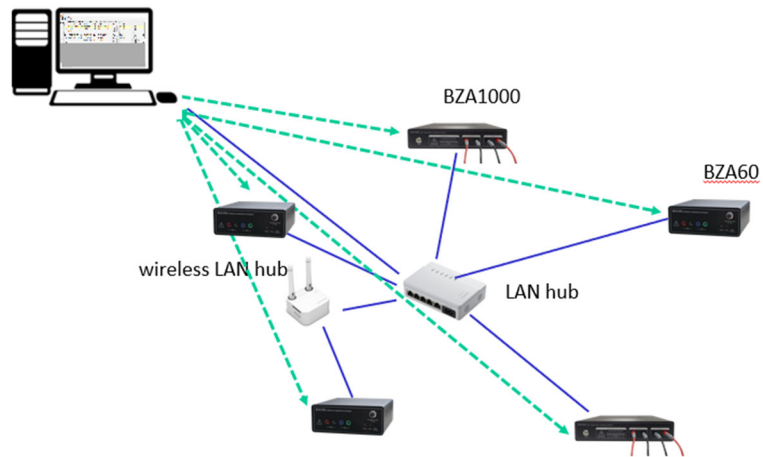
(2) BZA60 (BZA60M)



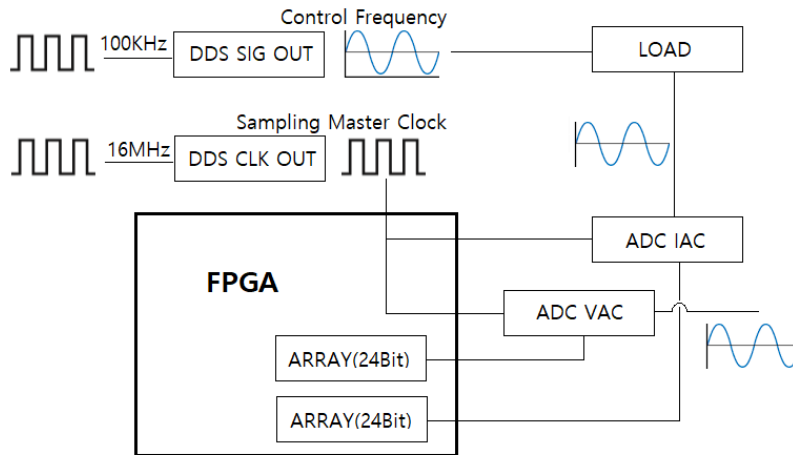
(3) BZA60/BZA1000 using portable option



(4) One PC can control several BZAs in network: Max 32channel



6. Impedance Measurement Flow.



7. Technique file and parameters

- Galvanostatic EIS
 - Initial frequency
 - Final Frequency
 - Density
 - Iteration
 - Current range
 - Max initial delay
 - Skip cycle
 - Cycle number
- HFR
 - Frequency
 - Interval
 - Load off at waiting
 - Total time
 - On delay
 - Current range
- Rs Psuedo Rp
 - Rs frequency
 - Psuedo Rp frequency
 - Psuedo Rp end frequency
 - Interval
 - Load off at waiting
 - Total time
 - On delay
 - Current range
- Eoc/Temperature monitor
 - ◆ Sample time
 - ◆ Total time
- Discharge Test
 - ◆ IR measurement option
 - ◆ IR measurement frequency

- ◆ IR measurement interval
- ◆ Sample time
- ◆ Discharge control current
- ◆ Cut-off voltage
- Quick galvanostatic EIS
 - Initial frequency
 - Final frequency
 - Current range

8. Control Panel

- Channel assignment and edit content of the experiment conditions file
- Assignment of experiment data file
- Start and stop of experiment
- Assign groups and start groups when using multi-channels
- Individual channel assignment is available for each device regardless of model
- Monitoring of the following parameter
 - ◆ Voltage range
 - ◆ Current range
 - ◆ Battery voltage value
 - ◆ Temperature value

9. Real time Monitor

- Status display
- Process display
- Time display for experiment progress
- Current/Voltage range
- Bias voltage
- Temperature
- Measured latest frequency
- Impedance Magnitude
- Impedance Phase
- Current impedance measurement frequency
- Real-time graph 1
 - AC waveform graph : Select between Voltage vs. Current or Voltage/Current vs. Time . It is not displayed during the voltage temperature monitoring.
- Real-time graph 2
 - Galvanostatic EIS: Nyquist
 - High frequency resistance measurement: Zre Vdc vs Time
 - Rs,Pseudo Rp monitor: Rs,P_Rp vs Time
 - Eoc, Temperature monitor : Eoc, Temperature vs Time
 - Discharge Test: Vdc, Temperature vs Time
- Real-time graph 3
 - Galvanostatic EIS: Bode
 - High frequency resistance measurement: Cs,Cp vs Time
 - Rs Pseudo Rp monitor : Cs, Cp vs Time

10. Communication malfunction and reconnection

If BZA' electric power was shut down or LAN cable was disconnected, The communication between PC and BZA was interrupted. In this case this software will detect communication disconnection promptly and shows the status in control panel and real time monitor as the follows.

A. Communication problem by BZA power failure

Control panel (Channel is running)

Channel	Group	Status			
		Status	Last error	Elapsed(s)	Range
1	☒	Running.	None...	00:00:00	200mA/ 6V

Control panel (Software detected communication disconnection)

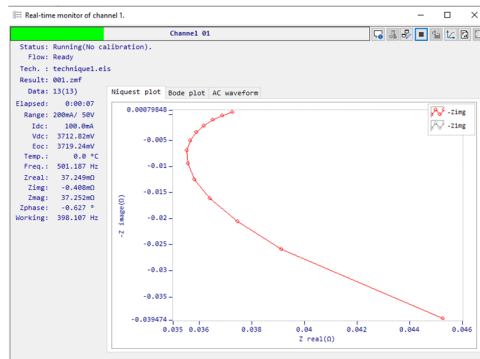
Channel	Group	Status
1	☒	disconnected

Control Panel (Software detected reconnection)

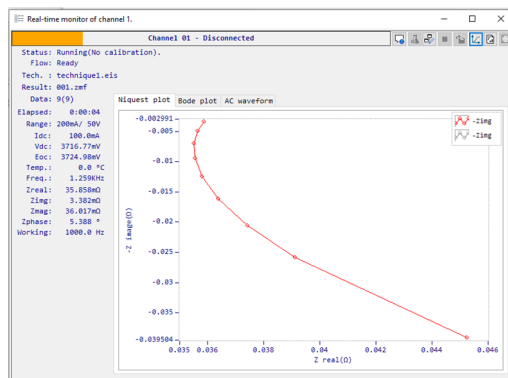
Channel	Group	Status
1	☒	Ready(No calibration).

In this case, BZA was turned off and turned on so the experiment was stopped and no data was stored inside of BZA. You need to run the experiment again.

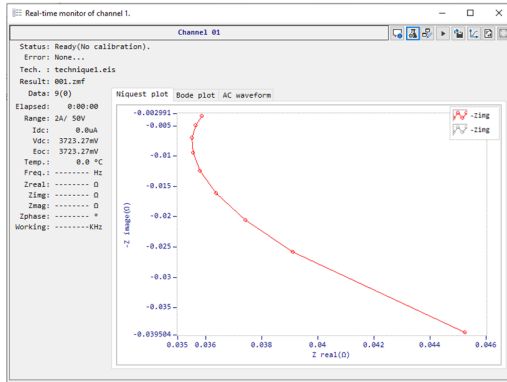
Real time monitor (Channel is running)



Real time monitor (Detection of communication failure)



Real time monitor (Reconnected)



B. Communication problem by LAN cable connection etc.

Control panel (channel is running)

Channel	Group	Status		
		Status	Last error	Elapsed(s)
1	☒	Running.	None...	00:00:03

Control panel (Software detected communication disconnection)

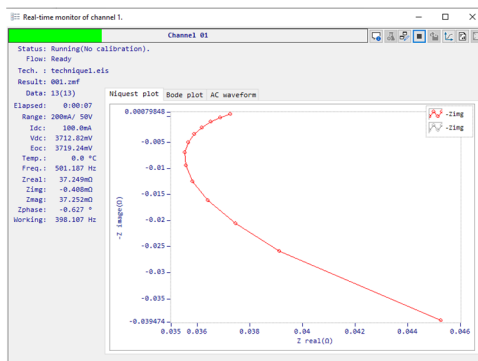
Channel	Group	Status		
		Status	Last error	Elapsed(s)
1	☒	disconnected	None...	00:00:28

Control Panel (Software detected reconnection)

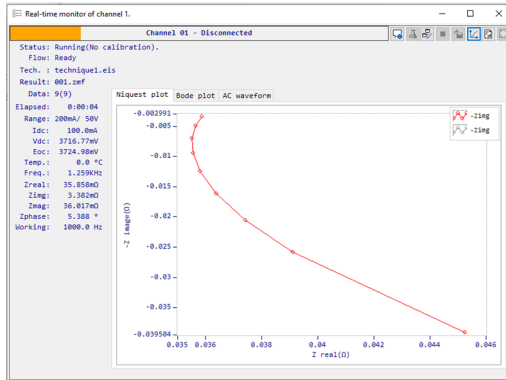
Channel	Group	Status			
		Status	Last error	Elapsed(s)	Range
1	☒	Running.	None...	00:00:14	200mA/

In this case, when connection is recovered, experiment will be continued. (The data points during communication failure were saved inside of BZA).

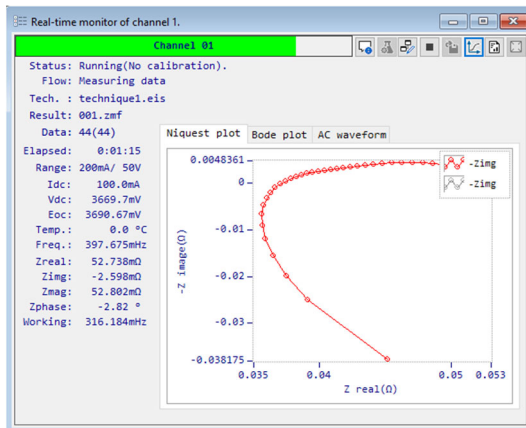
Real time monitor (Channel is running)



Real time monitor (Detection of communication failure)



Real time monitor (Reconnected)



11. Calibration

Refer to separated calibration manual if you had calibration training from WonATech and AC calibration resistor set which WonATech supplied.

If not, please do not try to calibrated BZA.

12. If you have trouble or questions;

If you have trouble on BZA series BATTERY IMPEDANCE ANALYZER or questions, please send the following file and explanation to [service@ WonATech.com](mailto:service@WonATech.com).

- ✓ Data file (*.zmf)

Chapter 1. Overview

1. Short description

The BZA60/BZA1000 are single channel battery impedance analyzers.

There are 4channel model BZA60M/BZA500M.

These can measure battery or battery pack impedance. Fixed frequency impedance or whole impedance spectra. Also, these can measure open circuit potential and battery temperature using optional PT100 sensor.

Independent impedance analysis software package ZMAN can read the data file and fit the equivalent circuit models automatically so user can determine the battery status.

PC connection is using ethernet communication so it can run other BZA device(s) in remotely.

Control/measurement software can display Bode and Nyquist plot with data list.

User can get detailed whole impedance spectra from 4kHz or 10kHz to 50mHz with max 100,000 points.

Quick galvanostatic EIS measurement for screening purpose can be done.

This software can support to determine Pseudo R_p measurement and Eoc/Temperature monitor and discharge test.



2. Model

Measurement Functions	BZA1000	BZA500M	BZA60(M)
Impedance ranges	500uΩ-50Ω	500uΩ-50Ω	500uΩ-50Ω
Frequency range	0.05-4kHz	0.05-10kHz	0.05-10kHz
Applicable battery Max Voltage	1000V	500V	60V
Temperature via ext. PT100	yes	yes	yes
Battery Voltage measurement	yes	yes	yes
Current ranges	8 ranges (2A-400uA)	8 ranges (2A-400uA)	8 ranges (2A-400uA)
Voltage ranges	1000/100 volt	500/50 volt	60/6volt
ADC	24bit 3ea	24bit 3ea	24bit 3ea
Cooling Wattage	60Watt	40Watt	20Watt

3. Functions

- ZMAN impedance analysis software: automatic model searching & fitting software
- Real time Bode & Nyquist Plot
- Export ascii format data file
- Technique file with parameter setting support.
- Galvanostatic EIS, HFR, Rs-Pseudo Rp, Eoc/temperature monitor, discharge test, quick galvanostatic eis
- Multichannel configuration available
- Searching BZA instruments under same network environment
- Various graph function

Chapter 2. General instruction

Please carefully read this operation manual before you start to use BZA series BATTERY IMPEDANCE ANALYZER.

1. Unpacking

Please check to ensure that the delivery is complete:

- measurement instrument
 - software setup USB memory
 - PDF manual on USB memory
 - LAN connection cable 2m
 - Power adapter cable
 - Cell cable set
 - PT100 sensor cable (option)
-
- Portable Option
 - Carrying baggage
 - Wireless Lan kit
 - Other options

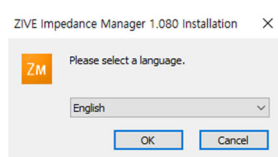
2. Installation:

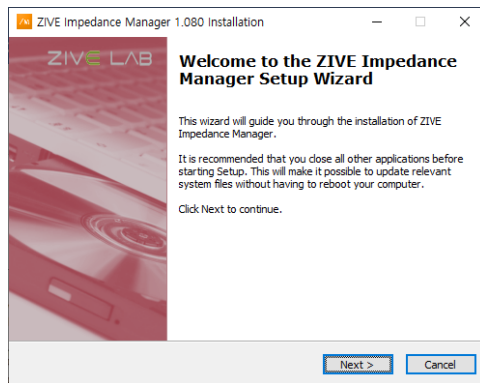
ZIVELAB Impedance Manager program can be run under Windows 7/8/10 environments. (32bit or 64bit OS available). The minimum specification of PC is PentiumIV CPU and RAM 1024MB and display card supporting 16bit high color and 1600x900 resolution or above. For multi-tasking, we recommend higher RAM size and memory size of video adapter.

If you have any queries regarding installation, please contact with WonATech service department. (service@WonATech.com)

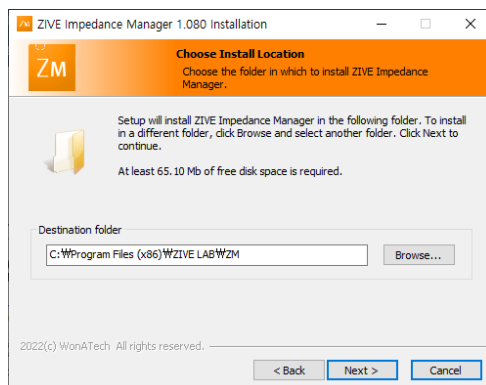
A. Software installation

To install Battery Impedance Analyzer software, run setup.exe in install CD. After installation, C:\program files(x86)\ ZIVELAB\ZM (for 32bit os)/ or C:\program files\ ZIVELAB\ZM (for 64bit OS) folder will be created

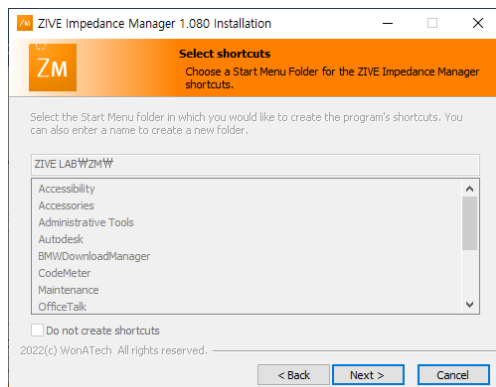




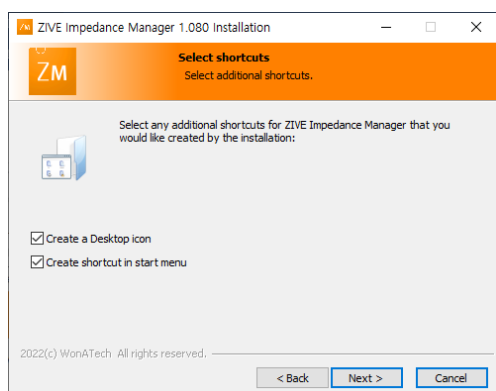
Click "next" button



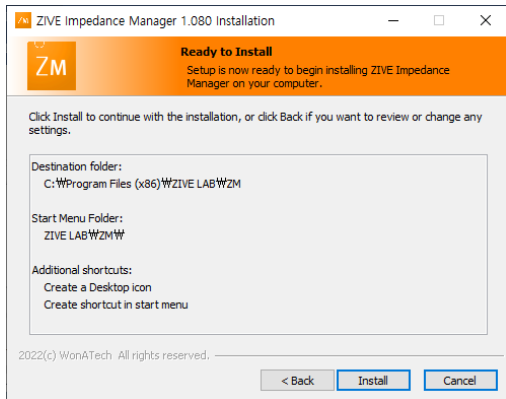
Default target folder is C:\program files(x86)\ ZIVELAB\ZM Click "Next" button



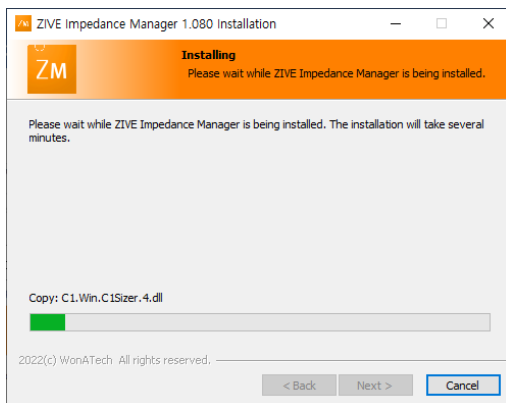
Click "next" button



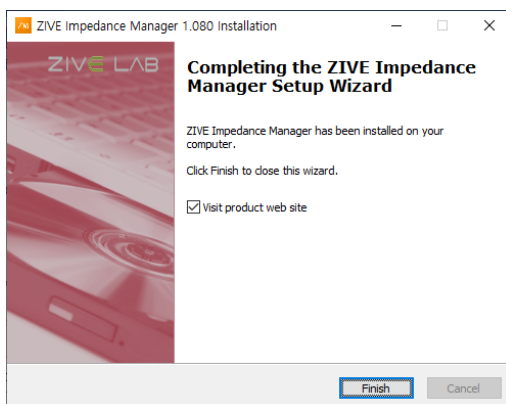
Click "next" button



Click "Install" button



It takes some time to install.



If installation is finished, c:\PROGRAM FILES(x86)\ZIVELAB\ZM should be created

If software installation is finished, you are ready to use BZA series BATTERY IMPEDANCE ANALYZER program.

You need to install ZMAN software to analyze data file.

ZMAN setup files are on setup media.

3. Max power



BZA device can be used up to maximum 60Watt (BZA1000), 40Watt(BZA500M), 20Watt (BZA60/BZA60M) for EIS measurement to protect instrument. This depends on battery voltage and max current. If setting parameter is over the above EIS power limit then software will decrease current range automatically.

For example, if battery voltage is 200 Volt and user set EIS current range as 400mA then calculated power is $200 \times 0.4 = 80\text{Watt}$ so software will change the current range from 400mA to 200mA automatically.

The heat should be dissipated properly. Housing surface for BZA1000 will work for this purpose

4. Connecting temperature sensor cable

For temperature measurement, Optional RTD (PT100) sensor can be connected to RTD connector at rear side(BZA1000) or front side (BZA60,BZA60M,BZA5600M).



This RTD sensor is surface measurement type. You can attach this sensor to battery surface to monitor the battery temperature.



5. Connecting the cell cable for EIS and OCV to the device



BZA1000 front panel



BZA500M front panel



BZA60 front panel



BZA60M front panel

On the front panel, there are 4 cell cable connectors. You can connect these connectors using 4 cell cables which we supplied with the device.

The cell cable supplied with BZA1000 is for voltage specification within 1000V, but since the crocodile clips are for 300V, crocodile clips for 1000V need to be purchased separately in experiments over 300V.

The cell cable supplied with BZA60 is available for 60V.



BZA1000/BZA500M cell cable with standard alligator clip



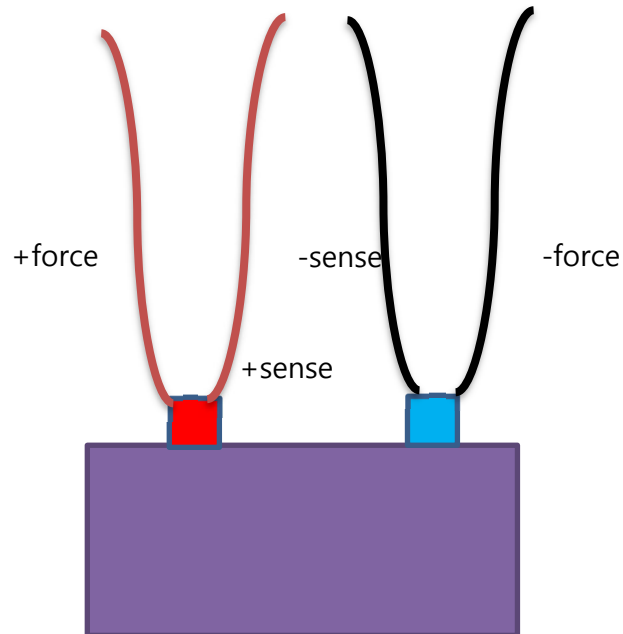
BZA60(M) cell cable with alligator clip

There are 4 labeled cables in BZA cell cable

- + force cable(red): For current flowing. Connect it to battery positive terminal
- - force cable(black): For current flowing. Connect it to battery negative terminal.
- + sense cable(red): For voltage measurement. Connect it to battery positive terminal
- - sense cable (black): For voltage measurement. Connect it to battery negative terminal

6. Connecting the cell cable to battery

Low impedance EIS measurement requires 4 probe kelvin type connection to minimize cell cable itself error.



7. Checking Point

Before running BZA series BATTERY IMPEDANCE ANALYZER, please check the following

- Connect LAN cable between BZA series BATTERY IMPEDANCE ANALYZER and PC.
- Power on BZA series BATTERY IMPEDANCE ANALYZER (Please check Busy green LED ON)

- Check charging status of battery pack if you use portable option
- Use power adapter which WonATech supplied with BZA series BATTERY IMPEDANCE ANALYZER.
- Virus infection of Software in your PC

8. Caution

- Don't remove cell cable lead during run
- Don't locate noise source (radio, monitor etc.) nearby instrument
- Don't power off during run
- Don't disassemble the instrument
- To minimize induction effect in cell cable, twist the cell cable with two pairs (twisting with I+,I- cables , twisting with V+,V- cables)

Chapter 3. BZA series Battery Impedance Analyzer Basic

Battery Manager software is to control ZIVELAB BZA series Battery Impedance Analyzer and it can support data acquisition and data view.

You can use independent analysis/tool software packages at free of charge as the follows.

- ZMAN software suite: For EIS data analysis

You can find the above software package on setup CD or you can download recent version on <http://www.WonATech.com>

- ZM(Impedance manager) software is for BZA instrument

1. ZIVELAB Impedance Manager(ZM) software feature

- When BZA SERIES BATTERY IMPEDANCE ANALYZER start the experiment, software will send the experiment parameters to SIF board in BZA SERIES BATTERY IMPEDANCE ANALYZER, then BZA series BATTERY IMPEDANCE ANALYZER will work experiment(control and data acquisition) independently from PC. So after starting experiment, BZA series BATTERY IMPEDANCE ANALYZER will work without PC control.
- Automatic BZA IP address searching function to find BZA on network.
- It can be used to control multi-channel devices and BZA on network can be controlled from one PC.
- User can upgrade firmware easily.
- Other PC on network connected to one specified BZA, user can not connect this BZA.
- User who has calibration training can calibrate BZA using ZM.exe.
- If there is PC power failure or LAN cable unplugged by accident, the experiment keeps going and and you

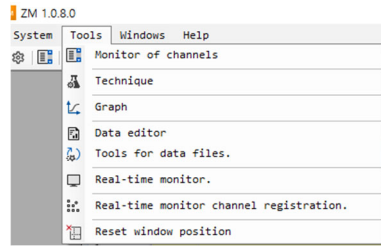
can transfer data in SIF control board memory. The data can also be transferred to the PC by click this  icon on the control panel at any time even if the data from the completed experiment is automatically transferred to the PC once a new device is connected.

- During experiment, there is a real-time graph function suitable for each type experiment such as Bode plot and Nyquist plot, etc. in real-time.
- User can see the waveform of voltage and current as Voltage vs. Current graph or Voltage/Current vs. Time graph.
- Impedance Manager software save impedance measurement data in a binary format(*.zmf) that is readable from ZMAN and user can also exports in a text(CSV, txt) format.
- Saved in a ZMAN format can be analyzed with ZMAN without license code.
- Instrument information can be displayed.

2. Data file

- There are 2 kinds of data files that you can save.
 - Data file
 - ZMAN format: EIS data will be saved as binary format(*.z#) which ZMAN software can read.
 - TXT format: Saved in a format assigned by EIS data by saving ascii files of graphs or reports.

- 3) CSV format: For special technique, software can convert data file with specific data in csv format in tools-tools for data files



The specific technique for this data conversion are;

- HFR [Process the HFR test results and save them to a text file.](#)
Including Cs(F) and Cp(F)
- PRR [Process the PRR test results and save them to a text file.](#)
Including Cs(F) , Cp(F) per frequency and (F3-F2).real(Ω), (F3-F1).real(Ω), (F2-F1).real(Ω)
- DCH [Process the DCH test results and save them to a text file.](#)
Including capacity data.

② Data file includes the following information.

- a. SIF board information
 - 1) Serial number
 - 2) Board version
 - 3) Firmware version
- b. ZIM board information
 - 1) Serial number
 - 2) Board version
 - 3) Firmware version
- c. Experiment technique file information
 - 1) File name
 - 2) Technique type
 - 3) Technique parameter setting
- d. Experiment start and end time
- e. Nominal battery capacity
- f. Battery ID
- g. Name of tester
- h. Number of data

③ Data parameter

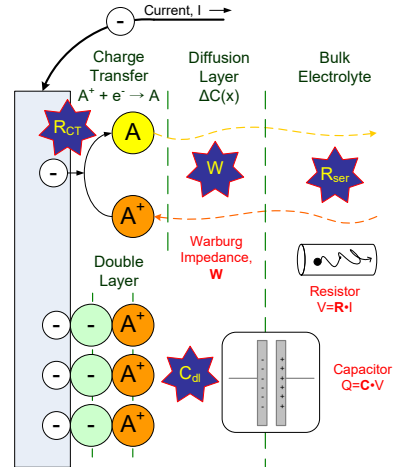
- a. Detailed data
 - i. Test time: Experiment time
 - ii. Cycle time: Time from the start of each cycle in case of iteration
 - iii. Frequency: Frequency value
 - iv. Zreal: Impedance real value
 - v. Zim: Impedance imaginary value
 - vi. Zmag: Impedance magnitude
 - vii. Zphase: Impedance phase
 - viii. Idc: DC current
 - ix. Vdc: DC voltage
 - x. Eoc: Open circuit voltage

- xi. Temperature: Temperature
- xii. Setting: Voltage and Current range

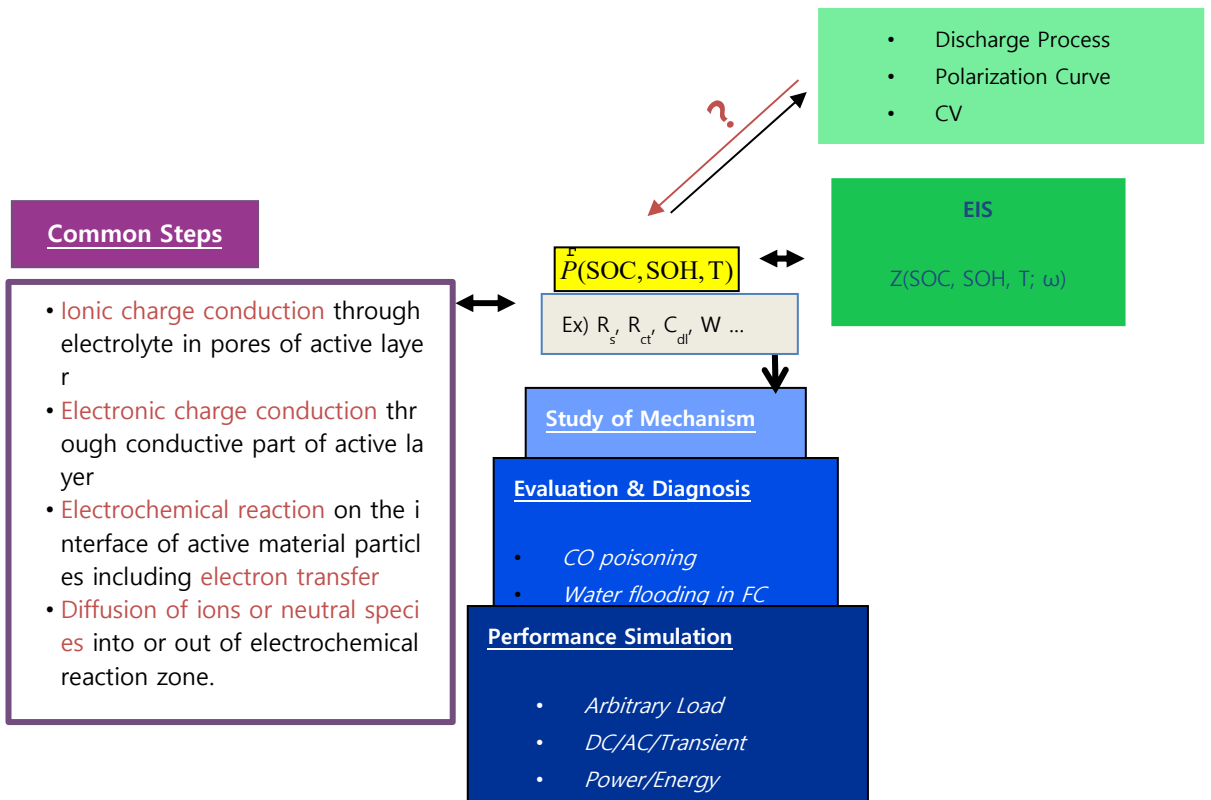
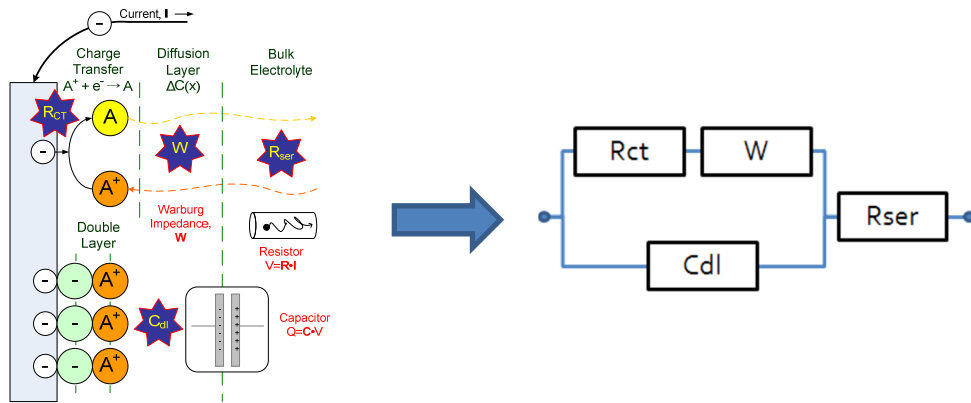
3. EIS Basic information

(1) Electrochemical Interface

- Everything happens at the interface
 - ✓ Charge Transfer $\Rightarrow R_{ct}$
 - ✓ $R_{ct} \sim 1/i_0$
- Butler-Volmer Equation
- Diffusion Layer $\Rightarrow W$
- Bulk Electrolyte $\Rightarrow R_{ser}, R_{\Omega}$
- Double Layer $\Rightarrow C_{dl}$
 - ✓ Non-Faradaic Process

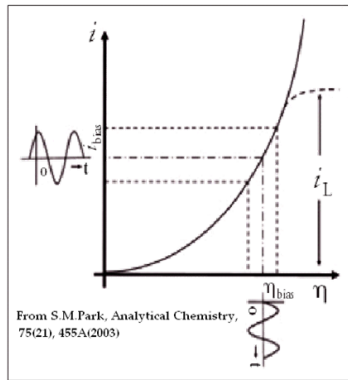


(2) Randle's Circuit



(3) What is Impedance?

- AC circuit theory describes the response of a circuit to an alternating current or voltage as a function of frequency
- Impedance is a totally complex resistance encountered when a current flow through a circuit made of resistors, capacitors, or inductors, or any combination of these
- Ohm's Law, $V = R \cdot I \rightarrow V = Z \cdot I$ (complex number Z)



- Circuit theory is simplified when the system is "linear".
- Z in a linear system is independent of excitation amplitude. The response of a linear system is always at the excitation frequency (no harmonics are generated).
- Look at a small enough region of a current versus voltage curve and it becomes linear.
- If the excitation is too big, harmonics are generated and EIS modeling does not work.

(4) Impedance Elements

Basic elements

Resistor		$I(t) = I_0 e^{j\omega t}$	
	$E = RI$	$\rightarrow$	$Z = R$
		$E = Z \times I$	
Inductor		$I(t) = I_0 e^{j\omega t}$	
	$E = L \frac{dI}{dt}$	$\rightarrow$	$Z = j\omega L$
		$E = Z \times I$	
Capacitor		$I(t) = I_0 e^{j\omega t}$	
	$E = \frac{Q}{C} = \frac{1}{C} \int Idt$	$\rightarrow$	$Z = \frac{1}{j\omega C} = -j \frac{1}{\omega C}$
		$E = Z \times I$	

Voltage $E(\omega) = E_o \cos(\omega t)$
 $= E_o e^{j\omega t}$, where $j = \sqrt{-1}$ & $\omega = 2\pi f$

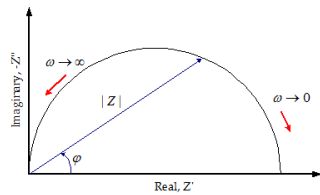
Current $I(\omega) = I_o \cos(\omega t - \varphi)$
 $= I_o e^{j(\omega t - \varphi)}$

Impedance $Z(\omega) = \frac{E(\omega)}{I(\omega)}$ ← **Ohm's Law**
 $= Z_o(\omega) e^{j\varphi(\omega)}$, where $Z_o = E_o / I_o$
 $= Z_o (\cos \varphi + j \sin \varphi)$ → **Modulus & Phase (Bode Plot)**
 $= Z' + jZ''$ → **Real & Imaginary part (Nyquist Plot)**

(5) Presentation of impedance plot

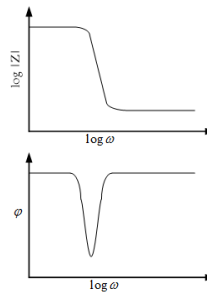
• Nyquist Plot

- Vectors of length $|Z|$
- Individual charge transfer processes are resolvable.
- Frequency is not shown.
- Small Z can be hidden by large Z.



• Bode Plot

- C may be determined graphically.
- Small Zs in presence of large Zs are usually easy to identify.



Basic circuit elements

Resistor



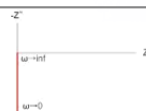
$$Z = R$$



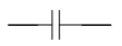
Inductor



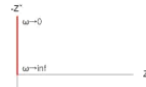
$$Z = j\omega L$$



Capacitor



$$Z = \frac{1}{j\omega C} = -j \frac{1}{\omega C}$$



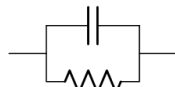
Combination of circuit elements

R-C



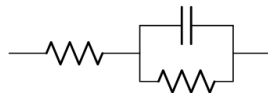
$$\rightarrow Z = R + \frac{1}{j\omega C}$$

R|C



$$\rightarrow \frac{1}{Z} = \frac{1}{R} + j\omega C$$

R-R|C



$$\rightarrow Z = R_s + \frac{1}{\frac{1}{R} + j\omega C}$$

- Rs-Ls

$$Z = R_s + j\omega L_s$$

$$Q \text{ Factor, } Q \equiv \frac{\omega L_s}{R_s}$$

Rs-R|C

$$Z = R_s + \frac{1}{\frac{1}{R} + j\omega C} = \left[R_s + \frac{R}{1 + \omega^2 R^2 C^2} \right] - j \left[\frac{R \times \omega R C}{1 + \omega^2 R^2 C^2} \right]$$

$$\equiv Z' + jZ''$$

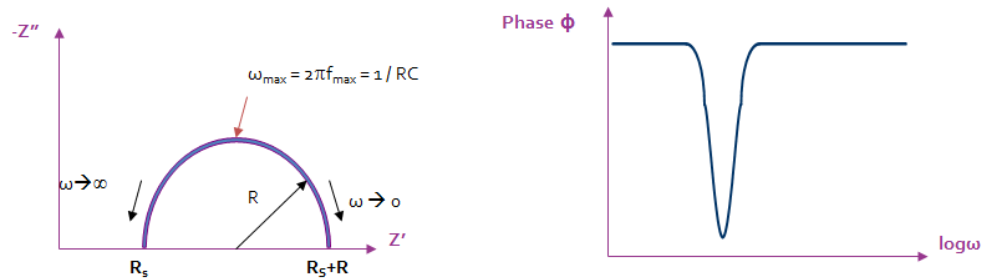
$$1. \omega \rightarrow 0, Z = R_s + R$$

$$2. \omega \rightarrow \infty, Z = R_s$$

$$3. Z = R_s + \frac{R}{1 + \omega^2 R^2 C^2}, Z' = -\frac{R \times \omega R C}{1 + \omega^2 R^2 C^2} \therefore \left\{ Z - \left(R_s + \frac{R}{2} \right) \right\}^2 + Z'^2 = \left(\frac{R}{2} \right)^2$$

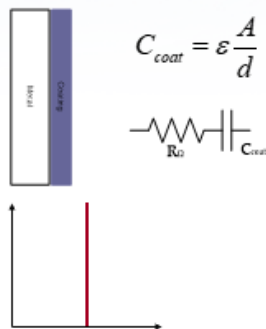
$$4. Z = R_s + \frac{R}{2} \Rightarrow \frac{R \times \omega_{\max} R C}{1 + \omega_{\max}^2 R^2 C^2} = \frac{R}{2}$$

$$\therefore \omega_{\max} = \frac{1}{RC} \Rightarrow -Z'' = -Z''_{\max}, \text{ phase } \phi = \phi_{\min}$$

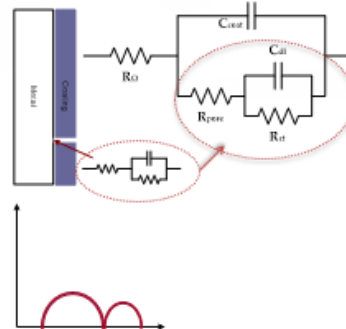


Coating Capacitance

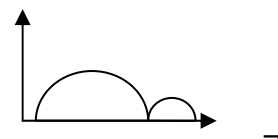
- Ideal Coating



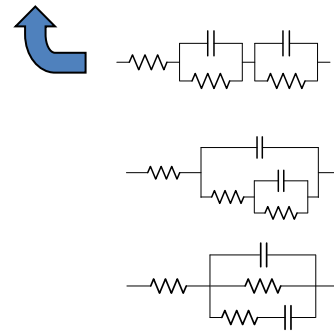
- Imperfect Coating



(6) Uniqueness of Models

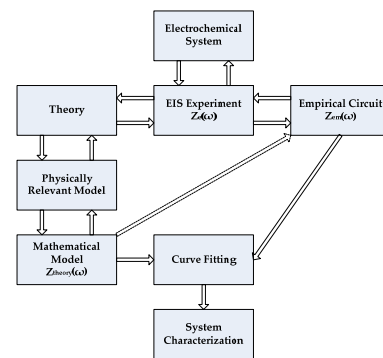


- There is not a unique equivalent circuit that describes a spectrum.
- Measuring Z is simple and easy, but analyzing it is difficult.
- **Physically relevant model** is important.
 - It can be tested by altering physical parameters.
- Be cautious in handling **empirical models** even if you get a good looking fit.
 - Use the fewest elements
 - Test it by T-test



(7) Advantage of EIS

- Relatively simple electrical measurement
- But analysis of complex material variables: mass transport, rates of chemical reactions, corrosion
- Predictable aspects of the performance of chemical sensors and fuel cells
- Providing empirical quality control procedure



(8) Disadvantage of EIS

- Ambiguities in interpretation
 - All cells have intrinsically distributed properties
 - Ideal circuit elements may be inadequate to describe real electrical response
 - Use of distributed elements (e.g. CPE)
- There is not a unique equivalent circuit describes measured impedance spectrum

(9) Equivalent circuit elements

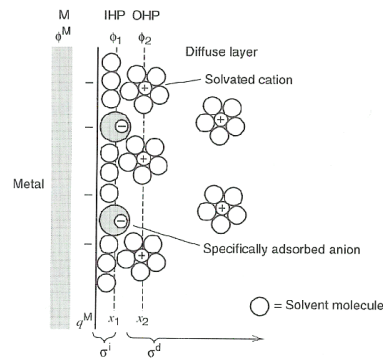
- **Electrolyte Resistance**
 - 3 electrodes: between WE and RE
 - 2 electrodes: all series R in the cell are measured including R of contacts, electrodes, solution, and battery separators
 - Depends on ionic concentration, type of ions, temperature, and geometry
- **Charge Transfer Resistance**

- Echem charge transfer reactions are generally modeled as resistances.
- When an EIS spectrum is measured on a corrosion cell at E_{corr} , the resistance at low-frequency is identical to the polarization resistance.

For a one step, multi-electron process, $O + ne \rightleftharpoons R$
small overpotential is given by

$$\eta = \frac{RT}{nF} \left[\frac{C_O(0,t)}{C_O^*} - \frac{C_R(0,t)}{C_R^*} + \frac{i}{i_o} \right] \quad \Leftrightarrow \quad R_{ct} = \frac{\partial E}{\partial i} \bigg|_{C_O(0,t), C_R(0,t)} = \frac{RT}{nFi_o}$$

• Double Layer Capacitance



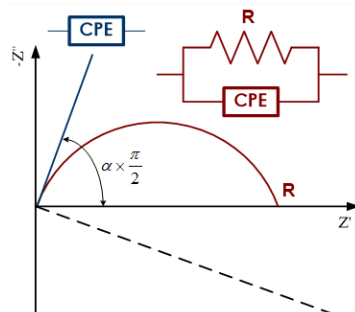
- An electrical double layer forms as ions from the solution “stick on” the electrode. There is an Å-wide separation between charge in the electrode and ionic charges in the solution.
- Charges separated by an insulator form a capacity. On a bare metal, estimate 20 to 40 μF of C for every cm^2 of electrode area.
- Depends on electrode potential, temperature, ionic concentrations, types of ions, oxide layers, electrode roughness, impurity adsorption, etc.

• Constant Phase Element (CPE)

- The CPE is basically an imperfect capacitor.
- Its phase shift is less than 90° .

$$Z_{CPE} = \frac{1}{A \times (j\omega)^\alpha}$$

- Unlike C, a CPE has 2 parameters
 - α is generally between 0.9 and 1.0
 - A is similar to C
- Possible Explanations
 - Surface roughness $\rightarrow$ Fractal Dimension, $D=1+1/\alpha$
 - Distribution of reaction rates on a surface
 - Varying thickness or composition of a coating



• Diffusion

- Diffusion processes can create impedance, which is small at high frequency and increases as frequency decreases.
- Warburg Impedance

- Warburg looks like a special CPE with $A=1/s$ and $\alpha=1/2$.
- However, remember that Warburg is derived from electrochemical kinetics.
Parameters you obtain with Warburg have physical meanings. It is only partly true for CPE.
- You can get a good fit, but how to interpret the resulting parameters?

For a one-step, multi-electron process

$$Z_w = \frac{\sigma}{\sqrt{\omega}}(1-j) = \frac{\sigma}{\sqrt{\omega}} e^{-\frac{\pi j}{4}} = \frac{\sigma}{(j\omega)^{1/2}} \quad \sigma = \frac{RT}{n^2 F^2 A \sqrt{2}} \left(\frac{1}{D_o^{1/2} C_o^*} + \frac{1}{D_R^{1/2} C_R^*} \right)$$

– Nernstian & Finite Diffusion Impedance

$$Z = \frac{\sigma}{\sqrt{\omega}}(1-j) \tanh(\delta \sqrt{j\omega/D}) \quad Z = \frac{\sigma}{\sqrt{\omega}}(1-j) \coth(\delta \sqrt{j\omega/D})$$

– Homogeneous reaction
(Gerischer)

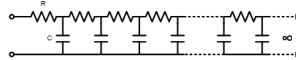
$$Z = \frac{1}{A\sqrt{B+j\omega}}$$

– Spherical Diffusion

$$Z = \frac{1}{A\sqrt{B+j\omega}}$$

- Diffusion ← Transmission Line Model

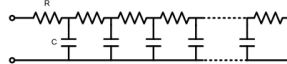
Warburg



$$Z = \frac{\sigma}{\sqrt{\omega}}(1-j)$$

W Element

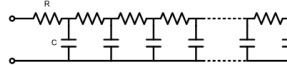
Nernstian Impedance: Diffusion by Constant Concentration



$$Z = \frac{\sigma}{\sqrt{\omega}}(1-j) \tanh(\delta \sqrt{j\omega/D})$$

O Element

Finite Diffusion Impedance: Diffusion by Phase Boundary



$$Z = \frac{\sigma}{\sqrt{\omega}}(1-j) \coth(\delta \sqrt{j\omega/D})$$

T Element

(10) Validation of impedance data

- Ideal impedance data must fulfill:
 - **Causality**: The output must be exclusively a result of the input
 - **Linearity**: The response must be a linear fn. of the perturbation
 - **Stability**: The system must not be changing during measurement
→ a serious problem for corroding systems
 - **Finite-Valued**: Impedance must be finite value at any frequency
- Kramers-Kronig Relation:
 - Validation Test
 - Low Frequency Extrapolation
 - The integration range includes the frequencies zero and infinity
 - Note pure capacitor cannot be calculated

a. $Z'' \rightarrow Z'$

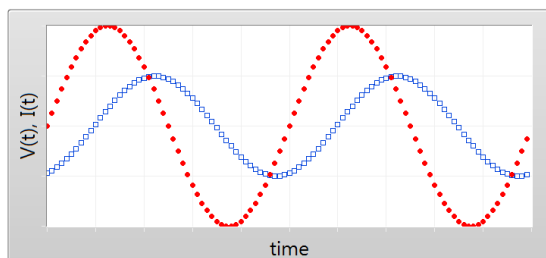
$$Z'(\omega) = Z'(\infty) + \frac{2}{\pi} \int_0^{\infty} \frac{x Z''(x) - \omega Z''(\omega)}{x^2 - \omega^2} dx$$

b. $Z' \rightarrow Z''$

$$Z''(\omega) = -\frac{2\omega}{\pi} \int_0^{\infty} \frac{Z'(x) - Z'(\omega)}{x^2 - \omega^2} dx$$

(11) Limitation of KK relation-ship

- The integration range includes the frequencies zero and infinity
- $|Z|$ and Phase are measured independently with different accuracy and sensitivity, but in theory, they are correlated with each other.



- Measuring EIS spectrum takes time (often many hours).
- The sample can change during the time the spectrum is recorded.
- If this happens, modeling results may be wildly inaccurate.
- To shorten the measuring time of impedance spectrum, use FFT EIS method.

Non-Stationary Conditions result in non-stationary spectra !

(12) How to extract model parameters

- Building equivalent circuit model
 - Physically relevant model
 - ◆ Each component is postulated to come from a physical process in the EChem cell based on knowledge of the cell's physical characteristics.
 - Empirical model
- Complex Nonlinear Least Square (CNLS) Fitting Algorithm
 - is used to find the model parameters that cause the best agreement between a model's impedance spectrum and a measured spectrum.
 - starts with initial estimates of model parameters.
 - Iterations continue until the goodness of fit exceeds an acceptance criterion, or until the number of iterations reaches a limit.
 - Please check the change of χ^2 after each iteration.
 - Sometimes, CNLS algorithm cannot converge on a useful fit because of
 - ◆ An incorrect model
 - ◆ Poor estimates for the initial values
 - ◆ Noise and etc.
 - Don't care if the fit looks poor over a small section of the spectrum.

(13) Common Parameters

- Frequency Setting
 - Initial Freq Final Freq: Frequency sweep will be done from Initial frequency to Final frequency. If you input same value for Initial and final frequency then it will measure impedance at one frequency (fixed frequency EIS).
- Amplitude value is peak to peak value. Value is half of current range value

- c. Iteration
 - Iteration is frequency scanning repeat time between initial frequency and final frequency.
Initial frequency-> Final frequency->Initial frequency-> final frequency->...
 - The data set is divided by cycle number (Iteration number will be saved as cycle number)
- d. Cycle: If the data is noisy, select "Auto". * data quality will be better but measuring time will be longer
To shorten the testing time, Decrease cycle number
- e. Density: This value is measurement point's number per decade to measure EIS. Max number is 65535
But it is limited by maximum data point 100,000

Chapter 4. ZM software setting/connection

1. Hardware installation

A. Preparation

- BZA instrument
- LAN cable
- Cell cable
- PT100 temperature sensor (option)
- Power adapter(C type cable)

B. connection

- Connect the USB C-type cable and power adapter to power
- Connect the power to the USB C-type inlet at the back of the BZA
- Connect the PT100 temperature sensor(option) at the front panel
- Connect the cell cable (The line connected to Force is the line in which the actual current flows , and the line connected to Sense is the cable for voltage measurement.
- Turn on the power by pressing the power button at the back of the BZA

2. PC-BZA direct connection

- ① Connect LAN cable between PC and BZA rear panel



- ② Run zm.exe

3. Connection via (wireless) switching hub

If your system has multiple devices, you need a switching hub.

(We do not recommend Router for this purpose).

- ① Connect LAN cable between PC and switching hub
- ② Connect LAN cables between switching hub and BZA rear panel's LAN connectors



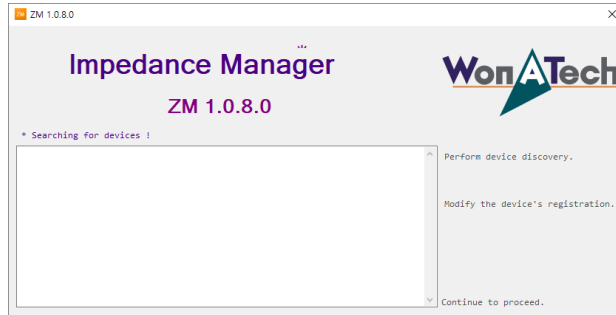
- ③ IP address will be determined automatically
- ④ Run zm.exe.

The connected devices appear in the list. If the channel is not registered, you need to enter the device registration menu and specify the channel.

Click the “perform device discovery” link and find the battery impedance analyzer. (It takes some minutes.)

4. Device discovery /Device Registration/ Firmware upgrade

BZA searching, registration and firmware upgrade



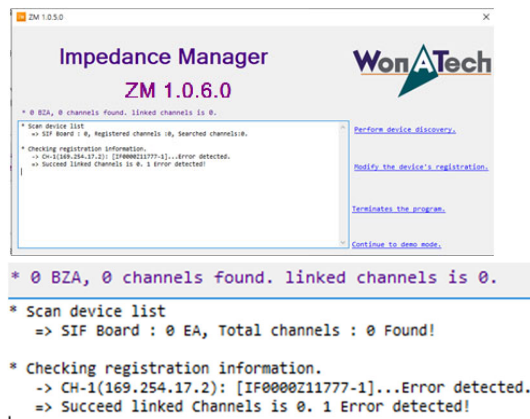
The initial menu as shown above appears when starting zm.exe, and it is under a search on the network when

this  symbol appears. The device connection result will appear on the left when the search is done. Perform device discovery

A. Perform device discovery

This link is function for find of the battery impedance analyzer on the network.

If a device is not found , displayed as 0BZA, 0channels found.



As the above screen capture, software can not detect BZA, please check the followings.

- . BZA's power switch on
- . Lan cable connection

If there is one reason which did cause the problem, fix it and try to search BZA by clicking Perform device discovery link

If there is no reason on the above two check points or if software can not detect BZA after fixing the problem, try to turn off the power of BZA and turn it on again.

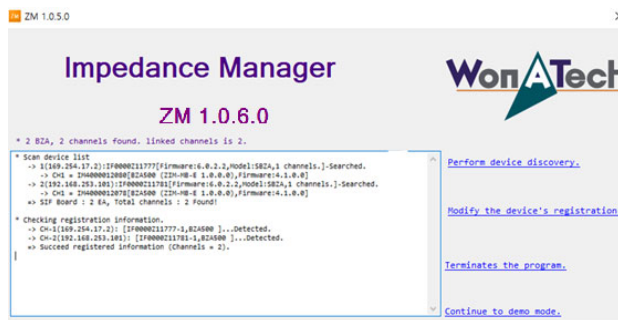
If the software can not detect BZA with above procedure, please contract with WonATech or WonATech distributor.

If the software detect BZA then the following display will show.

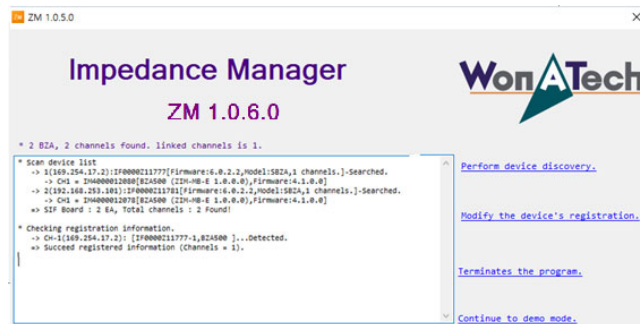


Single channel connection

Displayed as follows if more than one device is found on the network.

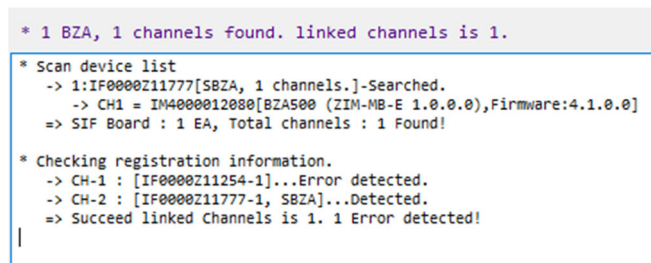


Multiple channel connection



The above means that although two devices are found, only one channel is set up, so a device needs to be registered for channel setting. Please refer to B. Modify the device's registration below.

If the connected device is turned off or communication problems occur,



display 'Error detected' as shown above, and the channel status is marked as 'no device' and 'unknown'. And the device will be normalized once reconnected.

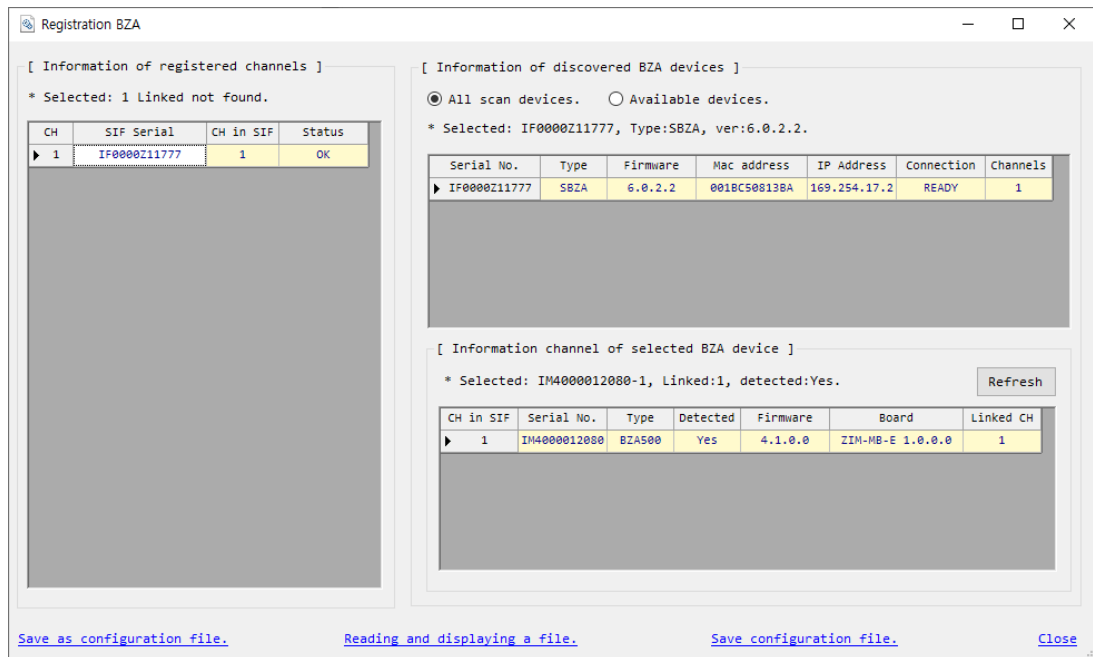
Click the link in 'Perform device discovery' to re-find devices connected to the network.

Assign each channel by entering the device registration link for multi-channel usage if multiple devices are available. Devices on the network will be automatically searched when entering zm.exe for the first time, but when the device is connected after starting the software or there is a change in connection, the connected device must be found by running the find device link again.

B. Modify the device's registration

This link is function for channel settings of the battery impedance analyzer on the network.

Click the 'Modify the device's registration' link to enter the following setting screen.



The device list set up in the channel display on the left, and SIF board information will display on the upper right as the search result.

- Serial No. : Serial number of SIF board
- Type : Type of SIF board. The single channel device or older version of SIF is displayed as SBZA and the multichannel SIF is displayed ZM.
- Firmware: Firmware version of SIF board
- MAC address : MAC address on the connected network
- IP address : Assigned IP address
- Connection status
 - Ready: Connection available but not connected yet
 - Busy: Found BZA but can not connect (other used this BZA etc)
 - Connected: User did connect this BZA
- Channels : Assigned channel number to the SIF board

Channel information will display on the lower right as the search result.

- Sireal No. : Serial number of Channel
- Type : Type of Channel (BZA60, BZA1000)
- Detected : Detect the connection status
- Firmware : Firmware version of channel

- Board : Board version of channel
- Linked CH : Linked channel number

The device registration screen is as follows.

[Information of discovered BZA devices]

☒ All scan devices. ☐ Available devices.

* Selected: IF0000Z11777, Type:SBZA, ver:6.0.2.2.

Serial No.	Type	Firmware	Mac address	IP Address	Connection	Channels
▶ IF0000Z11777	SBZA	6.0.2.2	001BC50813BA	169.254.17.2	CONNECTED	1
IF0000Z11781	SBZA	6.0.2.2	001BC50813BE	192.168.253.101	CONNECTED	1

The above photo shows two Battery Impedance Analyzer devices were found on networking. Direct connection show fixed IP address as 169.254.17.2. This address is fixed for direct connection. Other IP address shows Battery Impedance Analyzer devices which connected on TCP/IP connection.

Click the 'Save configuration file' link to save once the device(s) is registered to the channel. Connected device(s) will be automatically rescanned when exiting this screen by clicking 'Close'.

Displayed as 'Busy' on the device's SIF connection if someone else on the network is using the device. The device is connected but the channel registration is not completed if it is displayed as 'Ready,' and it will be marked 'connected' when both device connection and channel registration are successful.

[Information of discovered BZA devices]

☒ All scan devices. ☐ Available devices.

* Selected: IF0000Z11777, Type:SBZA, ver:6.0.2.2.

Serial No.	Type	Firmware	Mac address	IP Address	Connection	Channels
▶ IF0000Z11777	SBZA	6.0.2.2	001BC50813BA	169.254.17.2	CONNECTED	1
IF0000Z11781	SBZA	6.0.2.2	001BC50813BE	192.168.253.101	CONNECTED	1

If the device is an existing channel registered device, the channel is displayed as inactive on the main screen and cannot be used.

Single channel configuration or multichannel configuration with single channel model

(1) Assign channel number

```
* 2 BZA, 1 channels found. linked channels is 1.

* Scan device list
-> 1(192.168.253.101):IF0000Z11781[Firmware:6.0.2.2,Model:SBZA,1 channels.]-Connected.
-> CH1 = IM4000012078[BZA500 (ZIM-MB-E 1.0.0.0),Firmware:4.1.0.0]
-> 2(169.254.17.2):IF0000Z11777[Firmware:6.0.2.3,Model:SBZA,1 channels.]-Searched.
-> CH1 = IM4000012080[BZA500 (ZIM-MB-E 1.0.0.0),Firmware:4.1.0.0]
=> SIF Board : 2 EA, Total channels : 1 Found!

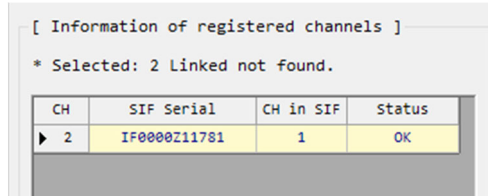
* Checking registration information.
-> CH-2(192.168.253.101): [IF0000Z11781-1,BZA500 ]...Connected.
=> Succeed registered information (Channels = 1).
```

On device searching, if the searching result is as the above, it means software found 2set of BZA and 1 linked channel only. 2nd device was not assigned to the channel yet.

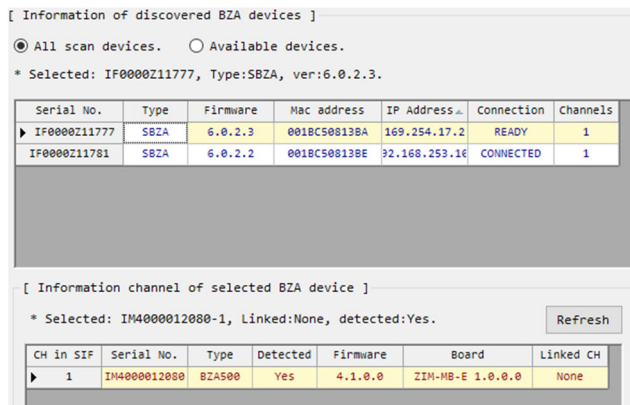
You need to assign the channel correctly.

Click “modify the device’s registration link.

Click the right mouse button in the lower right for the following menu to assign a channel number to a searched device.

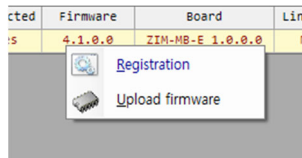


At left side, there is only 2nd channel.



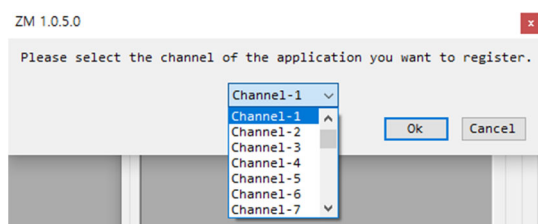
Brown colored channel was not assigned (not linked CH)

Click mouse right button then submenu will be displayed.

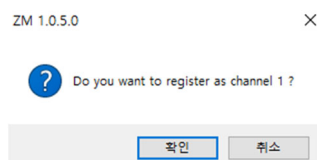


Click the 'Registration' to select the channel number for the assignment to assign a channel number.

the channel assigning menu will show the following which allows assigning the desired channel number.

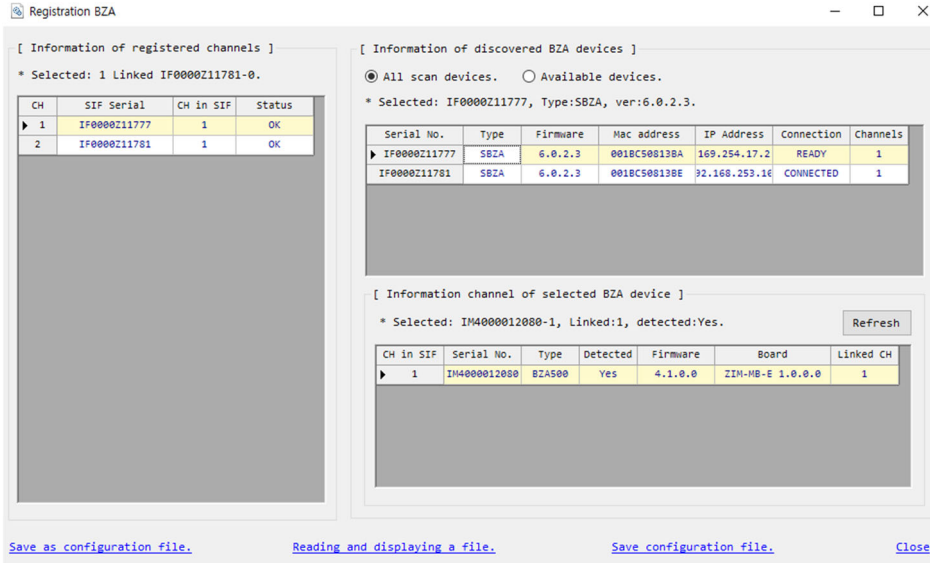


Click the 'OK' button.

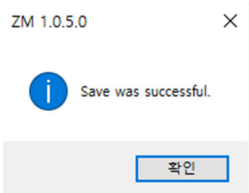


For the next step, when the click the 'OK' button, you can see that the device is

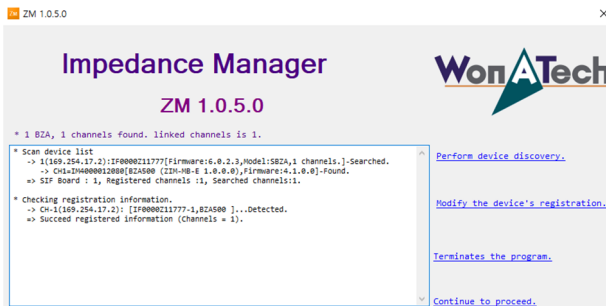
assigned to channel 1.



Click the 'Save configuration file' to save this setting.

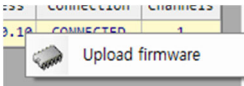


After this work, if you click close link , software will re-perform the task to discover devices connected to the network.



(2) SIF firmware upgrade

Click on the right mouse button to select upload firmware on the top or the bottom right side to update the



firmware to update the firmware of the SIF and channel.

SIF firmware file extension is *.sif.

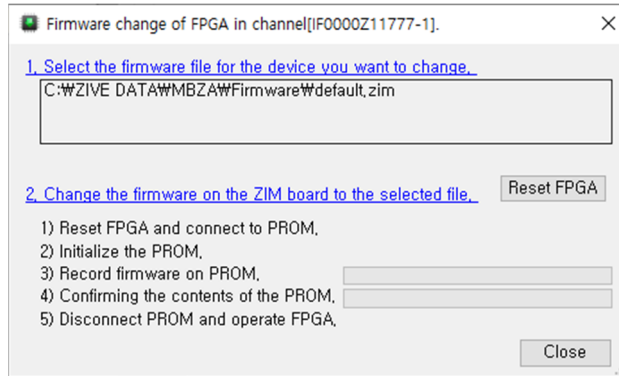
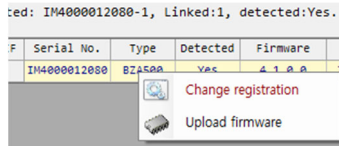
If the update SIF firmware, you can upgrade your system using firmware file.

Select a new firmware file and click the 'OK' button.

IMPORTANT: YOU MUST TURN OFF THE INSTRUMENT AND TURN ON IT AFTER UPGRADING SIF FIRMWARE

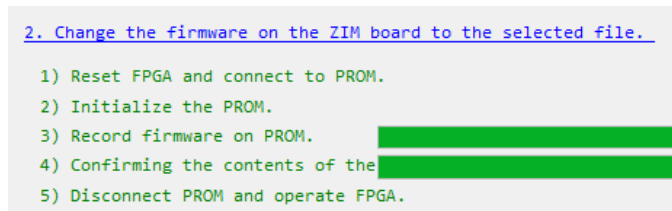
(3) FPGA firmware upgrade

Click on the right mouse button on lower right and click the 'Upload firmware' to update the FPGA firmware.



When you click this [1. Select the firmware file for the device you want to change.](#), you can select FPGA firmware file. This file extension is *.zim.

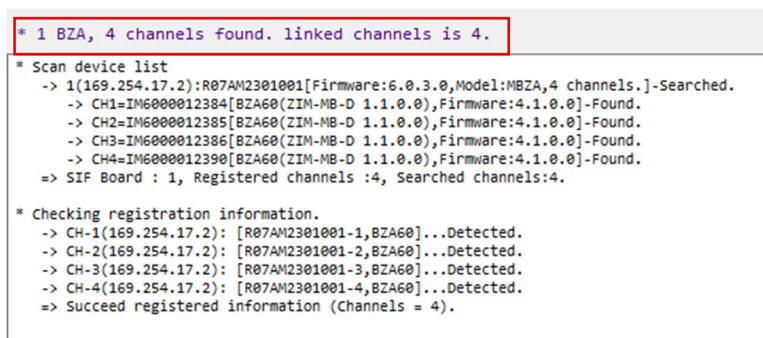
When you click this [2. Change the firmware on the ZIM board to the selected file.](#), you can proceed the update.



When the color changed to green as the above, firmware was successfully completed. Click the 'Close' button.

Multichannel configuration with multichannel model

(1) Assign channel number



On device searching, if the searching result is as the above, it means software found 1set of BZA and 4 linked channel

If you need to assign the channel differently

Click "modify the device's registration link.

Click the right mouse button in the lower right for the following menu to assign a channel number to a searched device.

[Information of registered channels]

* Selected: 1 Linked R07AM2301001-1.

CH	SIF Serial	CH in SIF	Status
▶ 1	R07AM2301001	1	OK
2	R07AM2301001	2	OK
3	R07AM2301001	3	OK
4	R07AM2301001	4	OK

At left side, there are 1st-4th channels.

[Information of discovered BZA devices]

☒ All scan devices. ☐ Available devices.

* Selected: R07AM2301001, Type:MBZA, ver:6.0.3.0.

Serial No.	Type	Firmware	Mac address	IP Address	Connection	Channels
▶ R07AM2301001	MBZA	6.0.3.0	0018C5001778	169.254.17.2	READY	4

[Information channel of selected BZA device]

* Selected: IM6000012384-1, Linked:1, detected:Yes. Refresh

CH in SIF	Serial No.	Type	Detected	Firmware	Board	Linked CH
▶ 1	IM6000012384	BZA60	Yes	4.1.0.0	ZIM-MB-D 1.1.0.0	1
2	IM6000012385	BZA60	Yes	4.1.0.0	ZIM-MB-D 1.1.0.0	2
3	IM6000012386	BZA60	Yes	4.1.0.0	ZIM-MB-D 1.1.0.0	3
4	IM6000012390	BZA60	Yes	4.1.0.0	ZIM-MB-D 1.1.0.0	4

If there are brown colored channel(s), That means not assigned (not linked CH) yet

Click mouse right button then submenu will be displayed.

cted	Firmware	Board	Lin
IS	4.1.0.0	ZIM-MB-E 1.0.0.0	N

Registration

 Upload firmware

Click the 'Registration' to select the channel number for the assignment to assign a channel number.

the channel assigning menu will show the following which allows assigning the desired channel number.

ZM 1.0.5.0

Please select the channel of the application you want to register.

Channel-1
Channel-1
Channel-2
Channel-3
Channel-4
Channel-5
Channel-6
Channel-7

Ok Cancel

Click the 'OK' button.

ZM 1.0.5.0

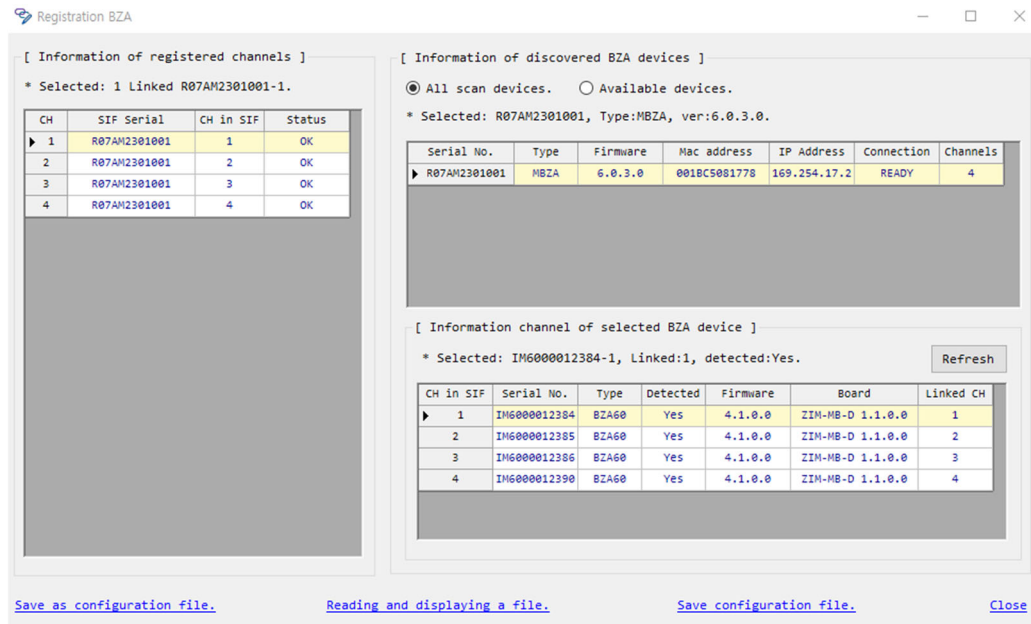
Do you want to register as channel 1 ?

확인

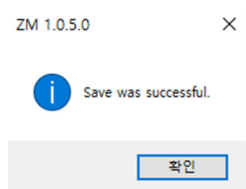
취소

For the next step, when the click the 'OK' button, you can see that the

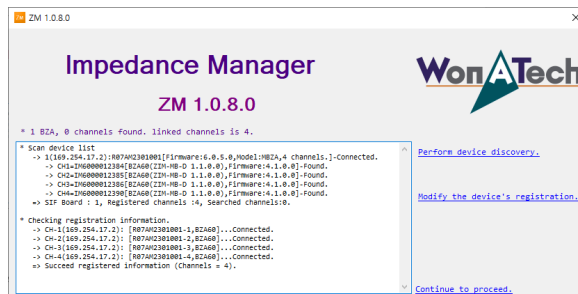
device is assigned to selected channel number.



Click the 'Save configuration file' to save this setting.

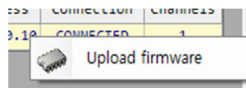


After this work, if you click close link , software will re-perform the task to discover devices connected to the network.



(2) SIF firmware upgrade

Click on the right mouse button to select upload firmware on the top or the bottom right side to update the



firmware to update the firmware of the SIF and channel.

SIF firmware file extension is *.sif.

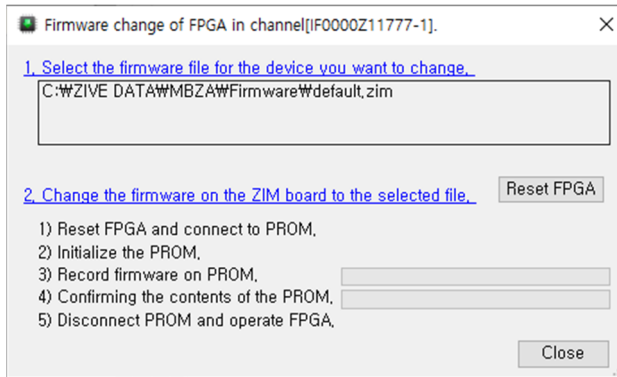
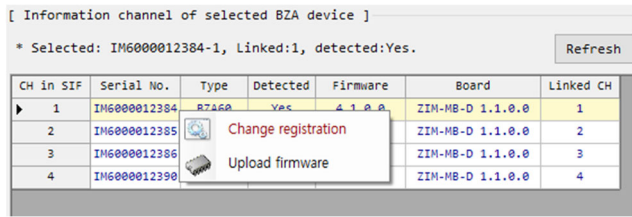
If the update SIF firmware, you can upgrade your system using firmware file.

Select a new firmware file and click the 'OK' button.

IMPORTANT: YOU MUST TURN OFF THE INSTRUMENT AND TURN ON IT AFTER UPGRADING SIF FIRMWARE

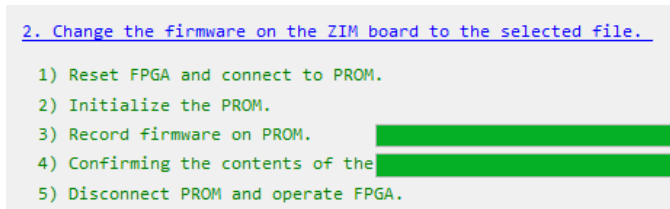
(3) FPGA firmware upgrade

Click on channel row with the right mouse button on lower right and click the 'Upload firmware' to update the FPGA firmware.



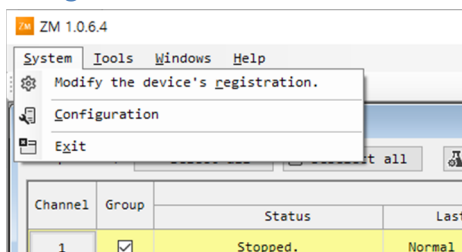
When you click this [1. Select the firmware file for the device you want to change.](#), you can select FPGA firmware file. This file extension is *.zim.

When you click this [2. Change the firmware on the ZIM board to the selected file.](#), you can proceed the update.

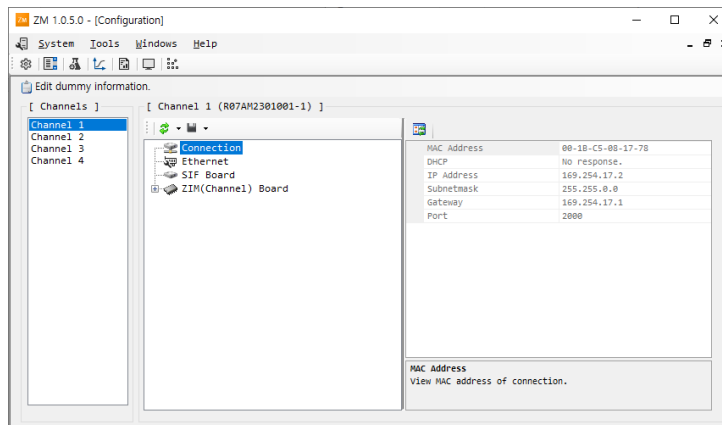


When the color changed to green as the above, firmware was successfully completed. Click the 'Close' button.

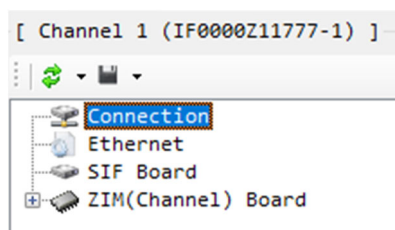
5. Configuration



Click the 'system-configuration' in the menu to view the system configuration or edit modifiable contents.

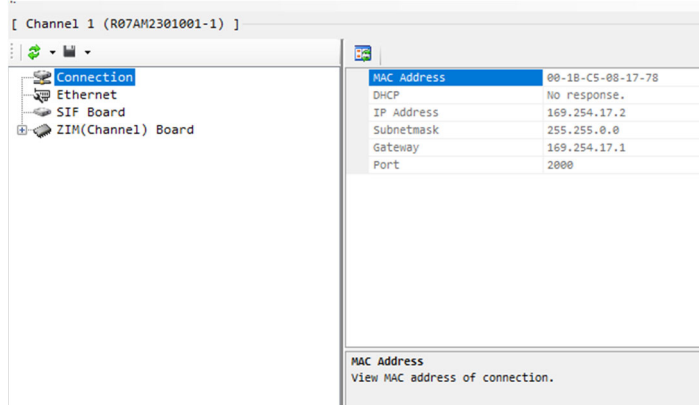


Items for configuration will display in the middle when selecting a channel for configuration in the left channel item.



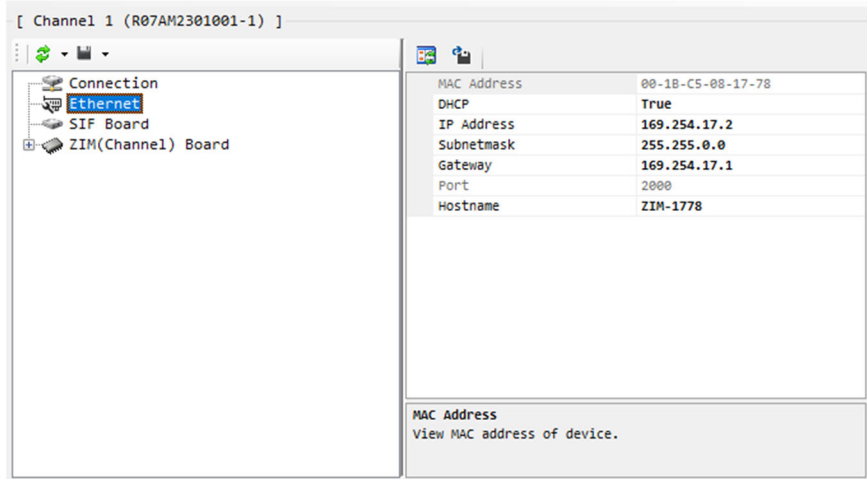
A. Connection

if you select 'Connection', you can see the information about the connection on the right.



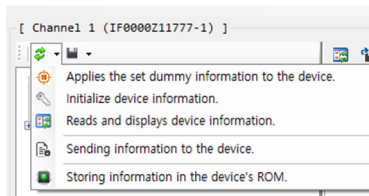
B. Ethernet

You can see the information about Ethernet and edit it if necessary. Items with dark black characters can be edited.



Click the 'apply'  icon to apply changes after the edit.

In this state, the changes are not applied to the device or saved in ROM, so click this  icon to activate the following menu.

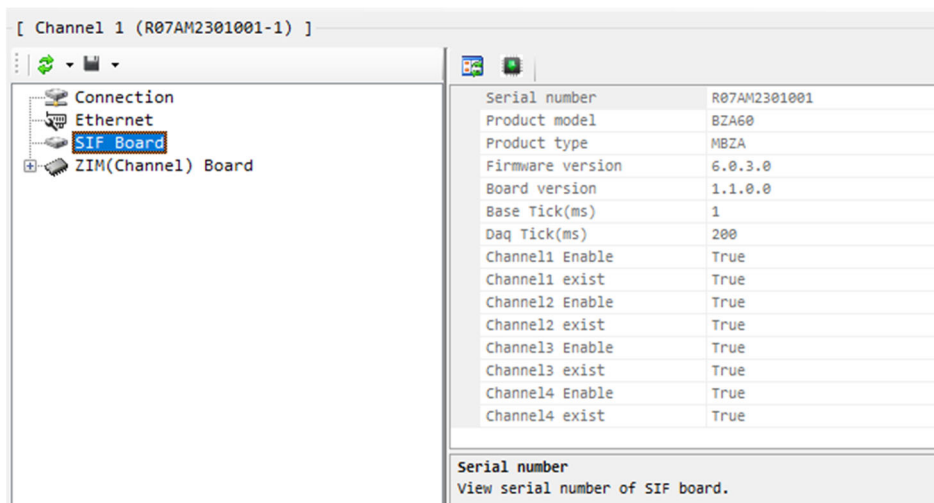


Selecting the 'Sending information to the device' in the menu above will transfer the edited contents to the device for activation, but the previous setting will be applied without saved contents if the device is turned off. Click the 'Storing information in the device's ROM' to save in device ROM.

(Caution) Before storing to the ROM, you must select 'Sending information to the device' to transfer the changes to the device.

C. SIF board

You can see the information set in the SIF.

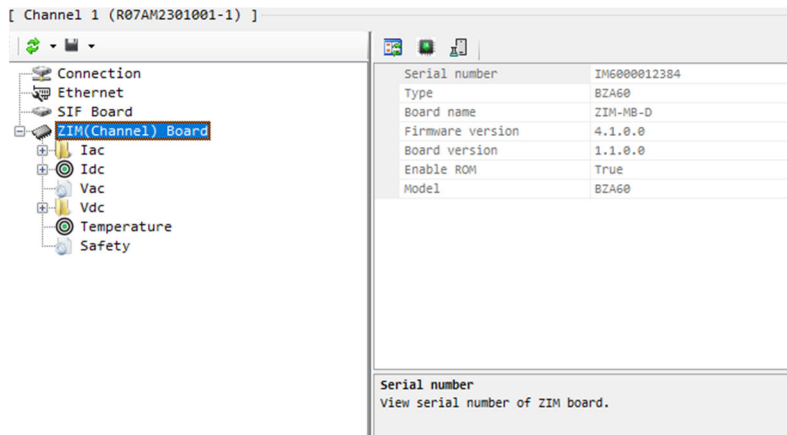


The meaning for each item are described below if click to select an item on the right.

Click this  icon, you can also upgrade the SIF firmware here.

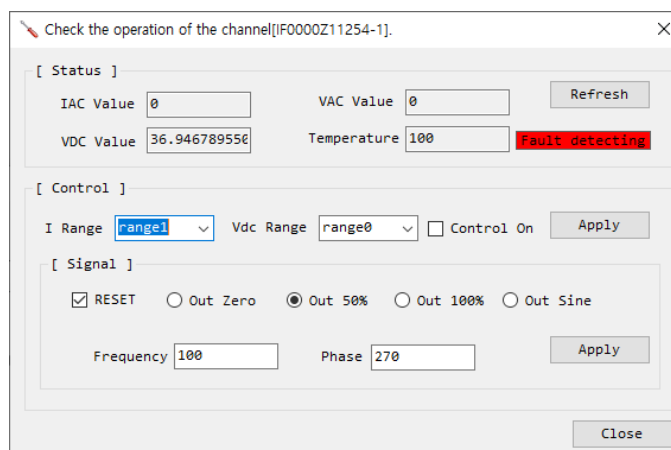
D. ZIM board

You can see the information about ZIM board(Channel board).



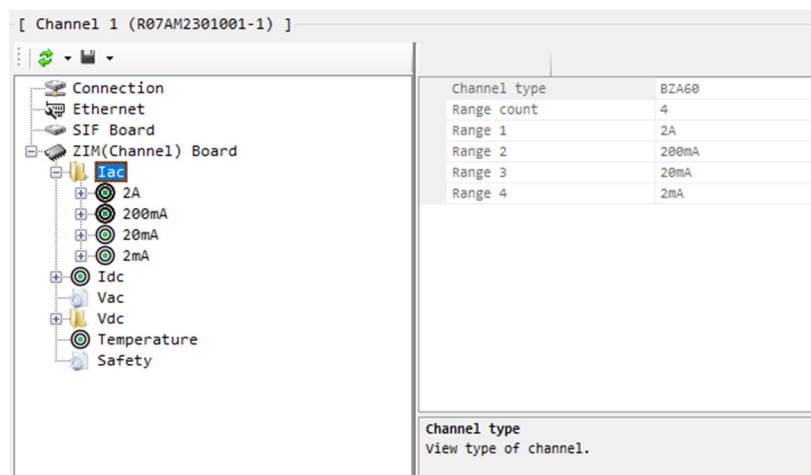
Click this  icon, you can also upgrade the ZIM(Channel hardware) firmware here.

Click this  icon, you can check the channel board status.
This menu is not available to user who are not trained.



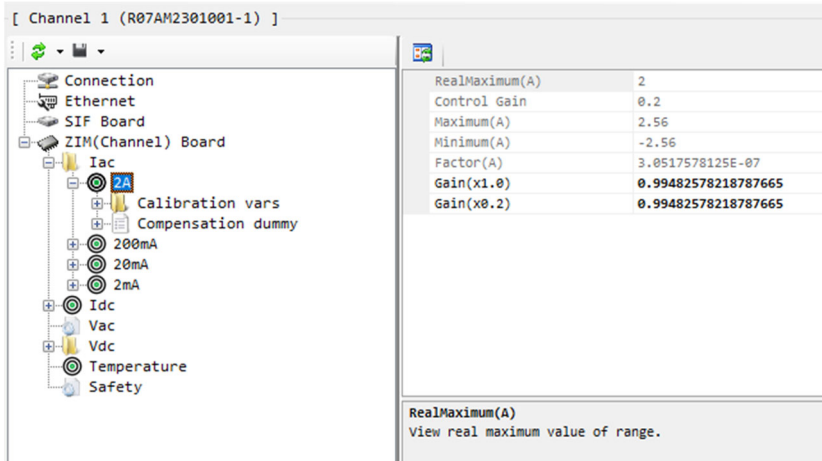
1) Iac information

You can see the information about the AC current.



Information by each current range.

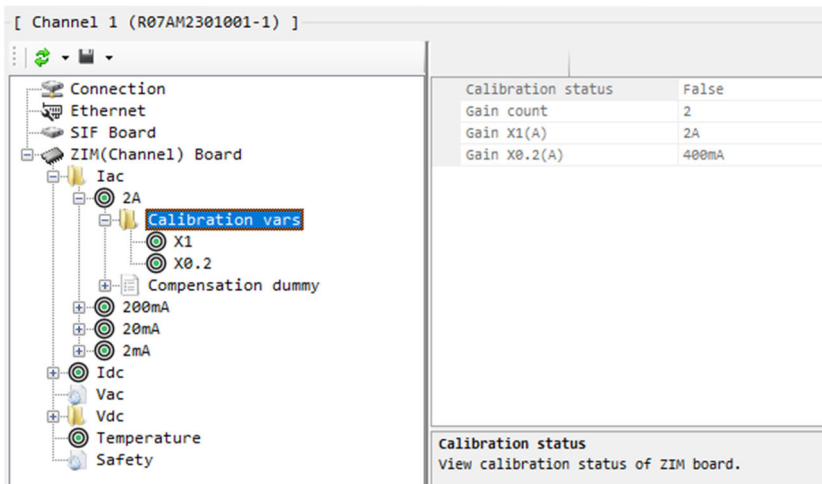
Information about each current range will display when clicking each current range as the following.



Parameter	Value
RealMaximum(A)	2
Control Gain	0.2
Maximum(A)	2.56
Minimum(A)	-2.56
Factor(A)	3.0517578125E-07
Gain(X1.0)	0.99482578218787665
Gain(X0.2)	0.99482578218787665

RealMaximum(A)
View real maximum value of range.

a. Calibration variables:



Parameter	Value
Calibration status	False
Gain count	2
Gain X1(A)	2A
Gain X0.2(A)	400mA

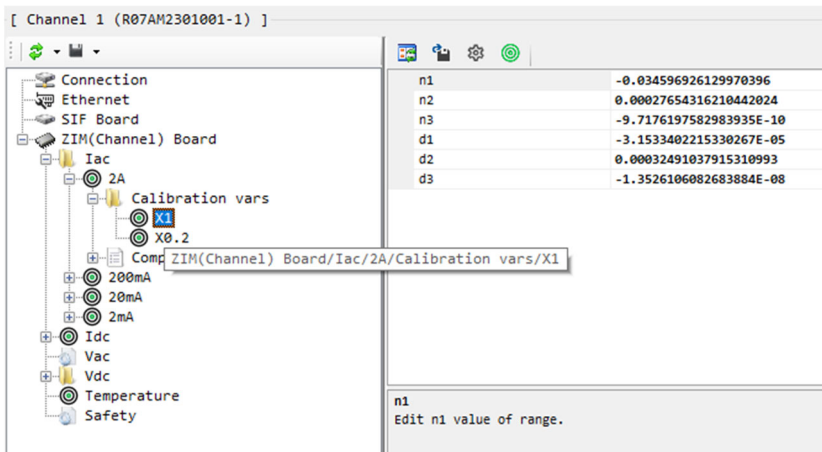
Calibration status
View calibration status of ZIM board.

You can see or manually edit the parameter values obtained through calibration.

This menu is not available to user who are not trained.

You can see the calibration values for once and 0.2 times for each current range and you can also be calibrated.

The status below is before the calibration.



Parameter	Value
n1	-0.034596926129970396
n2	0.00027654316210442024
n3	-9.7176197582983935E-10
d1	-3.1533402215330267E-05
d2	0.00032491037915310993
d3	-1.3526106082683884E-08

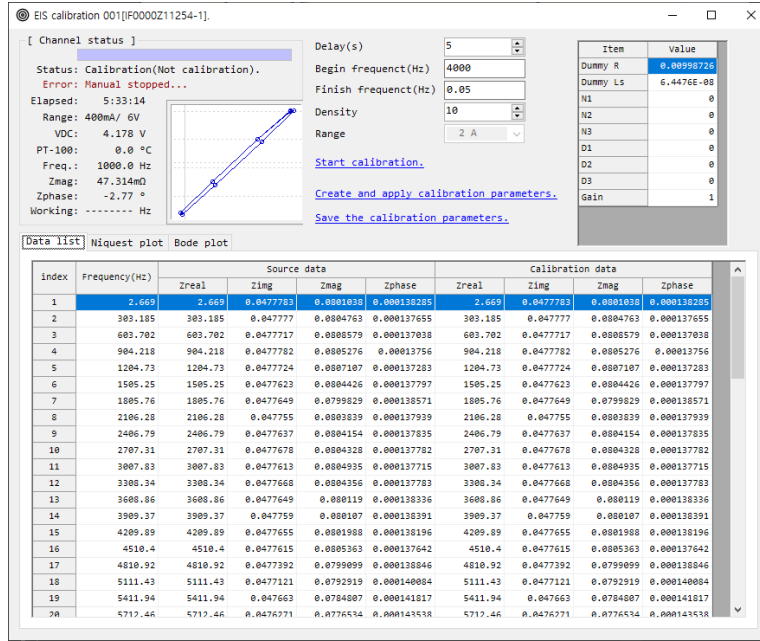
Calibration status
View calibration status of ZIM board.

Calibration vars
X1
X0.2

Comp ZIM(Channel) Board/Iac/2A/Calibration vars/X1

n1
Edit n1 value of range.

AC calibration can be performed by click this  icon .



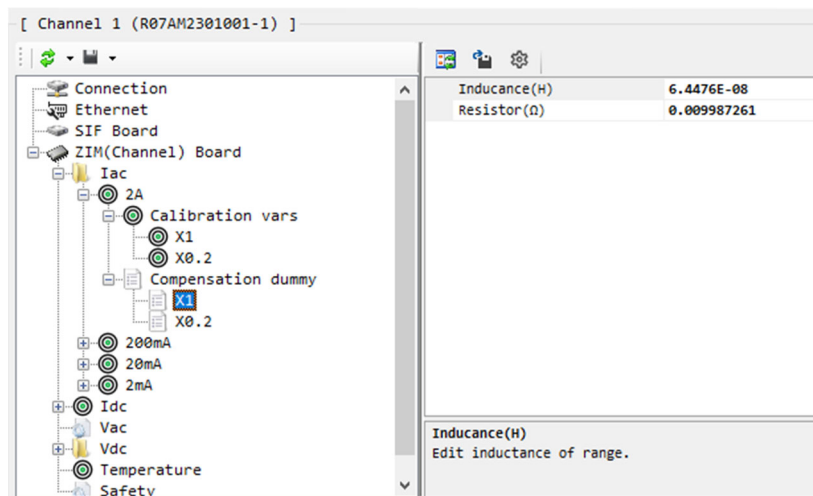
Please refer to separated calibration manual on how to calibration. A resistor for calibration is need for calibration.

Click this  icon to initialize the saved calibration information..

When click this  icon, the device only allows for uploading the parameter to the software. To send and save in the device, the click 'Storing information in the device's ROM' for save it in the device's ROM.
(Caution) Before storing to the ROM, you must select 'Sending information to the device' to transfer the changes to the device.

b. Compensation dummy

This function is to enter the values related to the resistance for calibration into X1 and X0.2 of each current range.

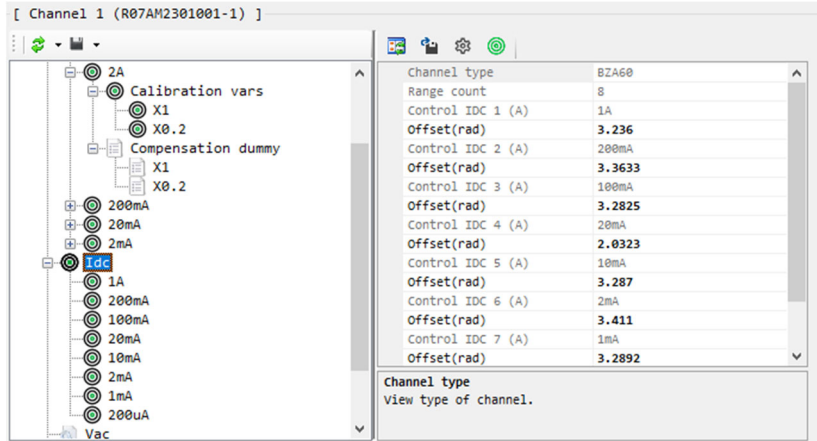


If you have purchased a calibration resistor from the manufacturer, you will be provided with an inductance value and resistance value for each calibration resistor.

When click this  icon, the device only allows for uploading the parameter to the software. To send and save in the device, the click 'Storing information in the device's ROM' for save it in the device's ROM.
(Caution) Before storing to the ROM, you must select 'Sending information to the device' to transfer the changes to the device.

2) Idc info

Current information for discharging test

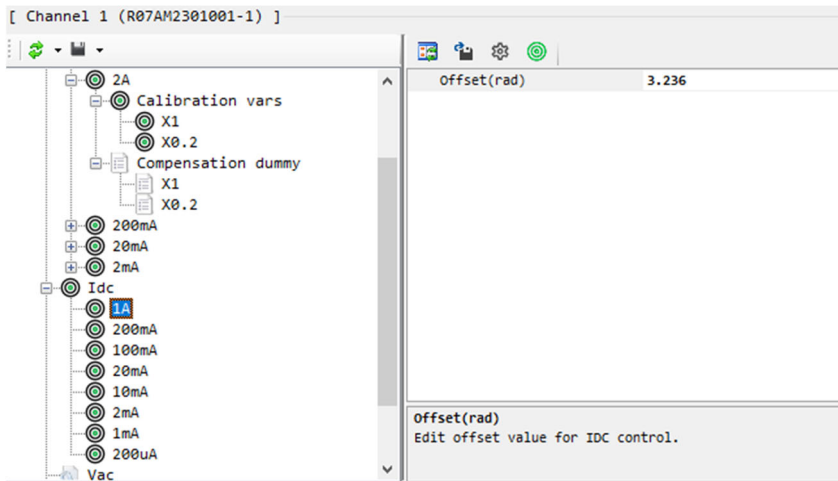


Channel type	BZA60
Range count	8
Control IDC 1 (A)	1A
Offset(rad)	3.236
Control IDC 2 (A)	200mA
Offset(rad)	3.3633
Control IDC 3 (A)	100mA
Offset(rad)	3.2825
Control IDC 4 (A)	20mA
Offset(rad)	2.0323
Control IDC 5 (A)	10mA
Offset(rad)	3.287
Control IDC 6 (A)	2mA
Offset(rad)	3.411
Control IDC 7 (A)	1mA
Offset(rad)	3.2892

Channel type
View type of channel.

Info per discharging current value

If you select discharging current value, you can see offset value at right sidfe



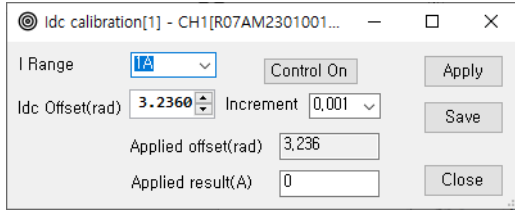
Offset(rad)	3.236
Offset(rad)	3.236

Offset(rad)
Edit offset value for IDC control.

c. DC current calibration:

Offset(rad)	3.236
Control IDC 2 (A)	200mA
Offset(rad)	3.3633
Control IDC 3 (A)	100mA
Offset(rad)	3.2825
Control IDC 4 (A)	20mA
Offset(rad)	2.0323
Control IDC 5 (A)	10mA
Offset(rad)	3.287
Control IDC 6 (A)	2mA
Offset(rad)	3.411
Control IDC 7 (A)	1mA
Offset(rad)	3.2892
Control IDC 8 (A)	200uA
Offset(rad)	3.4273

Click  icon to calibrate DCd current



Idc calibration[1] - CH1[R07AM2301001...

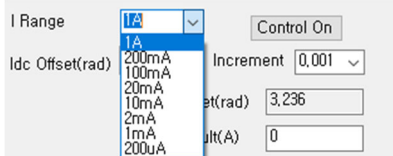
I Range: 1A Control On Apply

Idc Offset(rad): 3.2360 Increment: 0.001 Save

Applied offset(rad): 3.236

Applied result(A): 0 Close

You need ammeter to calibrate DC current.



I Range: 1A Control On

Idc Offset(rad): 3.2360 Increment: 0.001

Applied offset(rad): 3.236

Applied result(A): 0

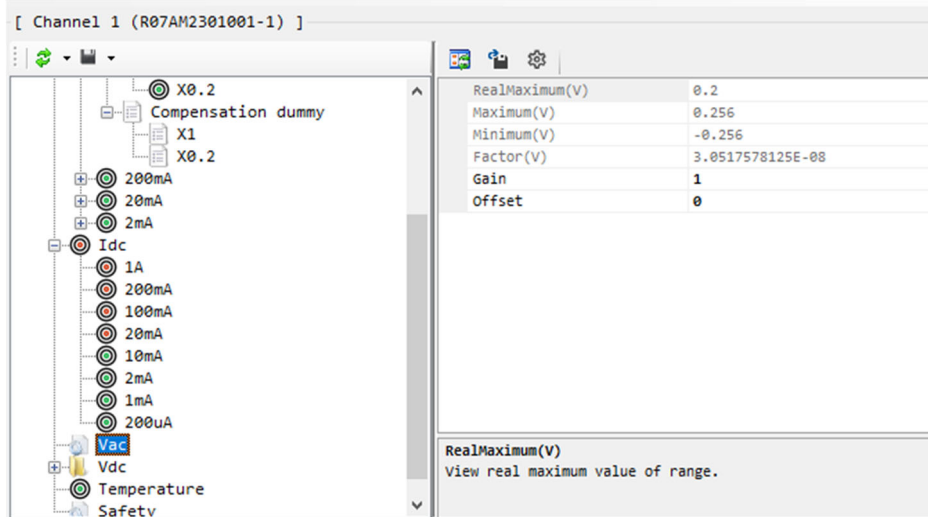
You can calibrate 8 DC current value.

Click this  icon to initialize the saved calibration information..

When click this  icon, the device only allows for uploading the parameter to the software. To send and save in the device, the click 'Storing information in the device's ROM' for save it in the device's ROM.

(Caution) Before storing to the ROM, you must select 'Sending information to the device' to transfer the changes to the device.

3) Vac information



[Channel 1 (R07AM2301001-1)]

RealMaximum(V): 0.2

Maximum(V): 0.256

Minimum(V): -0.256

Factor(V): 3.0517578125E-08

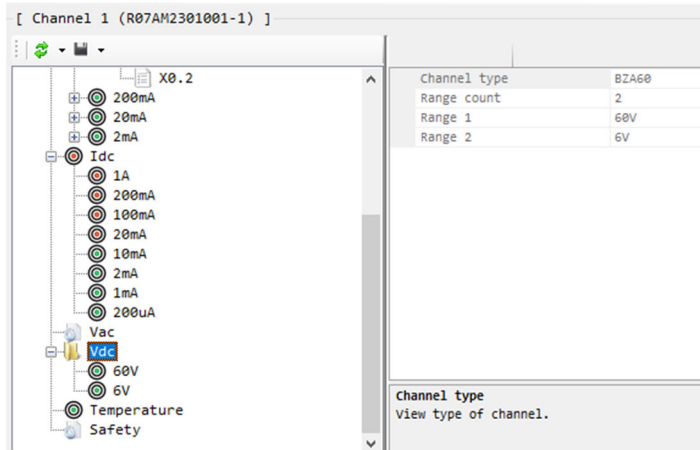
Gain: 1

Offset: 0

RealMaximum(V)
View real maximum value of range.

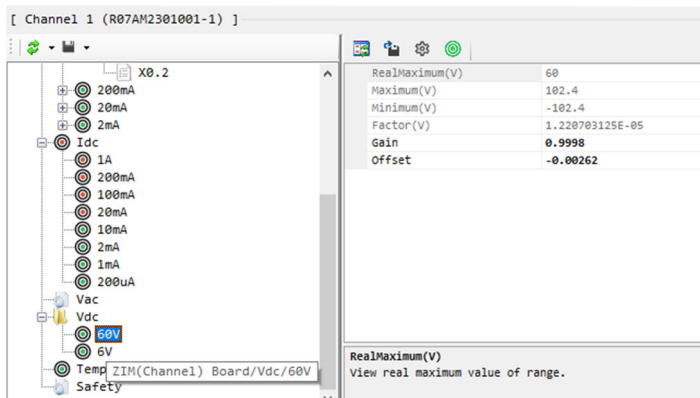
4) Vdc Information

You can see the information about Vdc.

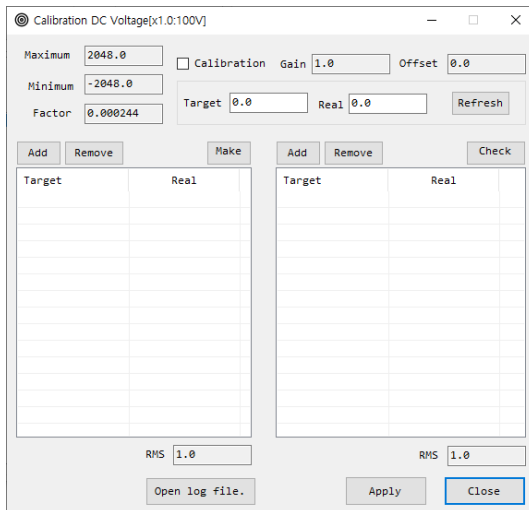


Information on each voltage range.

If you click on each voltage range, you can see the information about each voltage range as follow.



Vdc calibration can be performed by clicking this  icon.

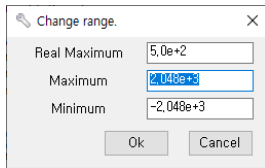


This menu is not available to user who are not trained.

The status above is before the calibration.

Please refer to separated calibration manual on how to calibration.

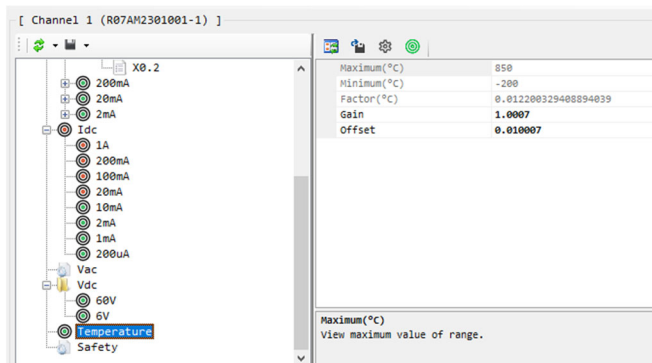
Click this  icon, the value for each voltage range is displayed. .



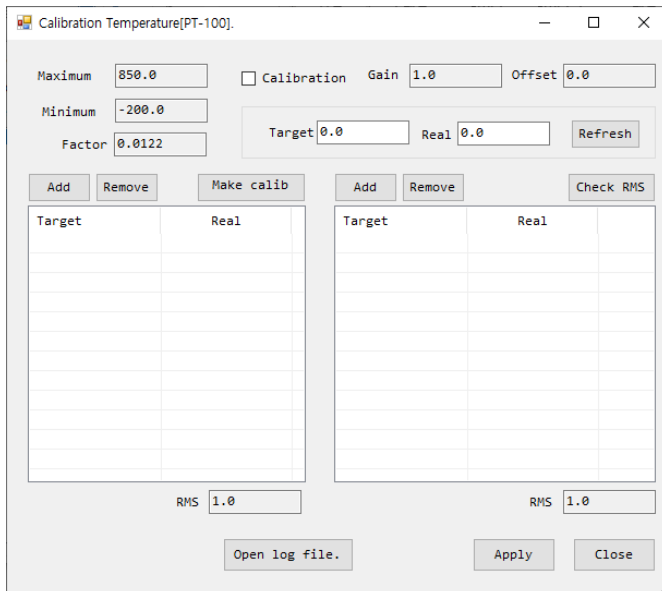
When click this  icon, the device only allows for uploading the parameter to the software. To send and save in the device, the click 'Storing information in the device's ROM' for save it in the device's ROM.
(Caution) Before storing to the ROM, you must select 'Sending information to the device' to transfer the changes to the device.

5) Temperature

You can see the information about temperature and you can calibration.



Calibration temperature can be performed by click this  icon .



This menu is not available to user who are not trained.

The status above is before the calibration..

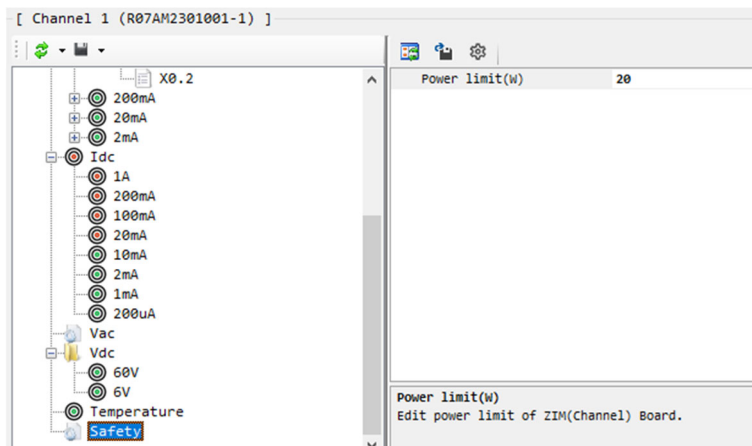
Please refer to separated calibration manual on how to calibration.

Click this  icon, the calibration temperature value is initialized.

When click this  icon, the device only allows for uploading the parameter to the software. To send and save in the device, the click 'Storing information in the device's ROM' for save it in the device's ROM

(Caution) Before storing to the ROM, you must select 'Sending information to the device' to transfer the changes to the device.

6) Safety setting



You can input the maximum Watt for each model.

The current range is automatically lowered one step down to the current range if a high wattage than this input is detected.

After changing the value, the device only allows for uploading the parameter to the software when click this



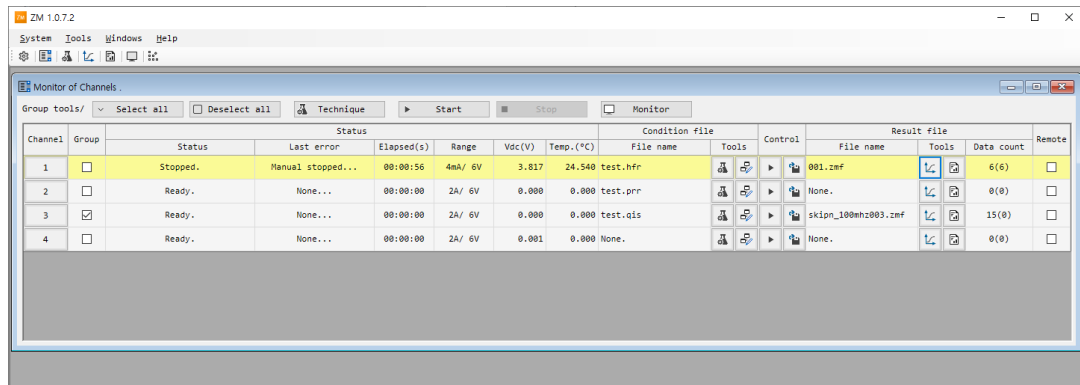
icon. To send and save in the device, the click 'Storing information in the device's ROM' for save it in the device's ROM.

(Caution) Before storing to the ROM, you must select 'Sending information to the device' to transfer the changes to the device.

Chapter 5. Control panel

Impedance Manager main software (control panel) for control and data collection display when clicking the 'Continue to proceed' link on the initial screen.

You can create/edit of technique condition files, channel assign of technique condition file, control experiments (start/stop, etc.), and process data.



Depending on the channel setting, all channels are displayed as above.

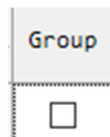
If the control panel is hidden, the control panel screen display when click this  icon. The control panel screen is always active while the software is running, the screen only hides or shows.

1. Group setting and group control

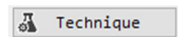
Simultaneous experiments are available by selecting multi channels as a group if it is a multi-channel device.



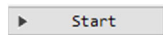
There are buttons for group control as above at the top of the control panel, and the desired channels must be selected for group control.



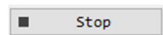
Manually check the group column as on the left to select or deselect, or click 'Select all' button to select entire channel, or click 'Deselect all' button to deselect the entire channel to select for group control of the desired channels.



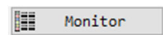
Click the 'Technique' button, experiment conditions of the channels assigned in a group are changed to the specified condition file when selecting the technique file to be assigned in a group.



Click the 'Start' button, the channels assigned in a group start the test simultaneously.



Click the 'Stop' button, the channels assigned in a group stop the test simultaneously.



Click the 'Monitor' button, the channels assigned in a group can see real-time test information and real-time graph simultaneously.

a. Start

This is a button to start the test of the assigned channels according to the condition files assigned for each channel.

If channel(s) have no assigned technique file, the following error message will appear.

ZM 1.0.4.3

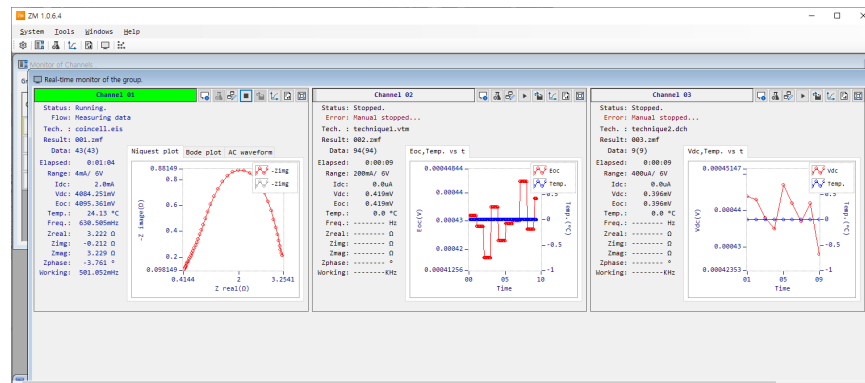


There is a problem with the condition file specified for channel 2.

If there is no problem on assigning technique file, the following start dialog will appear

- **Result file**
 - Select the folder: Click this icon to assign the folder where the data files will be saved. Default folder is **C:\ZIVE DATA\ZM\Data**
 - You can input the data file name to the right side of the 'Name'. A 3 digits number will display after the file you inputted(3 digits number means channel number), the file extension is zmf.
- **Information**
 - If check here ☒ **Use the parameters of the technique file.**, data you inputted in the condition file such as Battery id, Nominal capacity and User will be used as a value if check here. If unchecked, you can input and change it.
 - Memo: You can input a brief memo about the experiment. The line break is Ctrl+enter.
 - You can input the data file name to the right side of the 'Name'. A 3 digits number will appear after the file you inputted (3 digit number means channel number), the file extension is zmf.
 - The Battery id and nominal capacity, etc. for each channel are displayed and can also be edited if you select a channel in the 'Target' part on the right.
- **Real-time graph**

The display real-time information for an assigned channel if click the monitor button as follows,

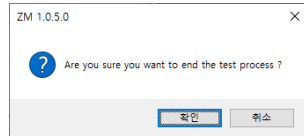


This real-time graph shows only the channels assigned as a group and can only be displayed within the main window.

For more information on real-time graph, please refer to separate real-time graph section.

b. Stop

The following message window will display if you click the stop test button.



Click 'OK' button to stop the test immediately on the assigned channels.

The status display of the real-time graph will display as **Status: Stopped** and the error below will show as **Error: Manual stopped...**

If the device is successfully turned off, displayed as 'Normal stopped' as shown

Status: Stopped(Not cal
Error: Normal stopped.

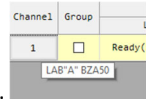
2. Single channel control

Channel	Group	Status				Condition file		Control	Result file			Remote				
		Status	Last error	Elapsed(s)	Range	Vdc(V)	Temp.(°C)		File name	Tools	File name		Tools	Data count		
1	☒	Running.	None...	00:00:01	4mA/ 6V	4.091	24.040	coincell.eis				001.zmf			0(0)	☐

Each line is designed to control and monitor each channel.

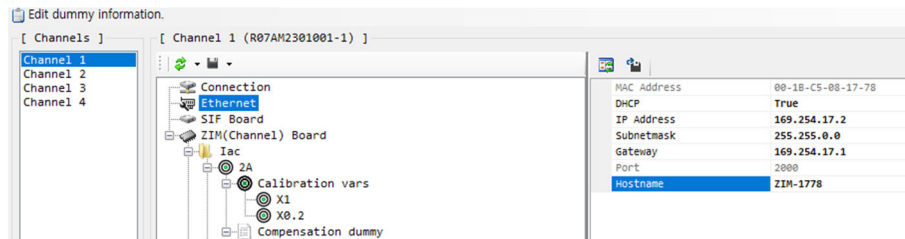
A. Channel

The channel number will appear with the tooltip to identify the location of the device with the name if you move



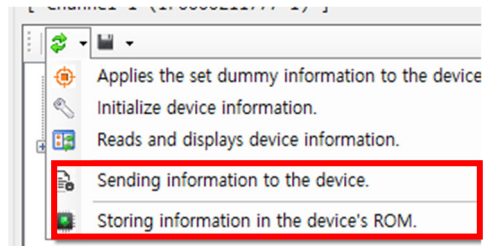
the cursor over the channel number.

This name can be edited in system-configuration-host name.



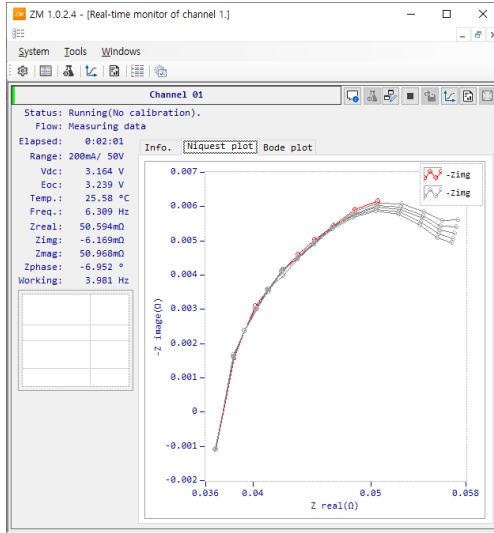
The maximum length of the name is 20 characters in a combination of English characters and numbers and within 10 characters in Korean.

Click the 'apply data' icon  after entering and,



Click 'Sending information to the device' and , 'Storing information in the devices ROM' in order.

Click the channel number, you can see the real-time monitor and graph screen for that channel.



For more information on real-time monitor and graph, please refer to separate chapter 6.

B. Group

Please check the channel you want to use as a group.

C. Status

(1) status

- Ready: After the power has been turned on, the experiment has not started yet and device is ready for the experiment.
- Running: During the experiment
- Calibration: on calibration process
- Finished: Normally experiment done
- Stopped: Stop experiment
(no calibration) tag will be added to the above status message before calibration
- Load data from device: Reading data from the device
- No device: Disconnected
- No Channel: fail to find channel
- Disconnected: Lost communication connection

(2) Last error

- None: No error
- Low voltage: Reading voltage is lower than low voltage setting value.
- Not ready: The device is not ready
- Communication error: fail to communicate
- NoZim: lost channel connection.
- Failed control: Fail to control channel
- Failed reset channel: software tried to reset channel but failed
- Heat sink is very hot: Stop by high temperature
- Unstable AC voltage: Stop or not started by unstable AC voltage
- Error technique: Failed to use technique file
- Normal stopped: Experiment was finished
- Manual stopped: User did stop experiment

- Failed FFT: FFT calculation failed
- Error memory: failed to load data to memory
- Error result file: Fail to save data into file

(3) Elapsed(s)

Duration for the recent experiment or elapsed time from the ongoing experiment

(4) Range

The current setting of the current range and voltage range display

(5) Vdc

Battery DC voltage

(6) Temp

Temperature value measured by the current temperature sensor

D. Technique file

(1) File name

Condition file name

(2) Tools

a. Assign technique file

You can assign it by selecting the experiment condition file assigned as a file in the technique menu.

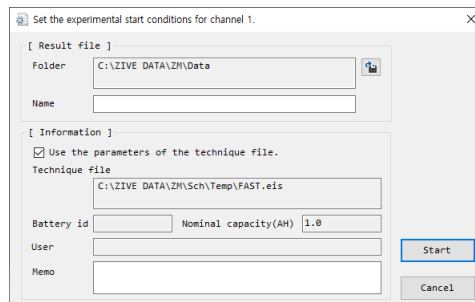
b. Technique file parameter editing

You can save it by editing the experiment condition file assigned as a file in the technique menu.

(3) Control

c. Start

This is a button to start the experiment according to the assigned conditions file.



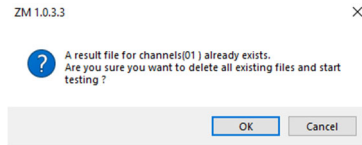
• **Result file**

- Select the folder: Click this  icon to assign the folder where the data files will be saved. Default folder is **C:\ZIVE DATA\ZH\Data**.
- You can input the data file name to the right side of the 'Name'. A 3 digits number will display after the file you inputted (3 digits number means channel number), the file extension is zmf.

• **Information**

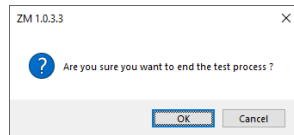
- If check here ☒ **Use the parameters of the technique file.**, the data you inputted in the condition file such as Battery id, Nominal capacity and user will be used as a value if check here. If unchecked, you can input and change it.

- Memo: You can input a brief memo about the experiment.
The line break is Ctrl+enter.
- You can input the data file name to the right side of the 'Name'. A 3 digits number will display after the file you inputted (3 digit number means channel number), the file extension is zmf.
- If you do not input a data file name, it is saved as a channel number and zmp extension.
- If there is data file with the same name exist, it is displayed as follows.



d. **Stop**

The following message window will display if you click the stop test button.



Click 'OK' button to stop the test immediately.

The status display of the real-time graph will display as **Status: Stopped** and the error below will show as **Error: Manual stopped...**

e. **Transferring data from BZA and data file saving**

You can receive the measured data from the device on a PC and save it as a file. This function allows saving the data file again if all data did not transfer to the PC due to a lost connection for any reason.

(4) **Result file**

a. **Data file name**

Display the previously inputted file names at the beginning of the experiment.

b. **Tools**

- Graph menu 
The data that is measured on the graph screen is plotted as a graph.
- Data editor menu 
The data that is measured on the report screen is displayed as a report.

c. **Data point number**

Display the number of the data points measured to date.

d. **Remote**

This function is used to control the BZA from the external device.

Chapter 6. Real time monitor & graph

There are 3 real-time monitor as follows.

A. Group real-time monitor

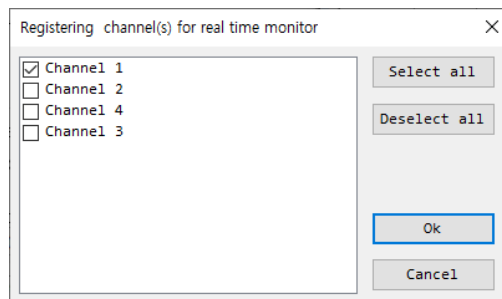
Click the 'Monitor' button and the real-time data and graph will display for the assigned channel if you want to see real-time data and graph for the assigned channels (group) in the control panel.

 Monitor You cannot exit the ZM window.

B. User defined channel(s) real-time monitor

The monitor icon to the right of the top toolbar is this  icon for monitoring the real-time content of the channel assigned by a user.

To channel assigned by a user, click this  icon to the right to set up the channels you want to see.



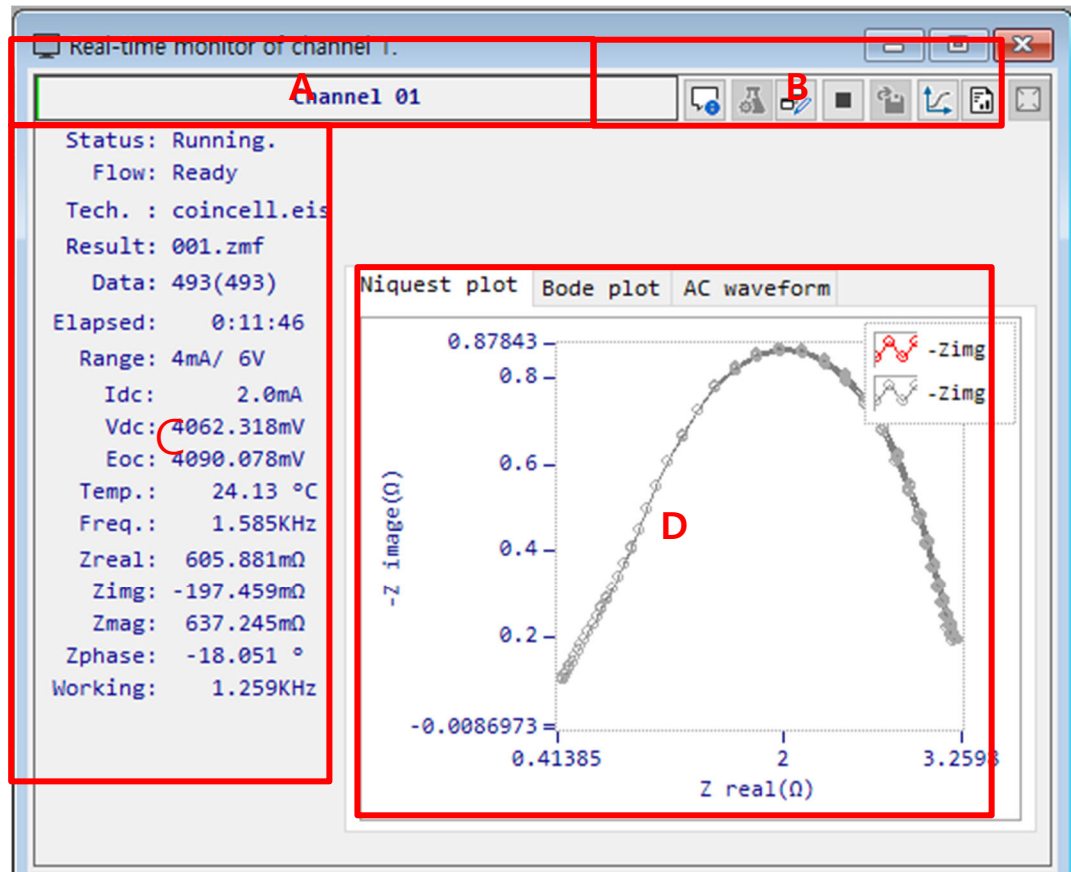
This window can be located outside of ZM window (floating windows)

C. Channel real time monitor

Real time monitor assigned to a channel : Click a channel number in the control panel to see the real time monitor for that channel. This only exists within the ZM window as the real time monitor of the group.

Channel	Group	Status			
		Status	Last error	Elapsed(s)	Rang
1		Running(No calibration).	None...	00:00:00	200mA/

The above three real-time monitor and graph windows can independently display the above three simultaneously.



The real-time graph is divided into 4 sections as above.

1. Channel number area

The channel number for the information display will appear.

2. Icon area

A. Channel information

Device information for the channel is displayed.

and Share Re	
Host Name	ZIM-13BA
Configure IP	Using DHCP
IP Address	192.168.0.11
Subnet Mask	255.255.255.0
Router	192.168.0.1
Port	2000
Mac Address	00:1B:C5:08:13:BA
Device Model	SBZA
Device S/N	IF000Z11777
Device Board	1 1 0 0

- Host name
- Network protocol
- IP Address
- Subnet Mask
- Router
- Port
- Mac Address
- Model
- Serial number
- Board version of channel
- Firmware version of channel

B. Assign technique file

You can assign the schedule file to be used for that channel.

C. Technique file parameter editing

You can edit the parameter in the schedule file scheduled to be used for that channel.

D. Start/Stop

You can start the experiment for the assign channel. The icon shape changes to  after the test begins, and click it again to stop the test.

E. Data transferring from instrument and save the data file

Click this icon and assign a new name when a user wants to save the data on the device with a different data file name on a PC, and the data file is saved with that file name.

This icon is active only when the experiment is stopped or finished.

F. Graph menu

The current data is loaded into the graph menu and applied when changing the graph axis.

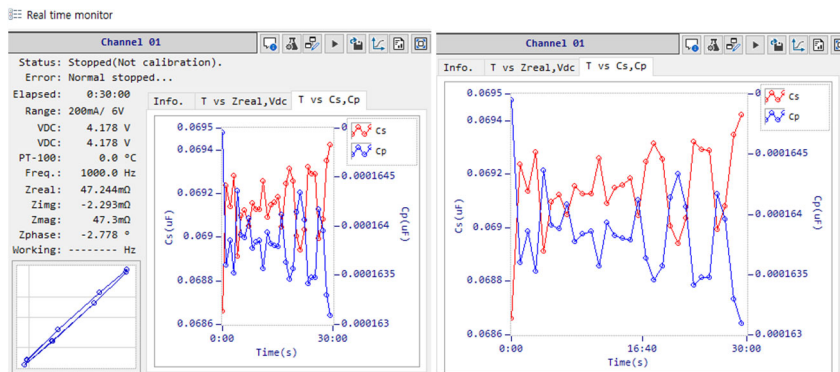
G. Data editor menu

The current data is presented in the report. Even during the experiment, data to date will be displayed.

H. Minimize/Maximize

Maximize screen  and minimize screen 

(Caution) This function is not available  in each channel number linked real time monitor. Minimization is available when switching to dual screen with real-time monitor and graph.



Dual screen display

Single screen display

For single screen and dual screen switching and double-click the cursor over the graph to toggle between the two modes.

3. Real-time monitor

This area displays the following information.

a. Status

Display the operation status of the current device.

b. Error

Display the error contents for any errors.

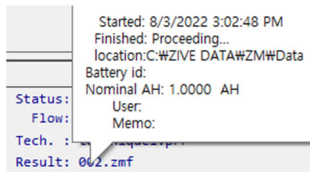
c. Technique

Technique file name. When moving the cursor over the technique file name, the tooltip below appears and shows the location.



d. Result

Result file name. When moving the cursor over the result file name display, the experiment start time, experiment end time, progress, file location, and battery id, capacity as shown below.



e. Data

The number of data in the data file in the PC (the number of data stored in the device's internal memory).

The data stored in the memory can be saved with a different data name if there is the data stored in the device's internal memory.

f. Elapsed

Display the elapsed time since the experiment started.

g. Range

Display the current working voltage range and current range.

h. Vdc

Display the bias voltage or the voltage at the time of discharge when measure the impedance(When rest time, the open circuit voltage is displayed simultaneously.)

i. Eoc

Display the open circuit voltage

j. Temp

Display the measured temperature

k. Freq

Display the recently measured frequency

l. Zreal

Display the real value of impedance

m. Zimg

Display the imaginary value of impedance

n. Zmag

Display the magnitude of impedance

o. Zphase

Display the phase of impedance

p. Working

Display the current working frequency

4. Real time graph area

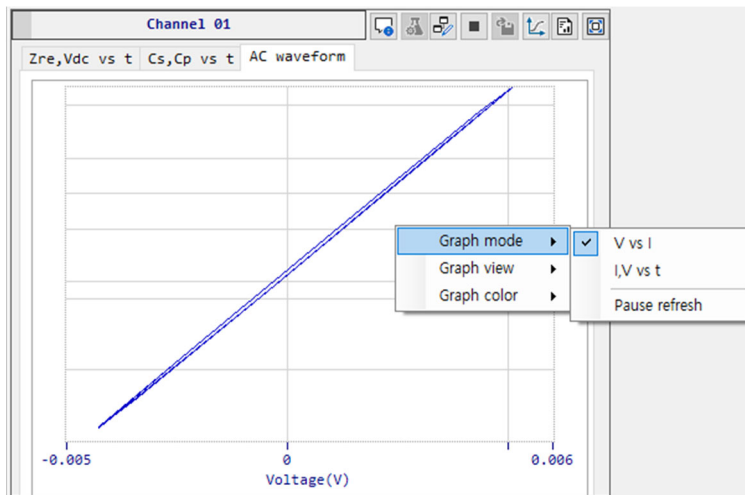
This area is divided into the information displayed on each tab as shown below.

A. AC waveform area

(1) Graph mode

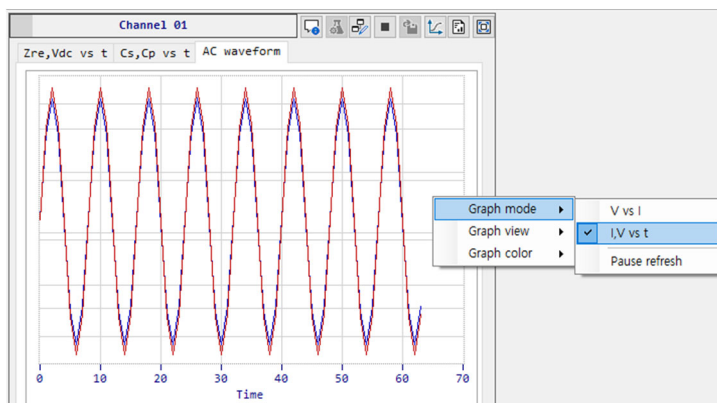
The AC waveform can be expressed by selecting one of the following two.

a. Lissajous plot (V vs I)

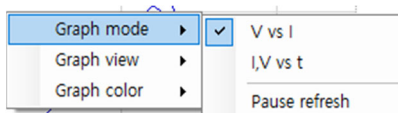


It can be displayed as a Voltage vs. Current graph.

b. Sine graph (I, V vs time)

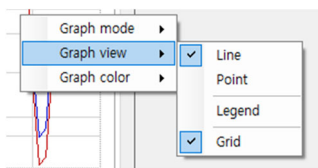


It can be displayed as a Voltage, Current vs. Time.



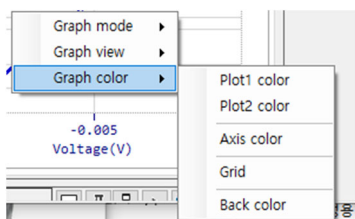
c. Pause refresh

(2) Graph view



You can select the Line, Point, Legend, Grid.

(3) Graph color.



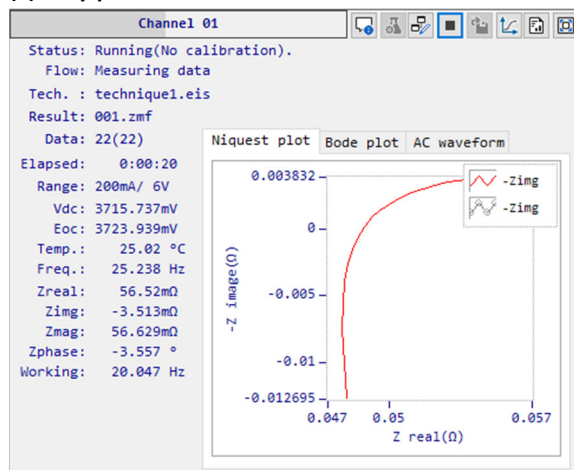
You can change the graph color.

The other two tabs change depend on the experiment.

For all real-time graph, if you double-click the screen and you can see the graph without information, and if you additional double-click at this state, you can see the experiment information on the left and the graph on the right side by original expression mode.

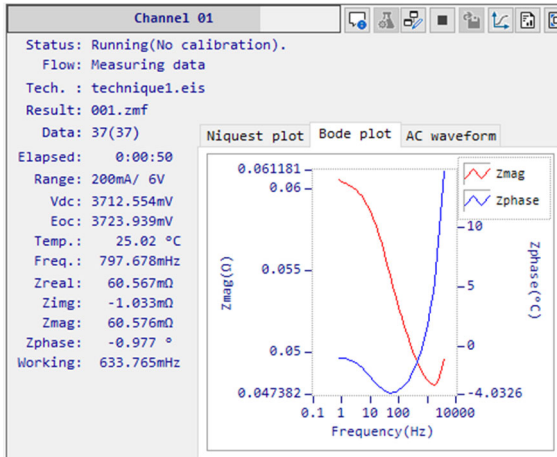
B. Galvanostatic EIS and Quick galvanostatic EIS

(1) Nyquist



Display the Nyquist plot of Zimg vs. Zreal.

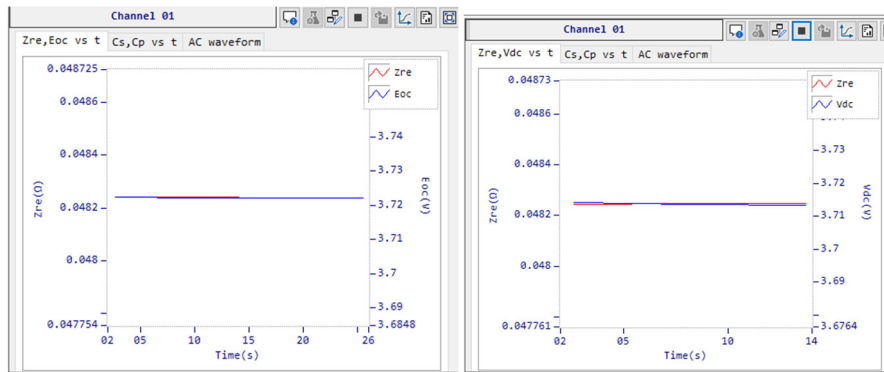
(2) Bode



Display the Bode plot of Zmag vs. Zphase.

C. HFR(Fixed frequency Impedance monitor)

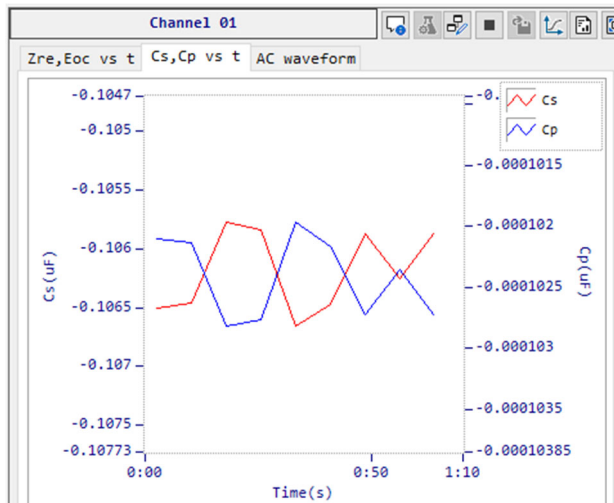
(1) Vdc(Eoc), Zre vs. time



- Vdc, Zre vs. time graph (discharge condition) on right side or Eoc, Zre vs. time graph (Eoc condition)

When 'Load off at waiting' is checked, EOC is measured not discharge while the load between impedance measurement and measurement is turned off. If unchecked, it will be discharged and the discharge current will flow to 1/2 of the current range.

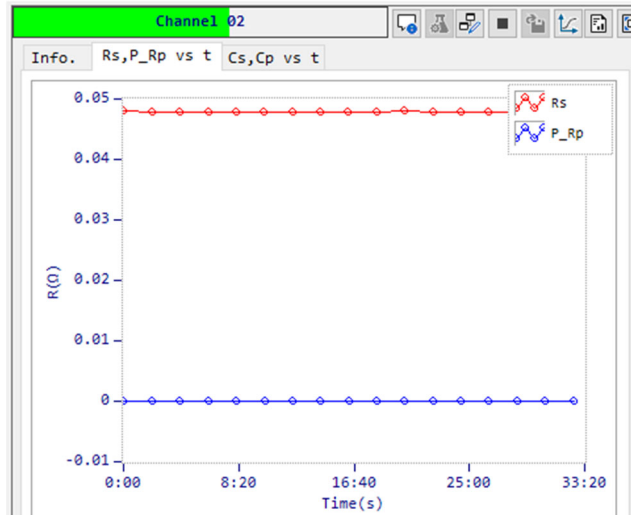
(2) Cs,Cp vs t ime



Display the Cs,Cp vs. Time graph.

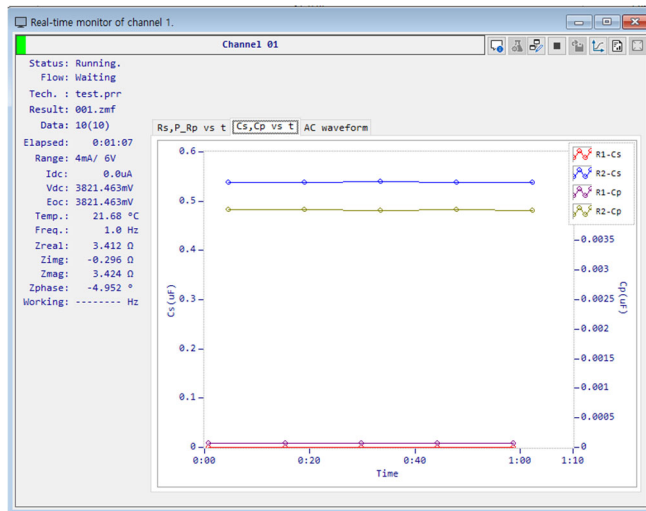
D. Rs , Pseudo Rp monitor

(1) Rs, Pseudo Rp vs. time

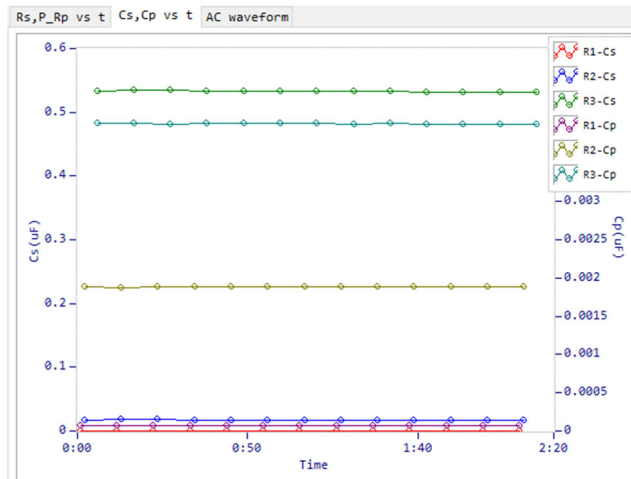


Rs,P_Rp vs t: Display the Rs, Pseudo Rp vs. Time graph.

(2) Cs,Cp vs time

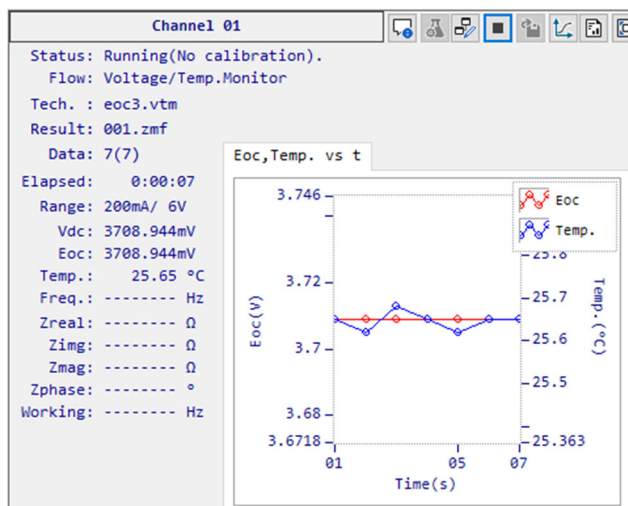


Display the Cs,Cp vs. Time graph.(2 frequency)



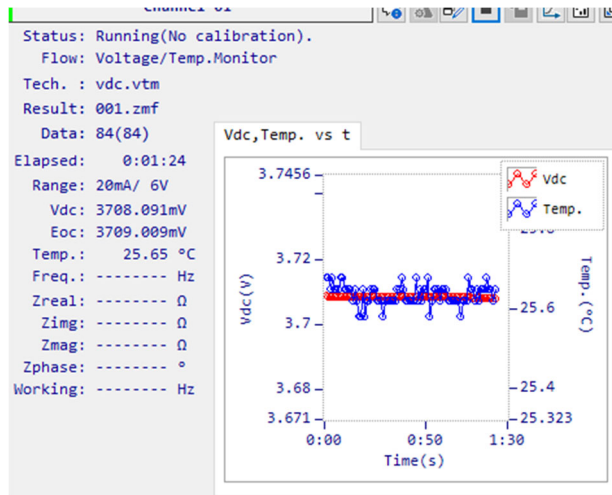
Display the Cs,Cp vs. Time graph.(3 frequency)

E. Eoc/ Temperature monitor



Display the Eoc, Temperature vs. Time (rest condition)

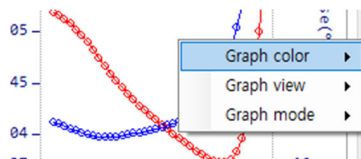
F. Discharge test



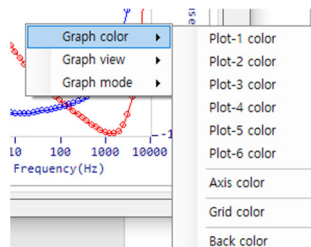
Display the Vdc, Temperature vs. Time (discharge condition)

G. Sub menu on real time graph display

Place the cursor on graph and click on the right mouse button, you can see the following submenu.



(1) Graph color

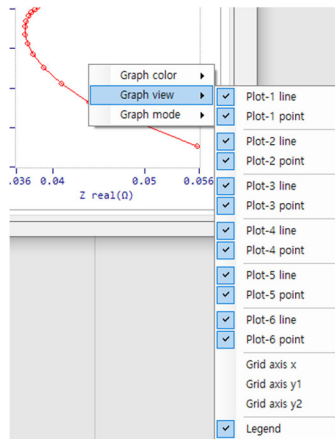


- Select the total of 6 Graph line colors.
- Select the Axis color
- Select the Grid color
- Select the background color

You can choose the color you want from the color selection window below.



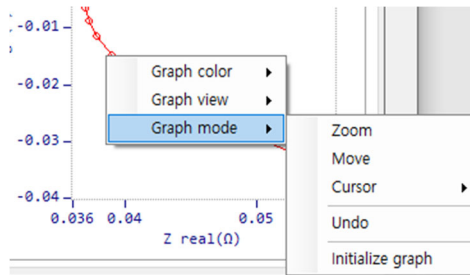
(2) Graph view



You can choose the item you want to see as shown above and it will be displayed on the screen.

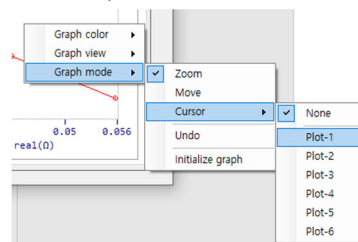
- Select 6 line display mode (Points/Line)
- Select to show or hide the grid for each axis
- Select to show or hide the legend

(3) Graph mode

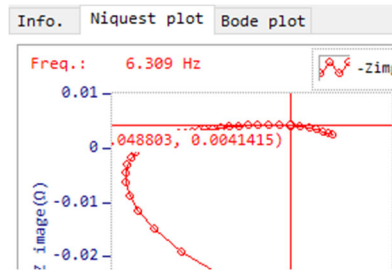


The following actions are activated when the graph update is stopped.

- Zoom: Drag to assign an area to zoom for enlargement
- Move: Drag to move the screen
- Cursor: If placed in cursor mode on each graph,

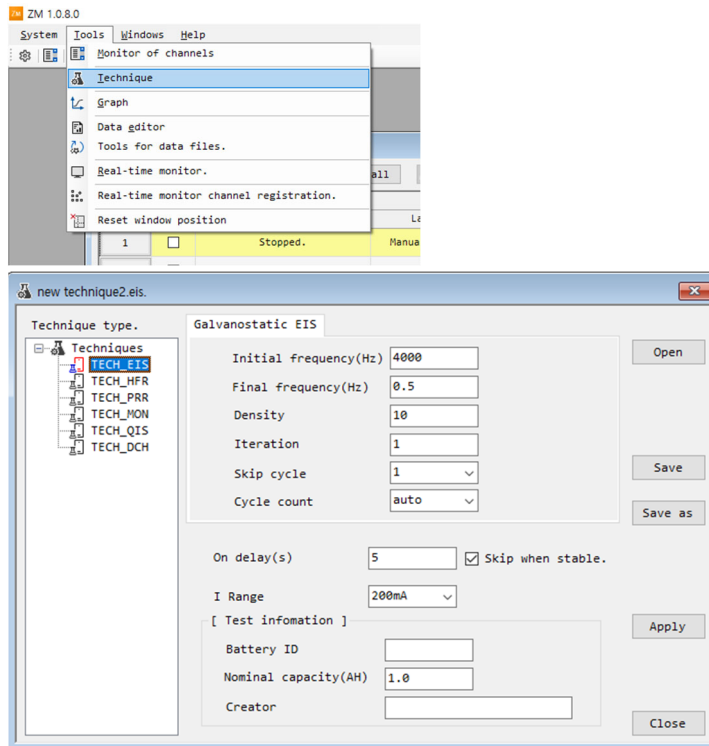


the coordinates of the line will display and the frequency value is displayed in the upper left corner for cases without frequency information like the Nyquist graph.



- Undo: Function to cancel the task when zooming or moving
- Initialize graph: Return the graph to its initial state

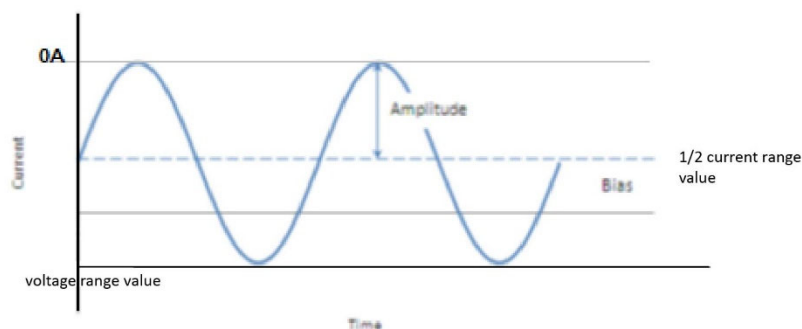
Chapter 7. Techniques



There are 6 techniques that ZM software supports for BZA.

- Galvanostatic EIS
- Galvanostatic HFR
- Rs PseudoRp Test
- Voc/ temperature monitor
- Quick Galvanostatic EIS
- Discharge test

1. Galvanostatic EIS



A. Description

If Initial frequency is different from final frequency, frequency scanning will be done. This works as galvanostatic EIS under current discharging.

This technique apply current sine wave adding to bias current(Half value with current range value) and measure impedance by scanning frequency from initial frequency to final frequency. Amplitude value is determined by current range setting automatically(Half value with current range value).

For example, if you set current range as 200mA then Bias discharging current is 100mA and amplitude is 100mA (peak-to-peak sine waves range between 0 to 200 mA of sine wave discharge).



During galvanostatic EIS measurement, battery will be discharged by bias current, bias current is half the current range. So, if you select 400mA current range, battery will be discharged with 200mA discharging during EIS measurement.

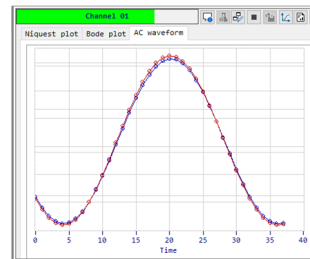
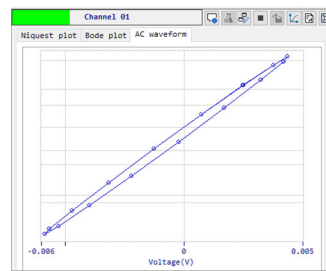


If the battery is connected to the device, battery will be discharged even if it does not EIS measurement. Disconnect battery from this device.

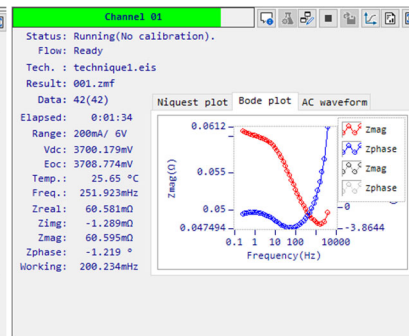
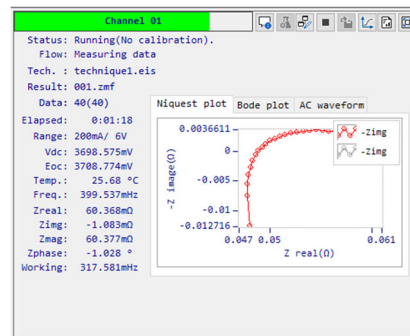
B. Real time monitor

The real-time graph according to experiment is as follows.

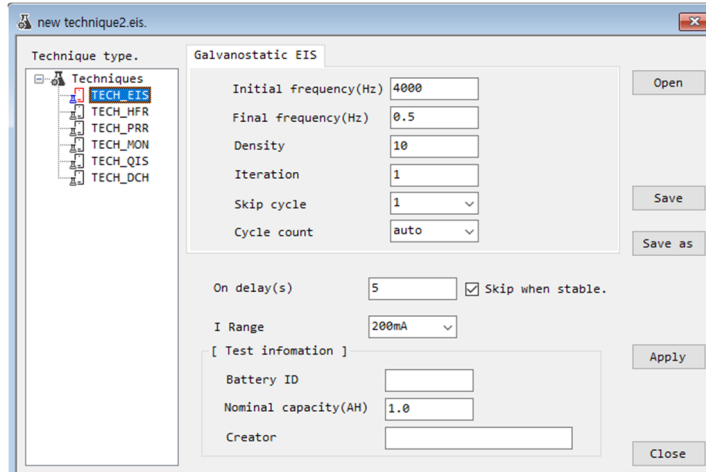
- Voltage vs. Current graph of AC waveform or Voltage/Current vs. Time graph of AC waveform



- Nyquist graph on the right
- Bode graph on the right



C. Technique menu



D. Technique Parameter input range

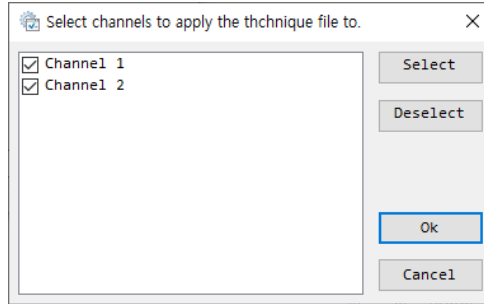
Parameters	Range	내용
Initial frequency(Hz)	50mHz - 10kHz(4kHz)	Initial frequency value
Final frequency (Hz)	50mHz - 10kHz(4kHz)	Final frequency value
Density	More than one	Frequency points per decade
Iteration	More than one	Iteration number
On delay (s)	More than a 0 second	Waiting time for the voltage to stabilize
Skip when stable	Check/uncheck	If check, the experiment start if it is stable within the specified time
Irange	2A,400mA,200mA,40mA, 20mA,4mA,2mA,400uA	Bias current is determined by half the current range and the waveform height is equal to bias current Current range= Ipp

E. Note::

You can create experiment conditions and save the condition file you created as a different name for experiment with the same conditions next time. (Click the 'Save' button on the right.)

To open an existing saved condition file, click the 'Open' button on the right to see files with the EIS extensions.

When click the 'Apply' button on the right for the experiment if the conditions are set you want, and the saving it as a new file shows if the changes are not saved. And the next window will appear to decide which channel to apply the conditions file if there are no changes made to the file.

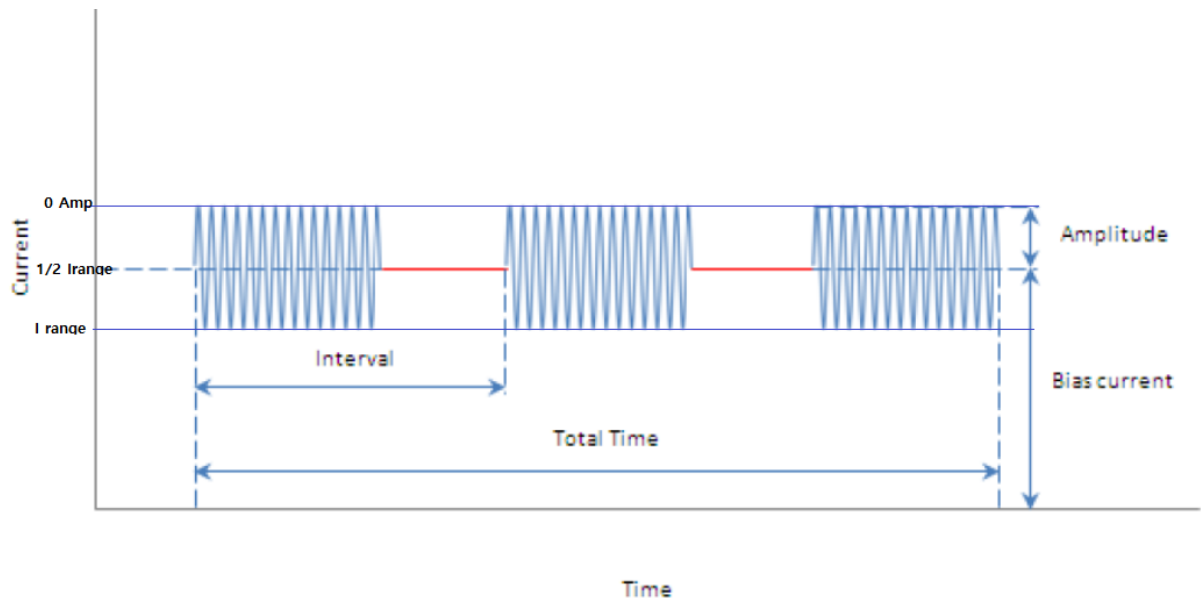


Click the 'Select' button, all channels are assigned and click the 'Deselect' button, all assigned channels are unselected. If you want to assign a specific channel, check the check box to the left of the channel number.

After the conditions file has been assigned, the conditions of technique file on the control panel screen will appear as set completed.

Condition file		
°C)	File name	Tools
000	technique_EIS3.eis	 

2. Galvanostatic HFR



A. Description

This technique that each section measures the impedance by applying current at a fixed frequency, and the interval time must be long enough to perform this technique at a low frequency.

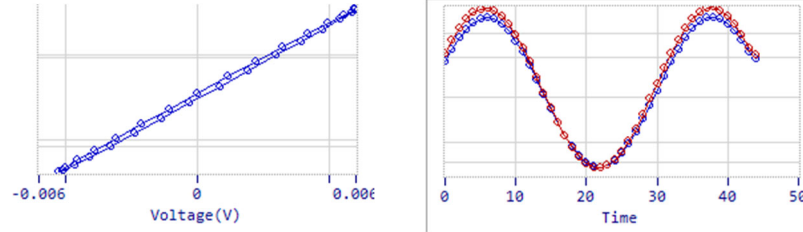
The bias current in the discharged status is half the current range and the current waveform height is equal to the bias current.

When 'Load off at waiting' is checked, EOC is measured not discharge while the load between impedance measurement and measurement is turned off. If unchecked, it will be discharged and the discharge current will flow to 1/2 of the current range.

B. Real time monitor

The real-time graph according to experiment is as follows.

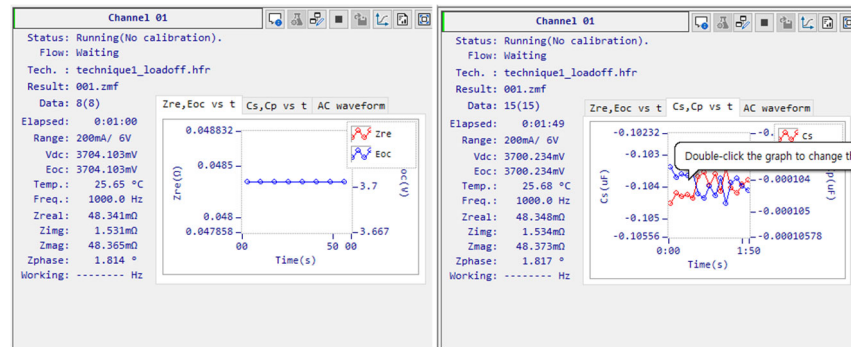
- Voltage vs. Current graph of AC waveform or Voltage/Current vs. Time graph of AC waveform



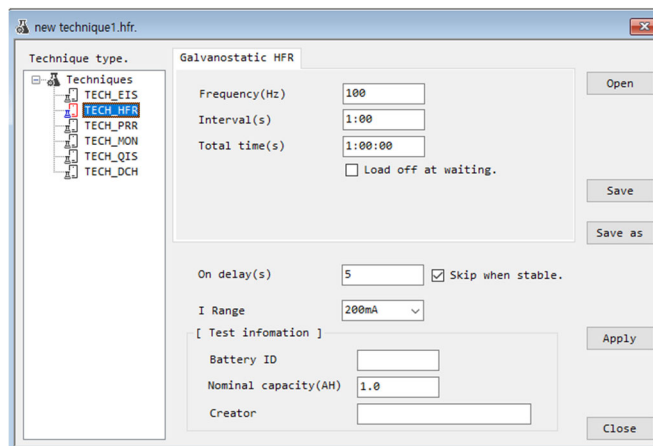
- Vdc, Zre vs. Time graph(discharge condition) on right side or Eoc, Zre vs. Time graph (Eoc condition)

When 'Load off at waiting' is checked, Eoc is measured not discharge while the load between impedance measurement and measurement is turned off. If unchecked, it will be discharged and the discharge current will flow to 1/2 of the current range.

- Cs,Cp vs t on right side : Cs. Cp vs. Time graph



C. Technique menu



D. Technique parameter input range

Parameters	Range	내용
Frequency(Hz)	50mHz - 10kHz (4kHz)	Initial frequency value
Interval	More than a 0 second	Time between measurement and measurement
Total time	More than a 0 second	Total measurement time
On delay (s)	More than a 0 second	Waiting time for the voltage to stabilize
Skip when stable	Check/uncheck	If check, the experiment start if it is stable within the specified time
Irange	2A,400mA,200mA,40mA,20mA,4mA,2mA,400uA	Bias current determined by half the current range and the waveform height is equal to bias current Current range= Ipp

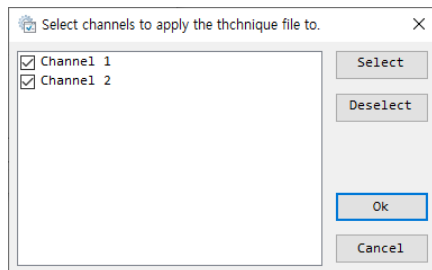
E. Note:

You can create experiment conditions and save the condition file you created as a different name for experiment with the same conditions next time. (Click the 'Save' button on the right.)

To open an existing saved condition file, click the 'Open' button on the right to see files with the HFR extensions.

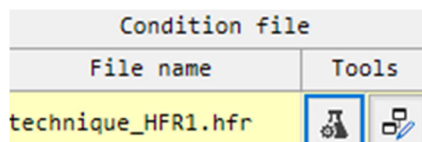
When click the 'Apply' button on the right for the experiment if the conditions are set you want, and the saving it as a new file shows if the changes are not saved.

And the next window will appear to decide which channel to apply the conditions file if there are no changes made to the file.

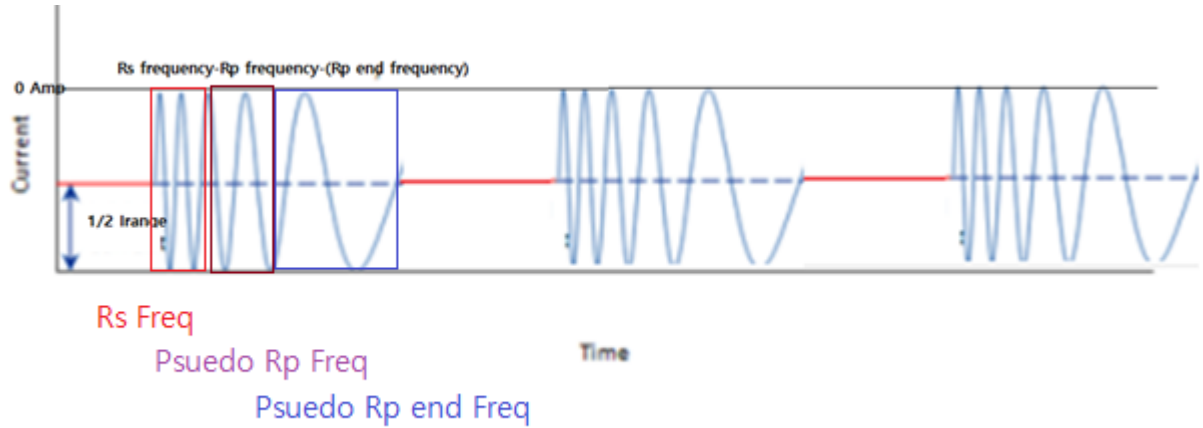


Click the 'Select' button, all channels are assigned and click the 'Deselect' button, all assigned channels are unselected. If you want to assign a specific channel, check the check box to the left of the channel number.

After the conditions file has been assigned, the conditions of technique file on the control panel screen will appear as set completed.

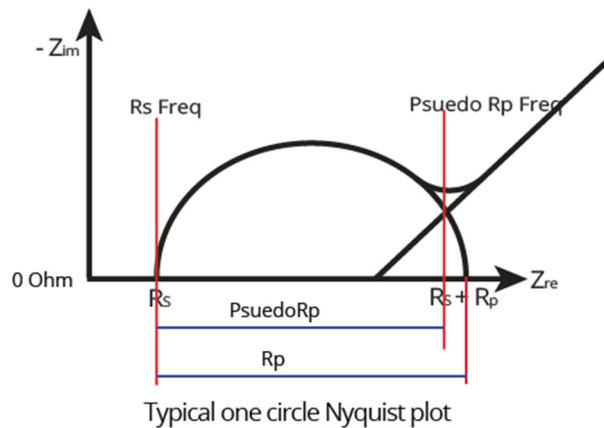


3. Rs_Pseudo Rp test



A. Description

This technique is to calculate the R_s value and the virtual R_p value, and before entering this test, conduct the Galvanostatic EIS experiment first to calculate the frequency to obtain the frequency corresponding to R_s and the virtual R_p and measure it with only the calculated frequency. The bias current in the discharged status is half the current range and the current waveform height is equal to the bias current. When 'Load off at waiting' is checked, EOC is measured not discharge while the load between impedance measurement and measurement is turned off. If unchecked, it will be discharged and the discharge current will flow to 1/2 of the current range.



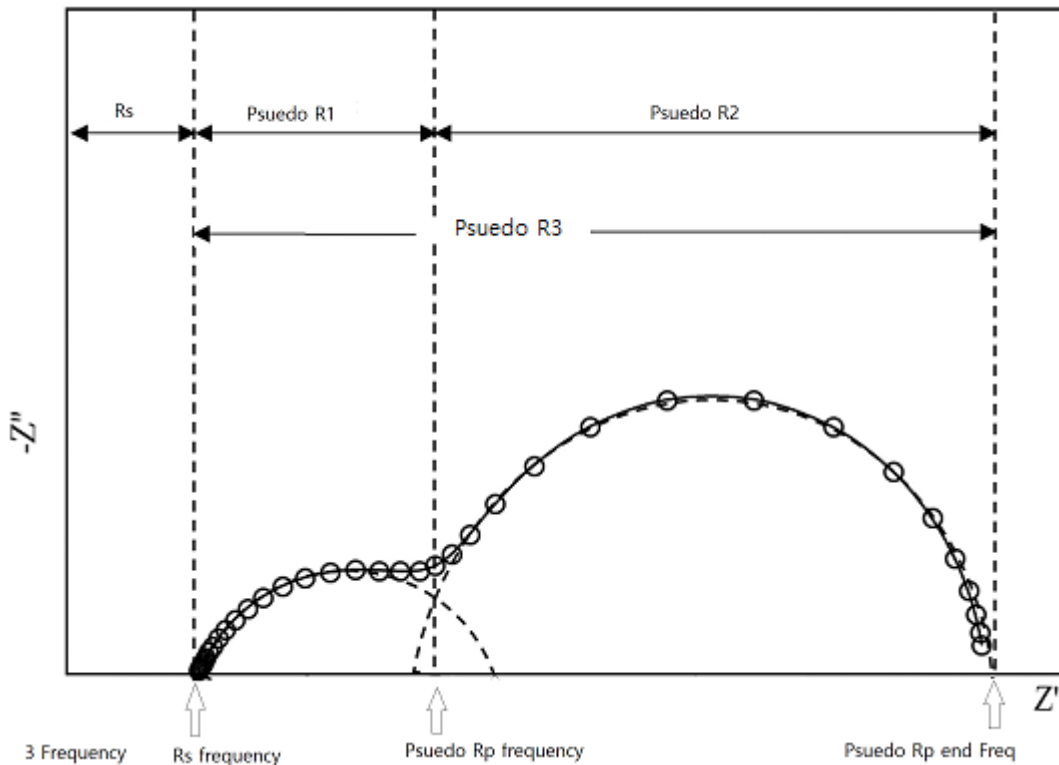
If the measured EIS graph has one semicircle as shown above and the 'Pseudo R_p end frequency' check box is unchecked,

☐ Pseudo R_p end Frequency(Hz)

it becomes the test with one semicircle

For this case,

- Input the frequency in which the $-Z_{im}$ value becomes 0 for R_s frequency and the Z_{re} value is minimized
- For Pseudo R_p frequency, input the frequency value that corresponds to the intermediate value where the spread occurs with the right side of the semicircle



If there is more than one semicircle as above,

check here ☐ Pseudo Rp end frequency(Hz) to activate the Pseudo Rp end frequency.

When you check on Pseudo Rp end frequency check box, then following display to select Pseudo Rp value.

☐ Pseudo Rp end frequency(Hz)

Interval(s)

=>

☒ Pseudo Rp end frequency(Hz)

Pseudo Rp calculation

Interval(s)

You can define Pseudo Rp value.

If you select $Zre(Rp\ end) - Zre(Rs)$ then Pseudo Rp will be Psuedo R3 on the above graph.

If you select $Zre(Rp\ end) - Zre(Rp)$ then Pseudo Rp will be Psuedo R2 on the above graph.

If you select $Zre(Rp) - Zre(Rs)$ then Pseudo Rp will be Psuedo R1 on the above graph

This is a function to measure two or more frequency, where it measures for a set of impedance after maintaining the discharge or rest during the interval time once one set of impedance measurements is completed.

For this case,

- Input the frequency in which the $-Z''$ value becomes 0 for Rs frequency and the Zre value is minimized.
- For Pseudo Rp frequency, input the starting frequency as shown in the picture example above if the Rp starting frequency is not Rs.
- For Pseudo Rp end frequency, input the frequency value that corresponds to the end value where the diffusion occurs with the right side of the semicircle.

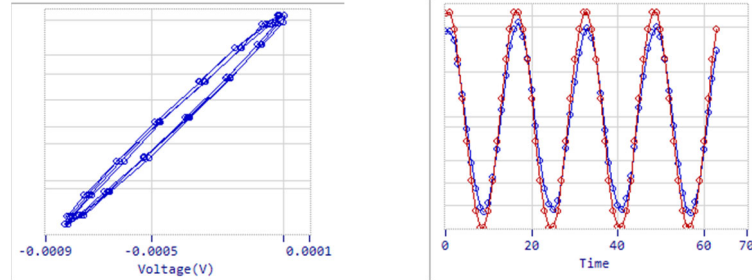
This function simply calculates the value corresponding to Rs and Pseudo Rp, and the real-time graph can be displayed as a monitoring for Rs, PsuedoRp and monitoring graph for Cs, Cp for time.

For example, this is a method that can be used to measure impedance after charging or discharging at a constant capacity or to measure impedance regularly.

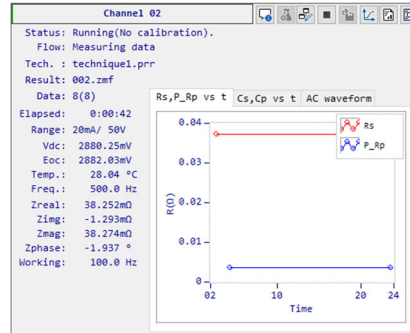
B. Real time monitor

The real-time graph according to this experiment is shown below.

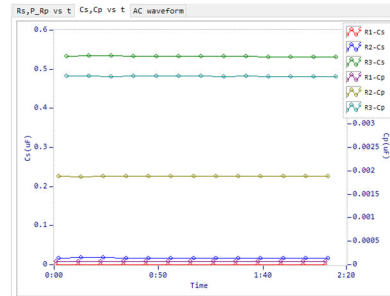
- Voltage vs. Current graph on lower left or Voltage, Current vs. Time graph



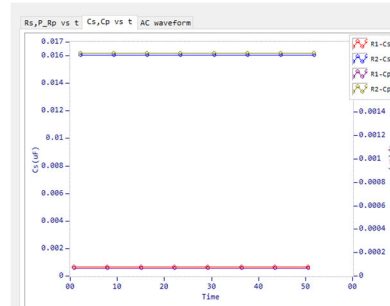
- Rs, P_Rp vs t on right side: Rs vs. Pseudo Rp vs. Time graph



- Cs, Cp vs t on right side : Cs, Cp vs. Time graph

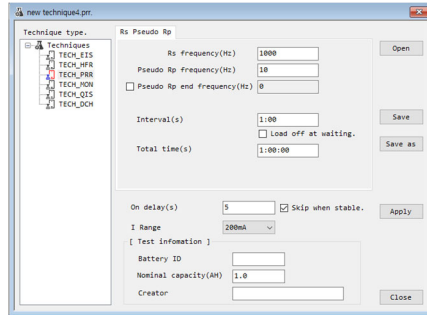


If there are three frequency

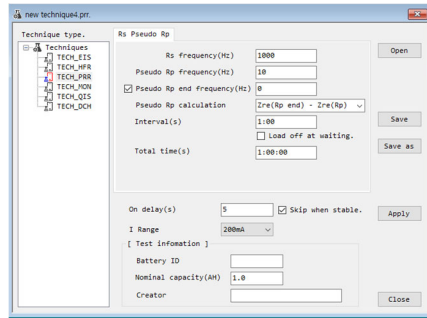


If there are two frequency

C. Technique menu



2 frequency selection



3 frequency selection

D. Technique parameter input range

Parameters	Range	내용
Rs Frequency(Hz)	50mHz - 10kHz (4kHz)	Frequency value corresponding to Rs
Psuedo Rp Frequency(Hz)	50mHz - 10kHz (4kHz)	Frequency value corresponding to Psuedo Rp
Psuedo Rp end Frequency(Hz)	50mHz - 10kHz (4kHz)	Frequency value corresponding to Psuedo Rp end
Psuedo Rp calculation	Zre(Rp end)-Zre(Rp) Zre(Rp end)-Zre(Rs) Zre(Rp)-Zre(Rs)	Psudo Rp calculation definition
Interval	More than a 0 second	Time between measurement and measurement
Total time	More than a 0 second	Total measurement time
On delay (s)	More than a 0 second	Waiting time for the voltage to stabilize
Skip when stable	Check/uncheck	If check, the experiment start if it is stable within the specified time
Irange	2A,400mA,200mA,40mA, 20mA,4mA,2mA,400uA	Bias current determined by half the current range and the waveform height is equal to bias current Current range= Ipp

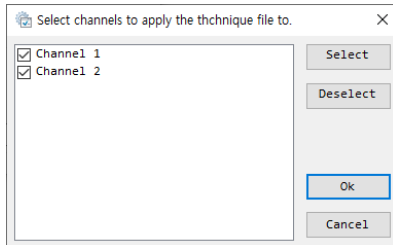
E. Note:

You can create experiment conditions and save the condition file you created as a different name for experiment with the same conditions next time. (Click the 'Save' button on the right.)

To open an existing saved condition file, click the 'Open' button on the right to see files with the PRR extensions.

When click the 'Apply' button on the right for the experiment if the conditions are set you want, and the saving it as a new file shows if the changes are not saved.

And the next window will appear to decide which channel to apply the conditions file if there are no changes made to the file.



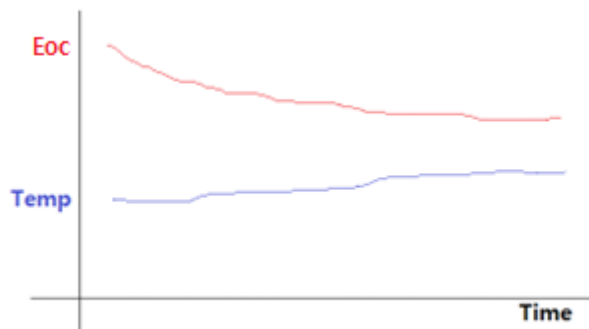
Click the 'Select' button, all channels are assigned and click the 'Deselect' button, all assigned channels are unselected. If you want to assign a specific channel, check the check box to the left of the channel number.

After the conditions file has been assigned, the conditions of technique file on the control panel screen will appear as set completed.

Condition file	
File name	Tools
technique_PRR1.prr	 

4. Voc / Temperature Monitor

A. Description

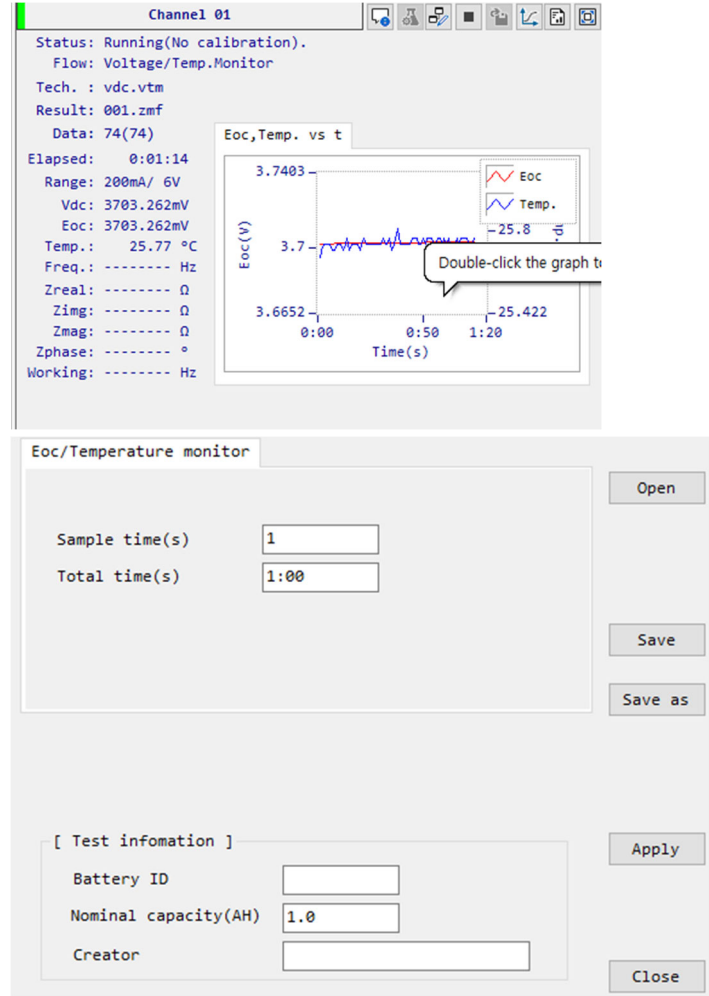


This technique monitors the voltage and temperature of the battery when the electronic load is not working (rest state), and it sets up the total experiment time and selects data sampling conditions.

B. Real-time monitor

The real-time graph according to this experiment is shown below.

- Eoc, Temperature vs. Time graph on the right (rest state)



C. Technique parameter input range

Parameters	Range	내용
Sample time(sec)	More than one	Sampling time
Total time(s)	More than one	Total measurement time

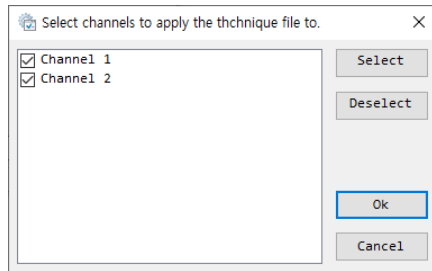
D. Note:

You can create experiment conditions and save the condition file you created as a different name for experiment with the same conditions next time. (Click the 'Save' button on the right.)

To open an existing saved condition file, click the 'Open' button on the right to see files with the VTM extensions.

When click the 'Apply' button on the right for the experiment if the conditions are set you want, and the saving it as a new file shows if the changes are not saved.

And the next window will appear to decide which channel to apply the conditions file if there are no changes made to the file.

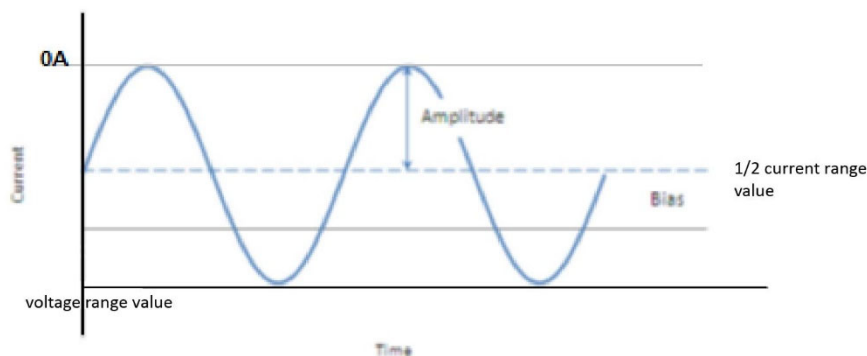


Click the 'Select' button, all channels are assigned and click the 'Deselect' button, all assigned channels are unselected. If you want to assign a specific channel, check the check box to the left of the channel number.

After the conditions file has been assigned, the conditions of technique file on the control panel screen will appear as set completed.

Condition file	
File name	Tools
technique3.vtm	 

5. Quick Galvanostatic EIS



A. Descriptin

This method to measure the impedance by changing the frequency from the initial frequency to the final frequency in quick time even if the data quality poor for screening purposes to comprehend the shape of the impedance spectrum in a specified frequency period. The data density is fixed at 4 points per decade. Performs as the constant current EIS when discharged.

This technique apply current sine wave adding to bias current(Half value with current range value) and measure impedance by scanning frequency from initial frequency to final frequency. Amplitude value is determined by current range setting automatically(Half value with current range value).

For example, the bias discharge current is 100 mA and the amplitude is 100 mA if the current range is set to 200 mA (peak-to-peak sine waves range between 0 to 200 mA of sine wave discharge)

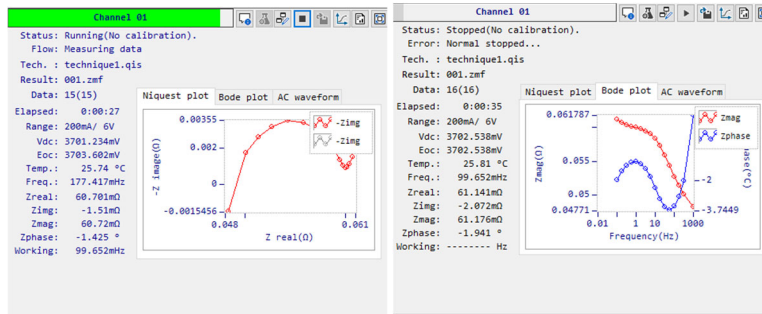


During Galvanostatic EIS measurement, battery will be discharged by bias current, bias current is half the current range. So, if you select 400mA current range, battery will be discharged with 200mA discharging during EIS measurement.

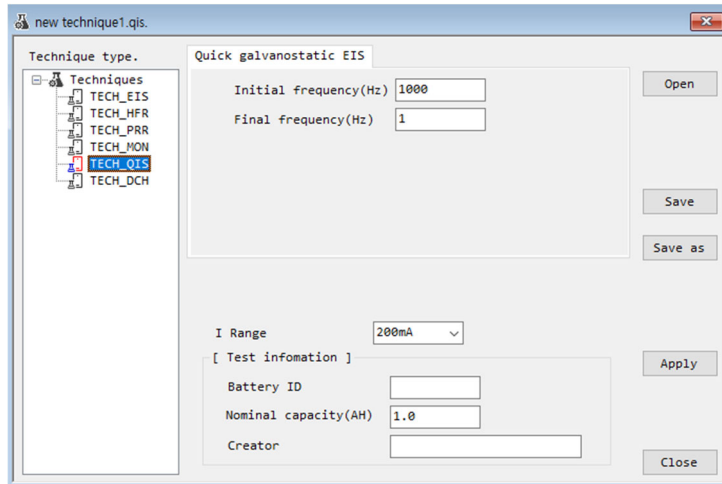


If the battery is connected to the device, battery will be discharged even if it does not EIS measurement. Disconnect battery from this device.

B. Real-time monitor



C. Technique menu



D. Technique parameter input range

Parameters	Range	내용
Initial frequency(Hz)	50mHz - 10kHz (4kHz)	Initial frequency value
Final frequency (Hz)	50mHz - 10kHz(4kHz)	Final frequency value
Irange	2A,400mA,200mA,40mA, 20mA,4mA,2mA,400uA	Bias current determined by half the current range and the waveform height is equal to bias current Current range= Ipp

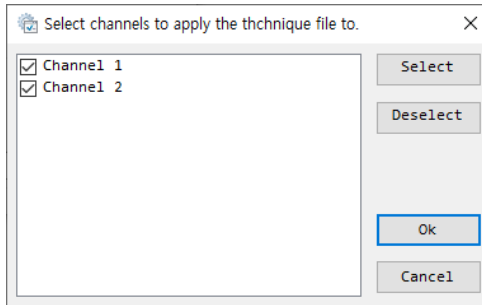
E. Note:

You can create experiment conditions and save the condition file you created as a different name for experiment with the same conditions next time. (Click the 'Save' button on the right.)

To open an existing saved condition file, click the 'Open' button on the right to see files with the QIS extensions.

When click the 'Apply' button on the right for the experiment if the conditions are set you want, and the saving it as a new file shows if the changes are not saved.

And the next window will appear to decide which channel to apply the conditions file if there are no changes made to the file.



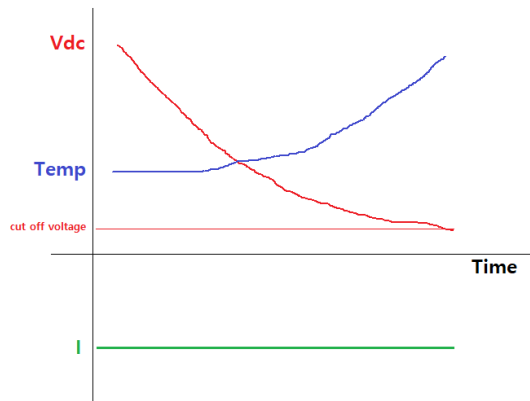
Click the 'Select' button, all channels are assigned and click the 'Deselect' button, all assigned channels are unselected. If you want to assign a specific channel, check the check box to the left of the channel number.

After the conditions file has been assigned, the conditions of technique file on the control panel screen will appear as set completed.

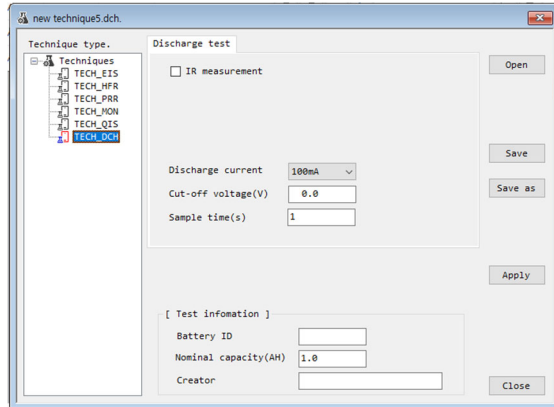
c(V)	Temp.(°C)	File name	Tools
3.493	28.700	technique2.qis	

6. Discharge test

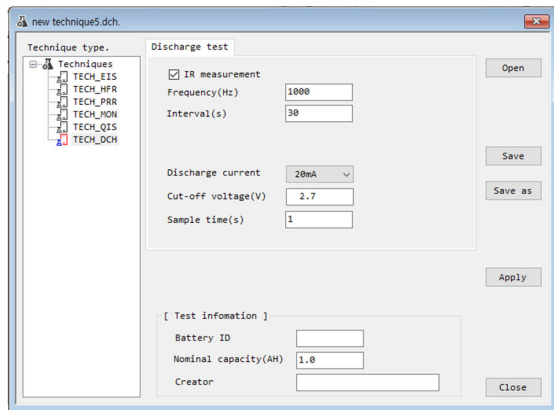
A. Description



Select the discharging current size from 1A, 200mA,100mA,20mA,10mA,2mA,1mA,200uA by using the discharge test technique for monitoring the battery voltage and temperature while discharging.



Discharge test without IR measurement



Discharge test with IR measurement

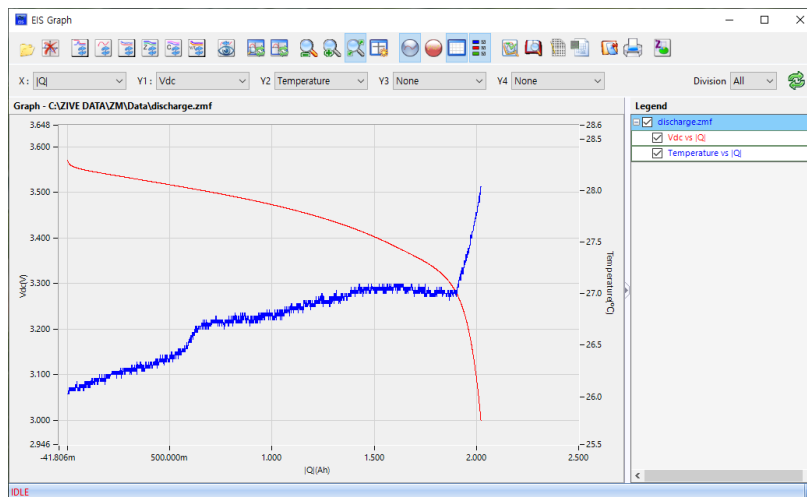
The experiment will be finished when the input cut-off voltage is reached, and the data sampling condition can be assigned by time. The experiment will be finished if the total time is reached even before the cut-off voltage.

The total time be set up for a long duration if you want to continue the experiment until the cut-off voltage.

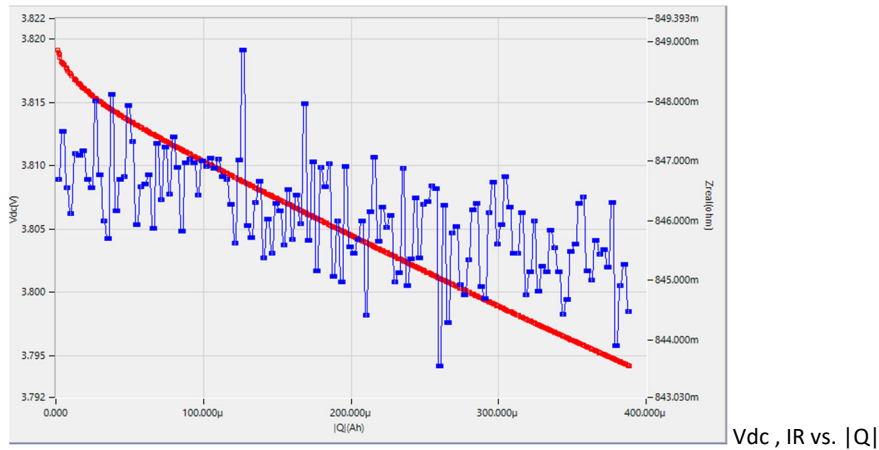
Vdc(V)	Idc(A)
4.04219e+0	100.0000e-3
4.04211e+0	100.0000e-3
4.04208e+0	100.0000e-3

In the discharged state, Idc display to the discharged current value and Vdc value display the voltage value.

You can see the discharge graphs by selecting |Q| on the X-axis and Vdc on the Y-axis for graphs.



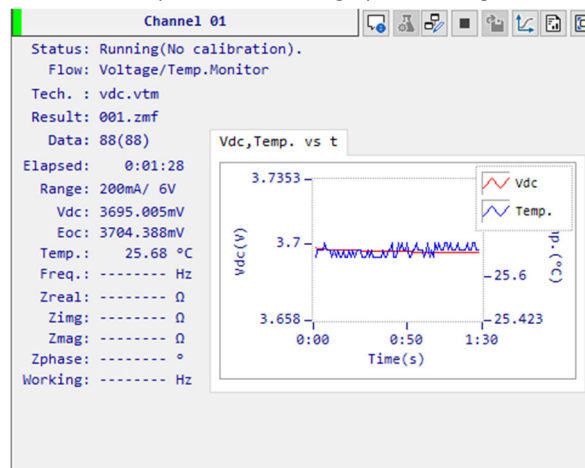
Vdc , temperature vs. |Q|



B. Real-time monitor

The real-time graph according to this experiment is shown below.

- Vdc, Temperature vs. Time graph on the right



C. Technique parameter input range

Parameters	Range	contents
IR measurement	Check /unchecked	IR measurement optio on/off
Frequency	50mHz-10kHz	IR measurement frequency
Interval	> 1/frequency * 5 (>1)	Interval between IR measurement
Sample time(sec)	More than one	Sampling time
Discharge current	1A, 200mA,100mA,20mA,10mA,2mA, 1mA,200uA	Discharge current value
Cut off voltage	Within device maximum voltage	Discharge end voltage

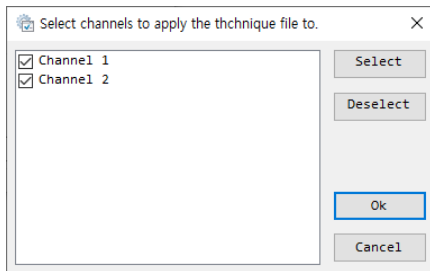
D. Note:

You can create experiment conditions and save the condition file you created as a different name for experiment with the same conditions next time. (Click the 'Save' button on the right.)

To open an existing saved condition file, click the 'Open' button on the right to see files with the dch extensions.

When click the 'Apply' button on the right for the experiment if the conditions are set you want, and the saving it as a new file shows if the changes are not saved.

And the next window will appear to decide which channel to apply the conditions file if there are no changes made to the file.



Click the 'Select' button, all channels are assigned and click the 'Deselect' button, all assigned channels are unselected. If you want to assign a specific channel, check the check box to the left of the channel number.

After the conditions file has been assigned, the conditions of technique file on the control panel screen will appear as set completed.

Condition file	
File name	Tools
technique2.dch	 

Chapter 8. EIS measurement parameter

1. Galvanostatic EIS parameter setting

- Initial Frequency: 0.05 – 10kHz
Frequency value that can be set as the initial frequency
- Final frequency: 0.05 – 10kHz
Frequency value that can be set as the final frequency
- Density: 1-65535 Pts/decade
Data density per decade. Total data is limited to 100,000 points.
- Iteration: >1
Iteration number: Number of times that frequency scanning is repeated from the initial frequency to the Final frequency. Each data set is separated by the number of cycles..
- Current range: 2A, 400mA, 200mA, 40mA, 2mA, 400uA
The bias discharge current value is half of the current range value and its same value is the current Amplitude(peak-to-peak) size.
- Max on Delay: >0sec option “skip when stable”
Waiting time for the voltage to stabilize at a discharge current
- Skip cycle: 0 – 10
Number of skip of initial waveforms
- Cycle count: Auto, 2, 4, 8... 128
Number of cycles averaging data from multiple waveforms to calculate a stabilized impedance value. When set as Auto, an appropriate number of cycles that corresponds to the frequency is automatically determined.

2. Setting

(1) Current range

The current range setting is to apply bias current and amplitude.

But measured AC voltage value should be within linear region. Also, it should be below 60 Watt(BZA1000).

If current range setting value and battery voltage is over the 60Watt(BZA1000), software will down the current range automatically..

We recommend the select lower current range. Higher current range can give high SNR and can be get clean data. But, at lower frequency range, impedance will be high. This will make higher voltage waveform. It is recommended to select the appropriate current since it becomes unreliable data outside the linear region of the current.

(2) Frequency

The frequency range from the initial frequency to the final frequency can be set to a maximum of 65535 values per decade from 10 kHz to 50 MHz in logarithmic step.

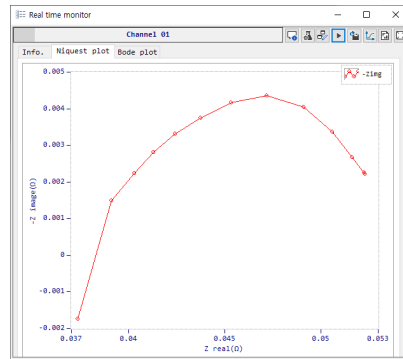
If you want to see diffusion region, you need to set low frequency but it takes longer time. To shorten the measurement time, you need to set final frequency as high as possible and short the measurement frequency point.

To fit EIS spectra accurately, you need large data point as possible. So, you have to decide the point number per decade and final frequency depending on your purpose.

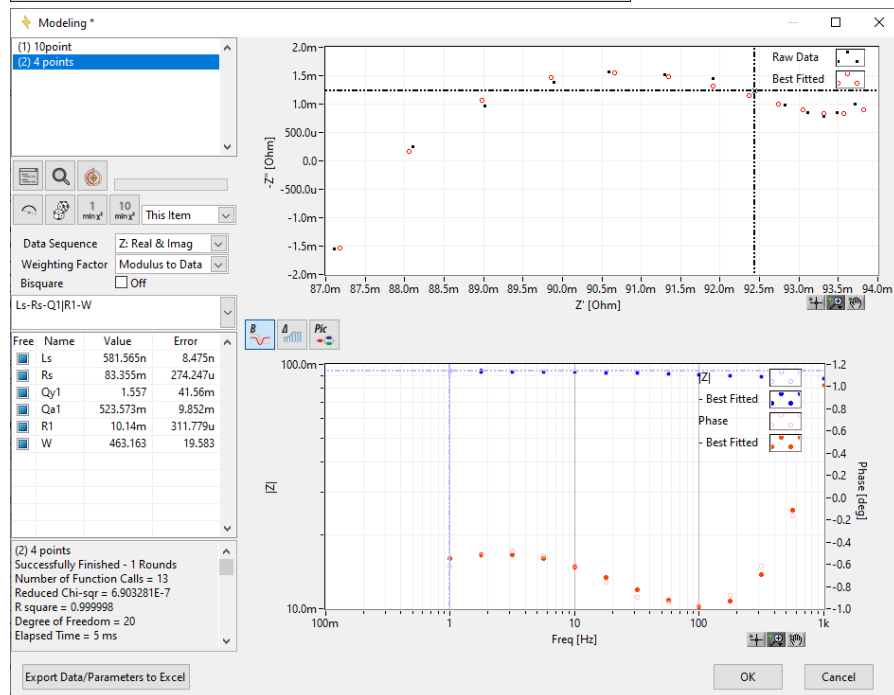
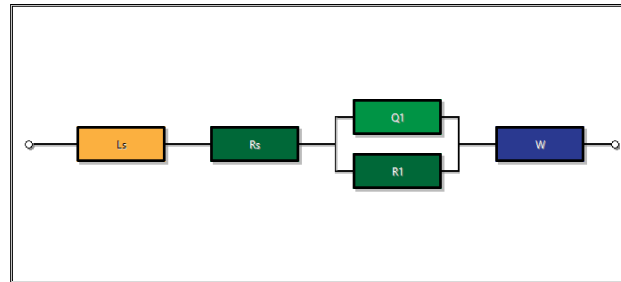
Quick GEIS recommend 1kHz to 1Hz with 4 points per decade with 1 cycle for screening purpose. For accurate result and initial EIS test, the Galvanostatic EIS technique recommend 1kHz to 50mHz with minimum of 10 points per decade with auto cycle. The lower the frequency, the longer the measurement time. It is therefore sensible to work with as high a frequency as possible.

- Screening condition (1kHz to 1Hz, 4 points per decade, 1 cycle)

Measurement time : about 7 sec

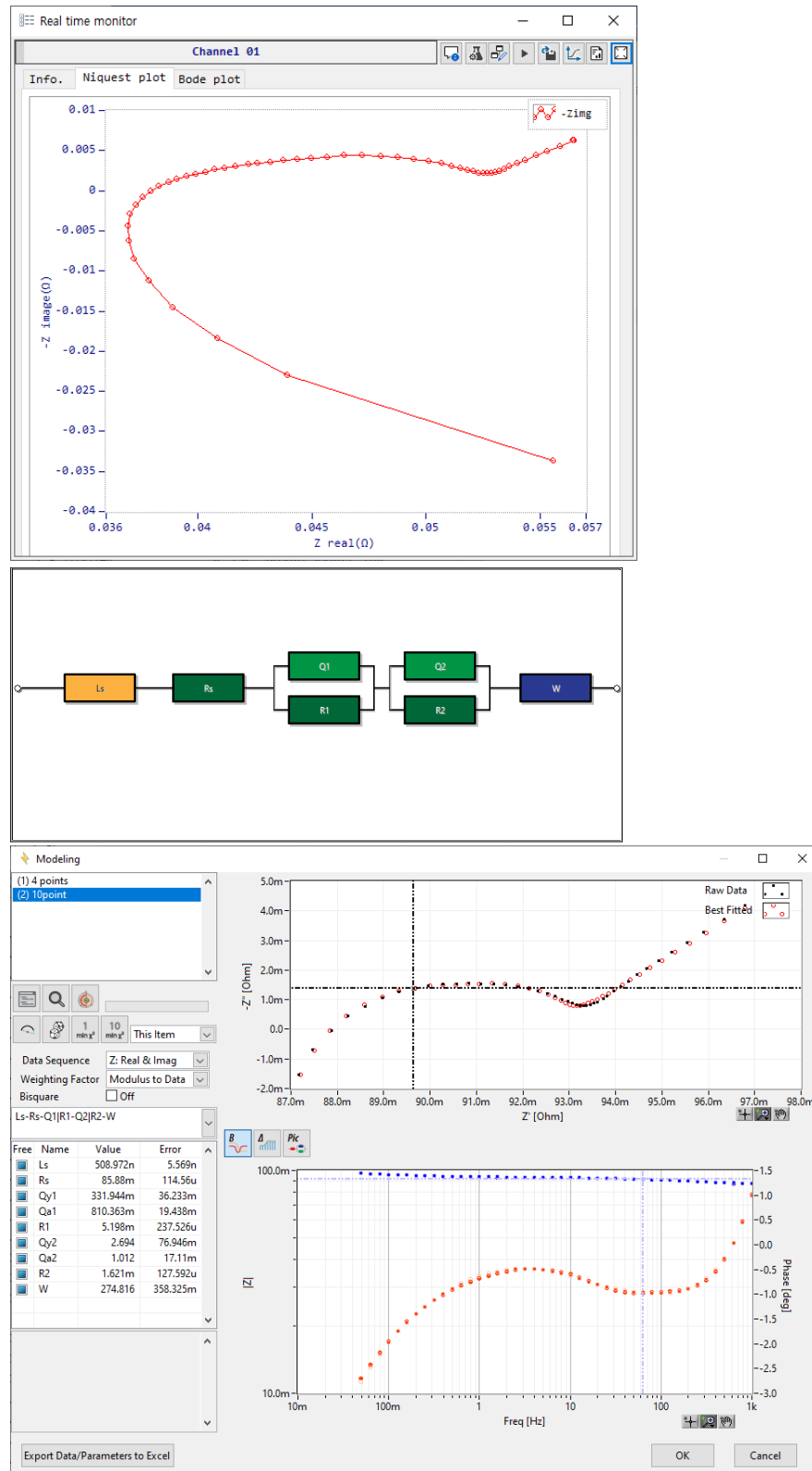


ZMAN automatic model searching using battery simple library.



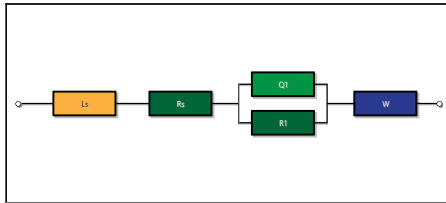
- Accurate measurement (10kHz to 50mHz, 10 points per decade)

Measurement time : about 5 minute 34 second



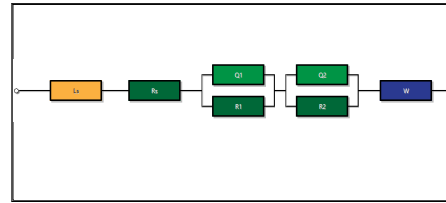
Comparison of fitting result

ZMAN selected model



Ls-Rs-Q1|R1-W

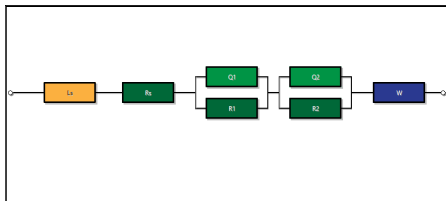
Free	Name	Value	Error
☒	Ls	560.219n	7.451n
☒	Rs	83.674m	207.418u
☒	Qy1	1.412	34.278m
☒	Qa1	549.836m	8.44m
☒	R1	9.719m	237.502u
☒	W	424.37	14.522



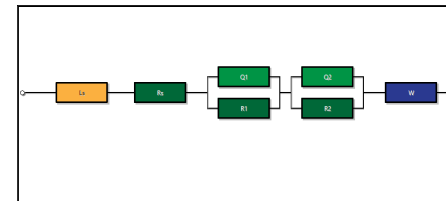
Ls-Rs-Q1|R1-Q2|R2-W

Free	Name	Value	Error
☒	Ls	508.972n	5.569n
☒	Rs	85.88m	114.56u
☒	Qy1	331.944m	36.233m
☒	Qa1	810.363m	19.438m
☒	R1	5.198m	237.526u
☒	Qy2	2.694	76.946m
☒	Qa2	1.012	17.11m
☒	R2	1.621m	127.592u
☒	W	274.816	358.325m

If user change mode for screen mode same as accurate mode



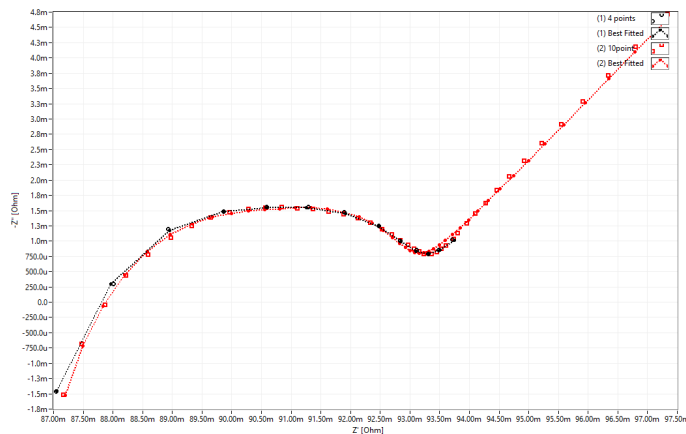
Free	Name	Value	Error
☒	Ls	535.527n	4.403n
☒	Rs	85.205m	118.848u
☒	Qy1	385.907m	36.418m
☒	Qa1	759.56m	16.784m
☒	R1	6.094m	247.429u
☒	Qy2	3.841	134.232m
☒	Qa2	960.069m	17.189m
☒	R2	1.563m	131.303u
☒	W	308.975	2.11



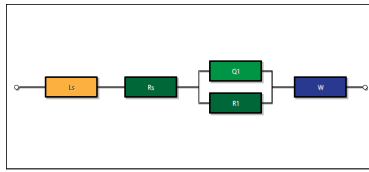
Free	Name	Value	Error
☒	Ls	508.972n	5.569n
☒	Rs	85.88m	114.56u
☒	Qy1	331.944m	36.233m
☒	Qa1	810.363m	19.438m
☒	R1	5.198m	237.526u
☒	Qy2	2.694	76.946m
☒	Qa2	1.012	17.11m
☒	R2	1.621m	127.592u
☒	W	274.816	358.325m

Screening

accurate



If user change mode for accurate mode same as screen mode



Ls-Rs-Q1|R1-W

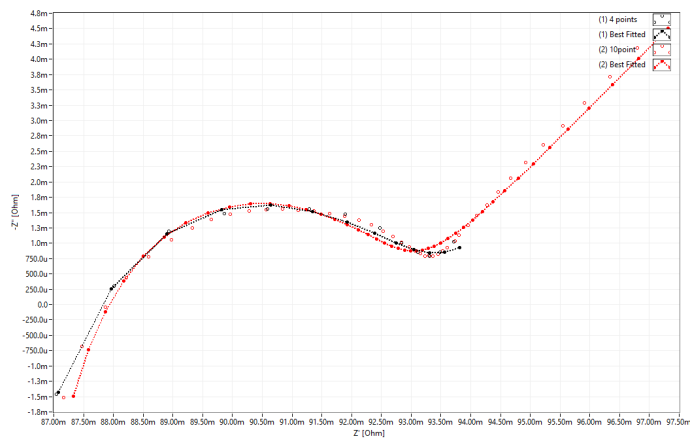
Free	Name	Value	Error
☒	Ls	560.219n	7.451n
☒	Rs	83.674m	207.418u
☒	Qy1	1.412	34.278m
☒	Qa1	549.836m	8.44m
☒	R1	9.719m	237.502u
☒	W	424.37	14.522

Ls-Rs-Q1|R1-W

Free	Name	Value	Error
☒	Ls	483.166n	5.783n
☒	Rs	85.762m	74.534u
☒	Qy1	1.126	20.491m
☒	Qa1	664.575m	5.028m
☒	R1	7.078m	80.62u
☒	W	281.824	888.705m

Screening

accurate



If the experiment can't be performed due to abnormal voltage or problems, the experiment automatically pauses with the following error.

Error: Failed control...

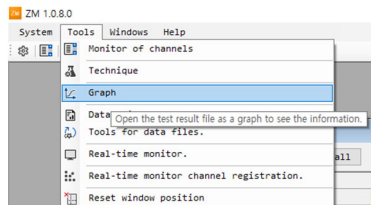
If the experiment is normally started, it will display as follow.

Status					Condition file		Control
Last Status	Elapsed(s)	Range	Vdc(V)	Temp.(°C)	File name	Tools	
Running(No calibration).	00:00:20	40mA/ 50V	3.734	0.000	technique_EIS1.eis		

For current-controlled impedance techniques, select the info/Nyquist/Bode tab to see the real-time information data list and Nyquist plot, Bode plot. For more information on real-time monitor and graph, please refer to separate chapter 6.

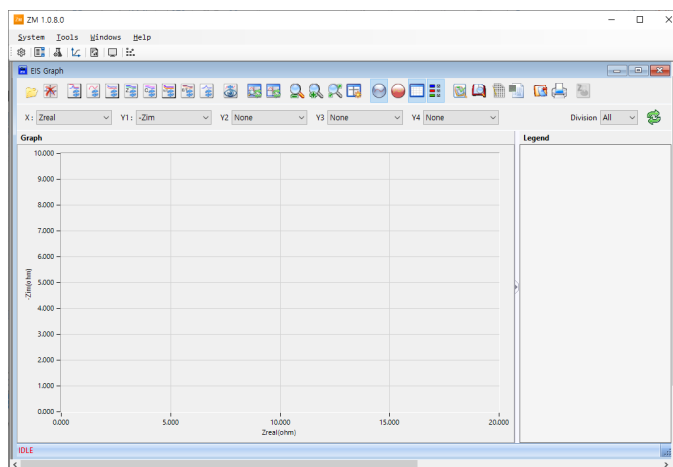
Chapter 9. Graph

1. Graph menu

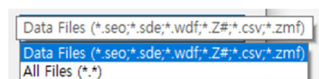


You can see the graph screen by selecting Tools

or click this  icon on the main toolbar.



1) Data file format



This software can read files having extension of seo, sde, wdf, z#, csv, zmf.

2) Axis parameters

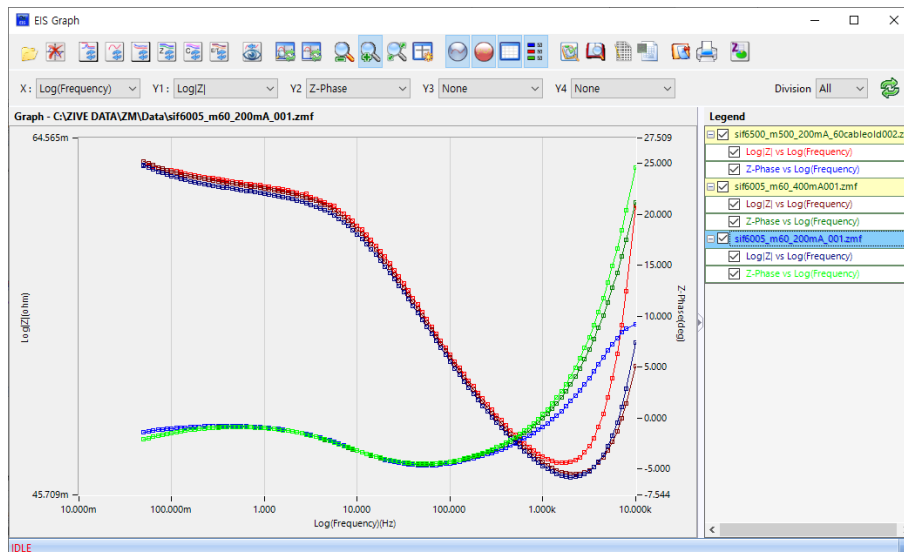
Parameter	Description	X axis	Y axis
Log(frequency)	Logarithmic value of frequency	Yes	No
Zreal	Real part of Impedance	Yes	Yes
Yreal	Real part of admittance	Yes	Yes
Z-phase	Phase shift of impedance	Yes	Yes
Y-phase	Phase shift of admittance	Yes	Yes
Rs(R-C)	Resistance value under Resistance-Capacitance serial connection	Yes	Yes
Cs(R-C)	Capacitance value under Resistance-Capacitance serial connection	Yes	Yes

Rp(R C)	Resistance value under Resistance Capacitance parallel connection	Yes	Yes
Cp(R C)	Capacitance value under Resistance Capacitance parallel connection	Yes	Yes
Rs(R-L)	Resistance value under Resistance-Inductance serial connection	Yes	Yes
Ls(R-L)	Inductance value under Resistance-Inductance serial connection	Yes	Yes
Q(R-L)	Q factor under Resistance-Inductance serial connection	Yes	Yes
time	Testing time	Yes	Yes
Vdc	DC potential value	Yes	Yes
Idc	DC current value	Yes	Yes
Temperature	Measured Temperature value	Yes	Yes
Aux1	Not available in BZA		
Aux2	Not available in BZA		
Aux3	Not available in BZA		
Frequency	Frequency value	Yes	No
Eoc	Open circuit potential	Yes	Yes
-Zim	Minus imaginary part of impedance	No	Yes
-Yim	Minus imaginary part of admittance	No	Yes
Z	Impedance magnitude	No	Yes
Y	Inductance magnitude	No	Yes
Log Z	Log impedance magnitude	No	Yes
Log Y	Log inductance magnitude	No	Yes
Q	Absolute value of capacity per cycle	Yes	Yes

3) Bode Plot



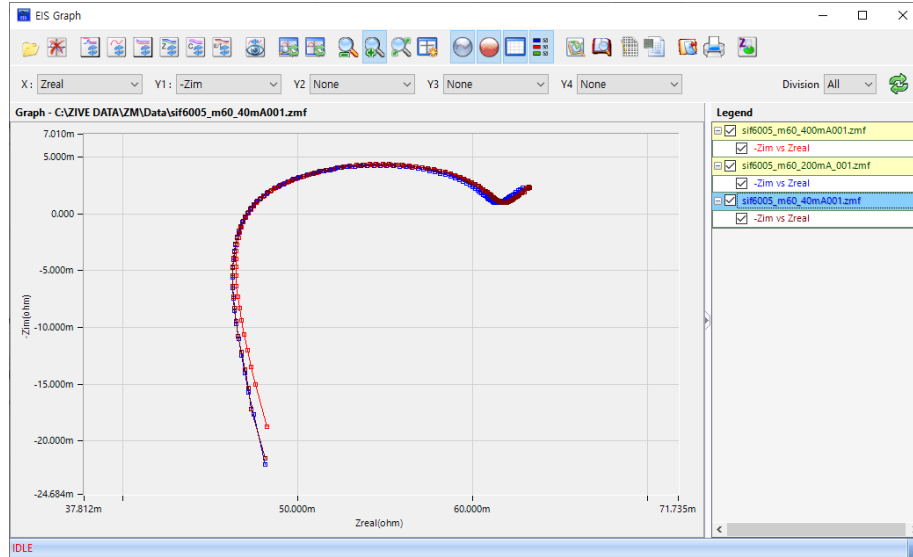
If you click this icon on graph tool bar, graph axis format will be changed as X axis: frequency, Y1 axis:Zph and Y2 axis: Magnitude(Log|Z|) automatically.



4) Nyquist plot



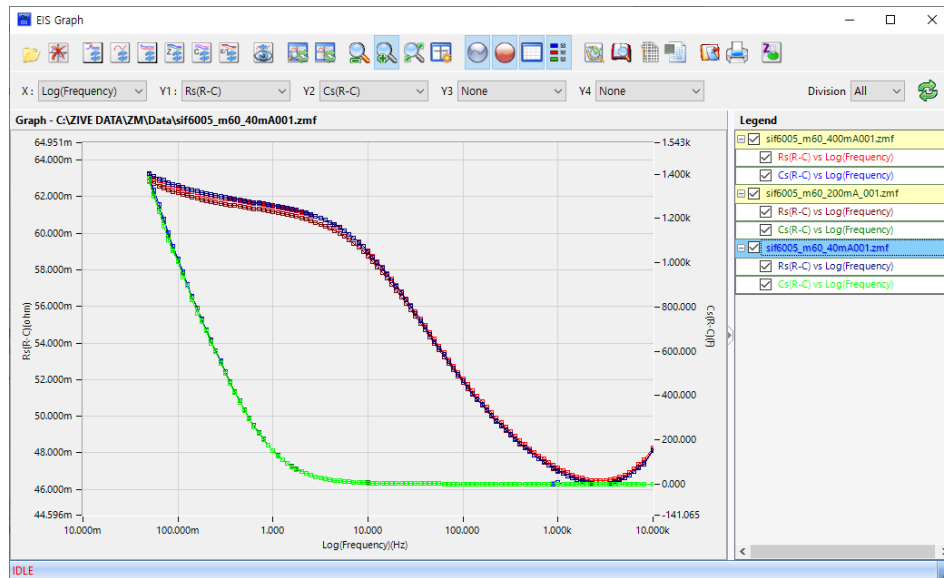
If you click this  icon on graph tool bar, graph format will be changed to X axis: Zreal(Zre), Y1 axis:-Zimaginary(-Zim) automatically.



5) Cs(R-C), Rs(R-C) vs. frequency plot



If you click this  icon on graph tool bar, graph format will be changed to X axis: log frequency, Y1 axis:Cs(R-C) and Y2 axis: Rs(R-C) automatically.



Type	x	y
Nyquist	Real part(Z')	-Z''
	Y'	Y''
Bode	f	y1
		Z
		y2
		Φ_z

$Z', -Z'', |Z|, \Phi_Z$

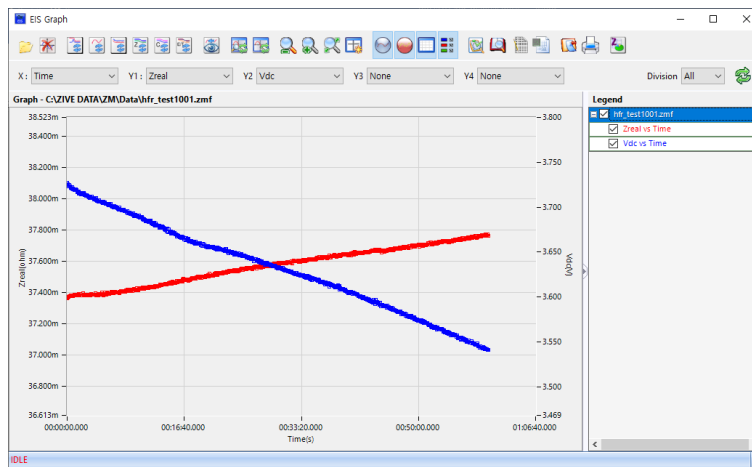
$Y', Y'', |Y|, \Phi_Y$

f is frequency, $Z = Z' + jZ'' = |Z| e^{j\Phi_Z}$, and $Y = \frac{1}{Z} = Y' + jY'' = |Y| e^{j\Phi_Y}$.

6) Zreal, Vdc vs time graph



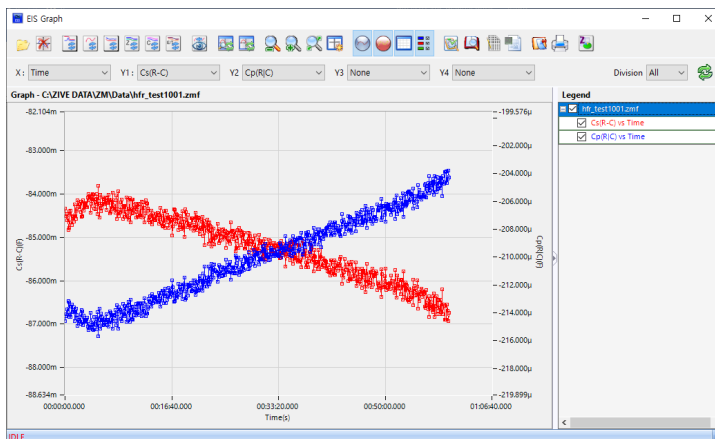
If you click this icon on graph tool bar, graph format will be changed to X axis: Tme, Y1 axis: Zreal and Y2 axis: Vdc automatically.



7) Cs, Cp vs. time graph



If you click this icon on graph tool bar, graph format will be changed to X axis: Tme, Y1 axis: Cs and Y2 axis: Cp automatically.

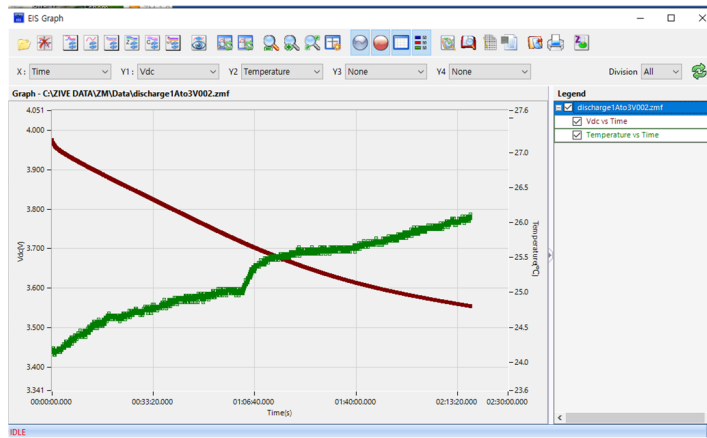


8) Vdc, Temperature vs time graph



If you click this icon on graph tool bar, graph format will be changed to X axis: Tme, Y1 axis: Vdc and Y2 axis: Temperature automatically.

It is used to see data to observe changes in voltage and temperature when discharging. You can use the data measured by voltage temperature monitor among techniques.

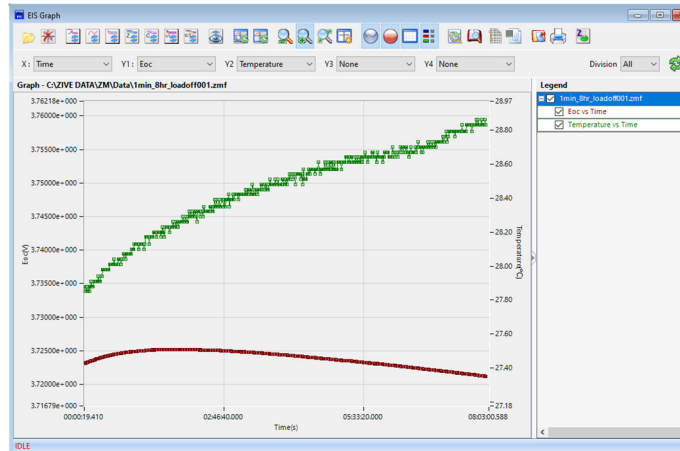


Changes in voltage and temperature when discharging.

9) Eoc, Temperature vs. time graph

If you click this  icon on graph tool bar, graph format will be changed to X axis: Tme, Y1 axis: Eoc and Y2 axis: Temperature automatically.

You can use the data measured by voltage temperature monitor among techniques for open circuit voltage and monitoring the temperature with the load off.



Changes in voltage and temperature when rest.

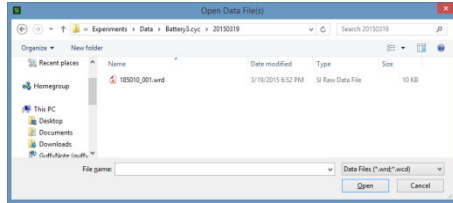
2. Graph common tool bar

1) Open data file.

If you click the following icon, you can select data file to open for graph.



If you want to overlay data file(s), you can use the above function. But if you want to see new graph, you must click graph icon on main tool bar.



Multiple file selection is available by dragging target file(s) or CTRL+ file selection

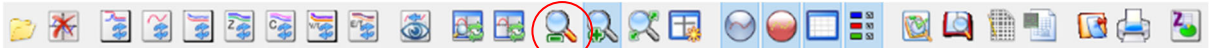
2) Clipboard copy

If you want to copy the graph, click the following icon and it will copy the image into clipboard



You can use copied image into other application software in Windows environment by click paste function (CTRL+V short key).

3) Graph initialize



If you want to come back initial scale after Zoom up etc., click this icon.

If you want to come back 1 step before, click mouse left button with pressing shift key..

4) Data file redraw



If you want to apply the updated file to the graph after updating the data file, click this icon.

If you want to update the data file in the file you are currently testing, click this icon to apply the updated status to the graph.

5) Graph zoom in



If you want zoom up the graph, click this icon and drag the cursor on target area.

If you want to come back 1 step before, click mouse left button with pressing shift key

6) Move



You can move graph by click this icon and move icon cursor.

7) Cursor



If you want to see data value on graph, click this icon.



Graph cursor is active.



Graph cursor is inactive



If you want to move the cursor, click cursor on + center part and dragging to target destination, graph cursor will move.

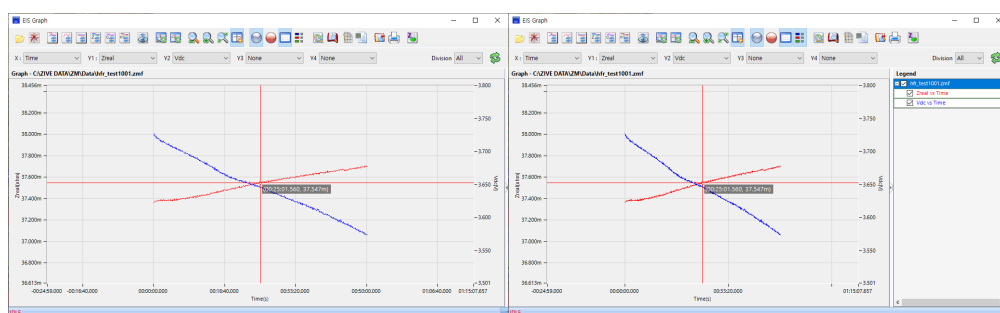
Current cursor value will be displayed right upper side. (Left value: X axis value, Right value: Y axis value; Unit will be displayed together.

If you did upload multiple files and/or multiple axis, activate legend by clicking check box and click axis name on legend area then cursor will be changed on selected data files' axis format.

8) Legend



You can see legend window by click the above icon.



- ① Each graph color is identical with legend character's color.
- ② By click on check box on legend, display will be on/off.

9) Dot/line selection

You can select dot graph and/or line graph.

- ①  Line graph activated.
- ②  Dot graph activated.

10) Grid

Grid on/off

- ①  Activate grid function
- ②  Inactivate grid function

11) Removing data file



If you uploaded multiple files for overlay, you can remove some data file from graph.

To remove the data file, click file name on right side legend box (If you click on data file name, back ground color will be changed) and click the above button, selected data file will be removed from graph.

Please note that this function will NOT erase the data file..

12) Print graph

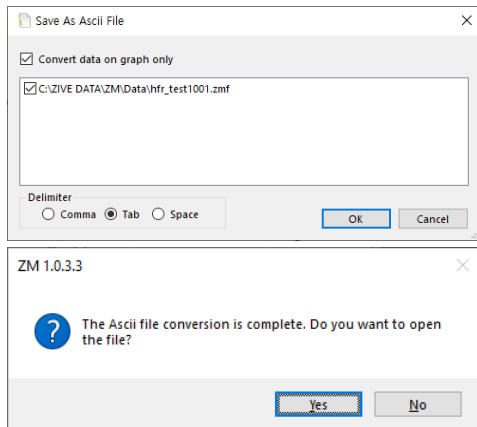


If you want to print graph, click this button..

13) ASCII file conversion



You can make ASCII file using the graph parameter value. Select “convert current view” and each specified axis value will be converted to ASCII format. If you select “convert selected file” selected file(s) will be converted to ASCII file with all data which the data file have.

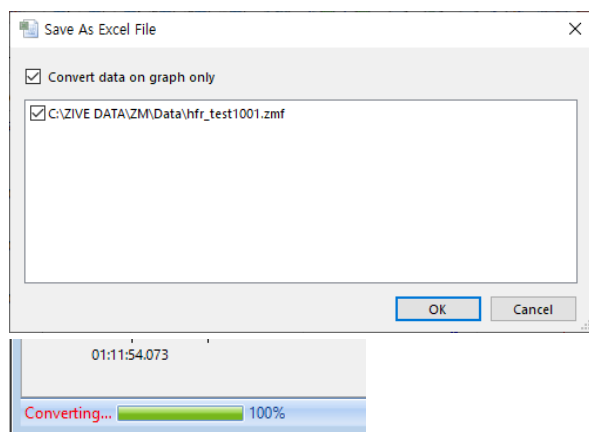


14) Excel file conversion

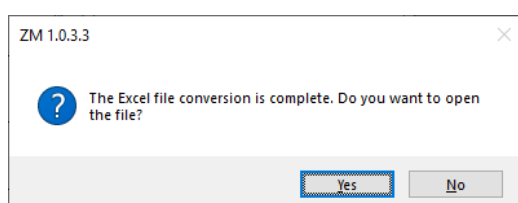


This function will convert graph data as Excel format.

If you select “convert current view” then each specified axis value will be converted to excel format. If you select “convert selected file” selected file(s) will be converted to excel file(s) with all data which the data file have.



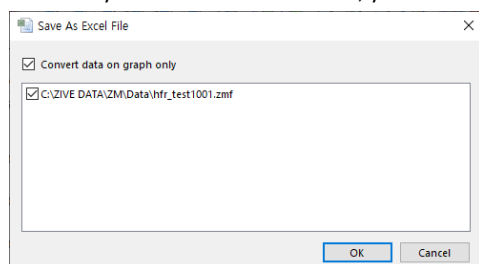
You can see progressing bar at left bottom side.



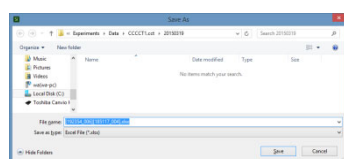
a. Convert Current View

If you want to convert X,Y1,Y2,Y3,Y4 axis value on graph only, select “convert current view”

When you click “convert” button, you can select target folder and input excel file name.



You can select target folder and the converted excel file(s) will be saved in this target folder with same named as original file name.



When you click OK button, Excel file will be saved as excel format (xlsx)

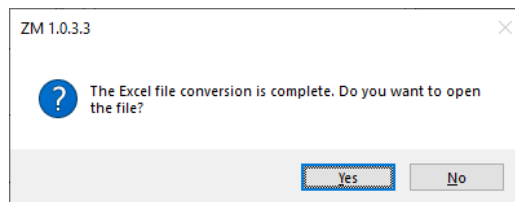
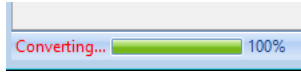
On Excel data file;

	A	B	C	D	E	F
1	EIS Data Report					
2	* Test Data File :	C:\ZIVE DATA\ZM\Data\hfr_test1001.zmf				
3	* Test Duration :	2022/07/19 15:49:39~2022/07/19 15:49:39				
4	* Data Count :	543				
5	* Tester :					
6	* Product No. :					
7	* Memo :					
8	* Schedule File Name :	C:\ZIVE DATA\ZM\Sch\5minhfr.hfr				
9	* Step Count :					
10						

There are file information on “File _Info” sheet. If you upload several data file for overlay on graph, Each data file is distinguished by File number.

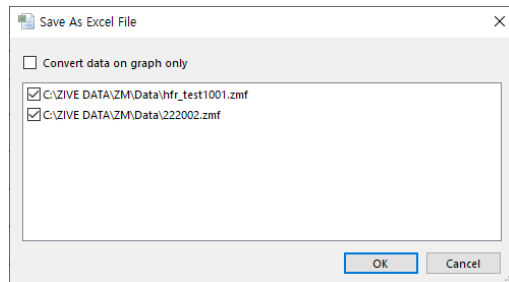
	A	B	C	D
1	Index	Time(s)	Zreal(ohm)	Vdc(V)
2	1	2.664	0.037364	3.72717
3	2	8.175	0.037368	3.72527
4	3	13.687	0.037371	3.72317
5	4	19.197	0.037379	3.72498

You can see converting progress bar on left bottom side



b. Convert all data set

If you want to convert all data information to an excel file, do not check on “Convert data on graph only”



If you did upload several files on graph, you can select file(s) which you want to convert

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	Index	Test_Time	File_No.	File_No.	Cycle_No.	Cycle_Tim	Step_No.	Step_Time	Frequency	Zre(ohm)	Zim(ohm)	Zmag(ohm)	Zph(deg)	Vdc(V)	Idc(A)	Vac(V)
2	1	2.664	1	2.664	0	2.664	0	2.664	999.9997	0.037364	0.001887	0.037412	2.891403	3.727173	0.2	
3	2	8.175	1	8.175	2	8.175	0	8.175	999.9997	0.037368	0.001884	0.037416	2.886803	3.725269	0.2	
4	3	13.687	1	13.687	3	13.687	0	13.687	999.9997	0.037371	0.001883	0.037419	2.885148	3.723169	0.2	
5	4	19.197	1	19.197	4	19.197	0	19.197	999.9997	0.037379	0.001881	0.037426	2.880999	3.724976	0.2	
6	5	24.707	1	24.707	5	24.707	0	24.707	999.9997	0.037375	0.001882	0.037423	2.883282	3.722729	0.2	
7	6	30.216	1	30.216	6	30.216	0	30.216	999.9997	0.037383	0.001876	0.03743	2.873143	3.721851	0.2	
8	7	35.726	1	35.726	7	35.726	0	35.726	999.9997	0.037384	0.001877	0.037431	2.874935	3.720703	0.2	

15) View data list



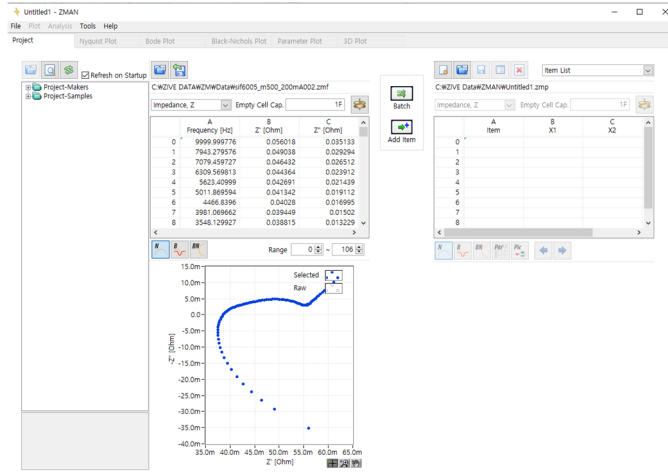
Refer to chapter 10 data editor.

16) Link to ZMAN eis data analysis software



If you want use ZMAN, a separate independent analysis program, click this icon.

You can analyze the data measured by ZM software without license code if ZMAN software is installed. This link provides the currently uploaded graphs on ZMAN. ZMAN will open only the latest data file you uploaded if you upload multiple overlapping data files. For more information, please refer to separate ZMAN manual.



17) Advanced setting



Additional graph function menu.

Graph Axes and Data Filter Setting

[Set Axes]					
Item	X Axis	Y1 Axis	Y2 Axis	Y3 Axis	Y4 Axis
Name	Discharge ...	Discharge P_s	None	None	None
Direction	☐ Invert	☐ Invert	☐ Invert	☐ Invert	☐ Invert
ScaleType	☒ Logarithm	☒ Logarithm	☐ Logarithm	☐ Logarithm	☐ Logarithm
Color					

[Data Filter]	
☐ Test Time	00:00:0 ~ 00:00:0
☐ Cycle number	
☐ Step number	

Refresh Cancel

a. Axis direction

If you want to change axis direction, please check on "invert" for each axis.

Graph Axes and Data Filter Setting

[Set Axes]					
Item	X Axis	Y1 Axis	Y2 Axis	Y3 Axis	Y4 Axis
Name	Discharge ...	Discharge P_s	None	None	None
Direction	☐ Invert	☐ Invert	☐ Invert	☐ Invert	☐ Invert
ScaleType	☒ Logarithm	☒ Logarithm	☐ Logarithm	☐ Logarithm	☐ Logarithm
Color					

b. Axis scale type

You can select axis display scale as logarithm or linear scale.

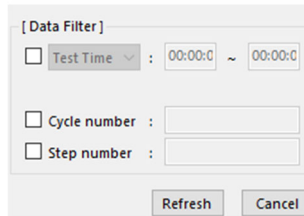
[Set Axes]					
Item	X Axis	Y1 Axis	Y2 Axis	Y3 Axis	Y4 Axis
Name	Discharge ...	Discharge P_s	None	None	None
Direction	☐ Invert	☐ Invert	☐ Invert	☐ Invert	☐ Invert
ScaleType	☒ Logarithm	☒ Logarithm	☐ Logarithm	☐ Logarithm	☐ Logarithm
Color					

c. Axis color setting

You can change axis color by clicking the color.

d. Setting data range

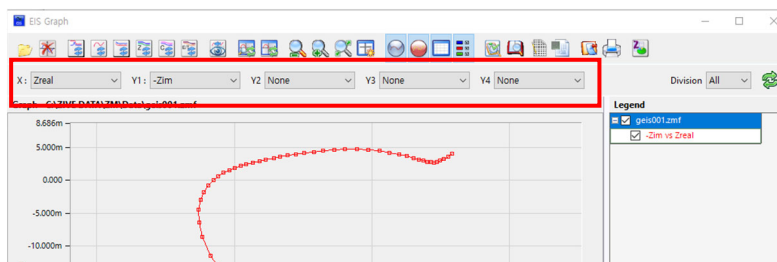
You can define data range to show on graph.



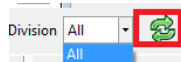
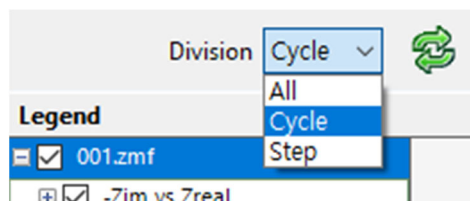
For File number, Cycle number, you can input numbers using comma(,) and/or region (-) mark
ex) If you want to display the graph with 1,3,4,5,6,7,10,15th cycles data only, input 1,3-7,10,15

3. Axis parameter setting**A. Axis selection**

You can select axis parameter for X axis: 1 Y axis 4.

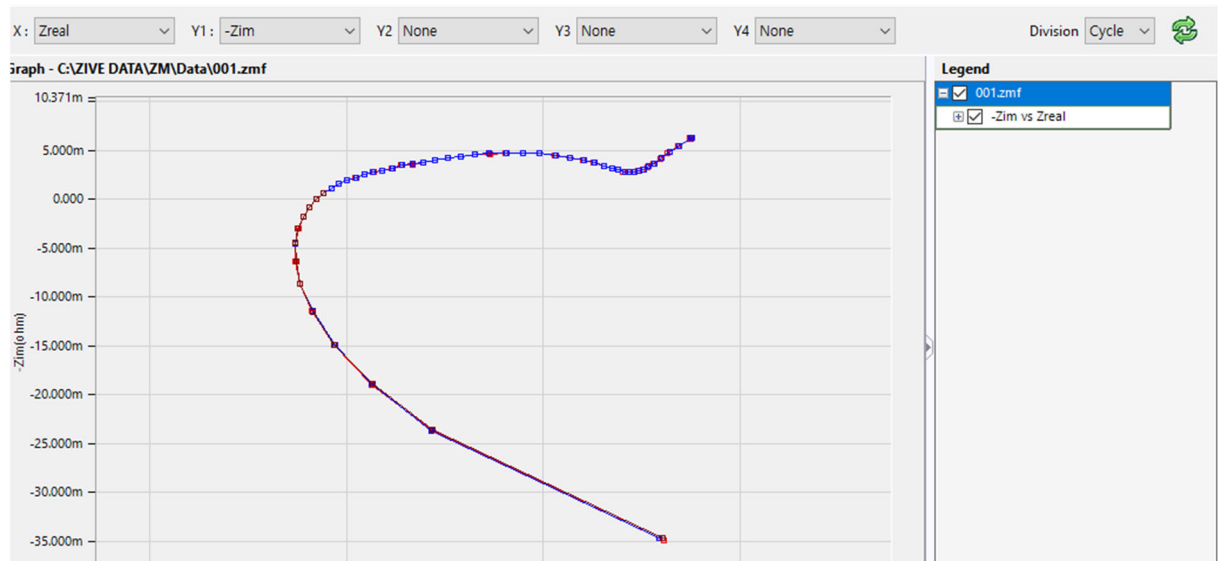


After selection, you must click Refresh button to apply the selection change.

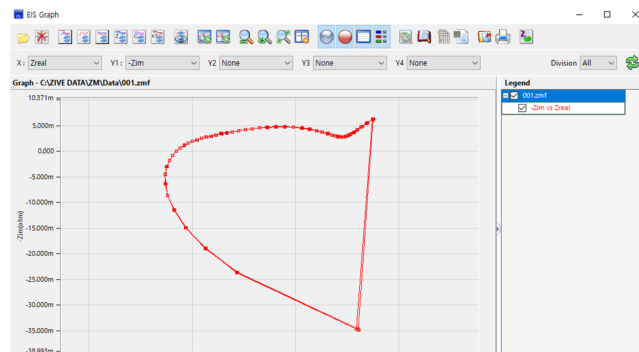
**B. Data line split**

You can split data by file, cycle number. Split data set's color was changed and graph line will be separated.

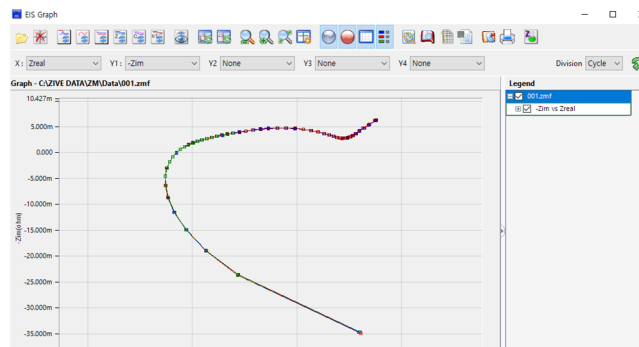
If you test eis test using multiple Iteration, You can use cycle to distinguish each eis graph.



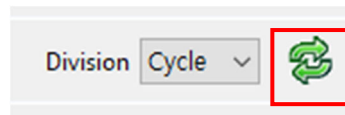
Before divided



After Cycle separation



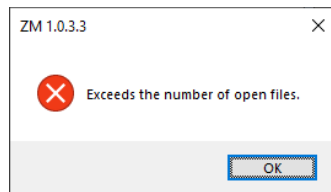
Note: when you click refresh button, this separation will work.



4. Other functions

A. File overlay

You can overlay max. 20 files on graph. If you tried to open more than 20 data files, following error message will display

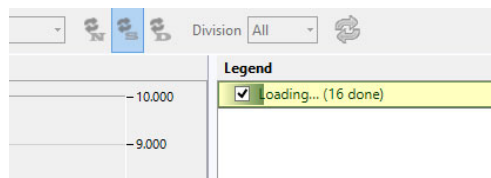


You can add the file(s) by opening the data file to overlay on current graph.

Multi files selection is available on opening the file(s). (You can drag the file area or CTRL + file selection)

If you want to make a new graph, click one of the graph icon  on main tool bar.

B. Data file loading for graph



If data file size is too big and it takes time to load, you can see the above progress bar.

You can click cancel button during loading data process, then you can see the data till loading cancel time.

It will take longer time when you try to load for dot graph.

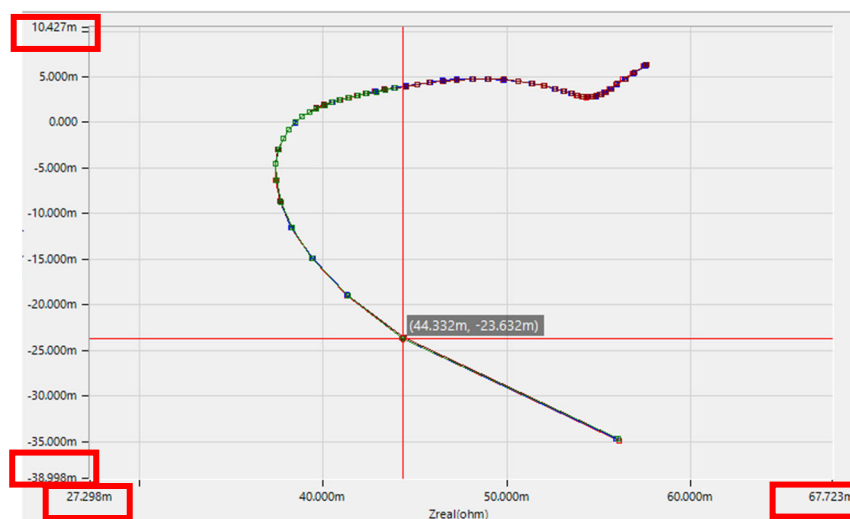
C. Cursor selection

You can change the cursor by clicking right mouse button when you locate cursor on graph.

The cursor can be changed  normal 3 modes by clicking right mouse button

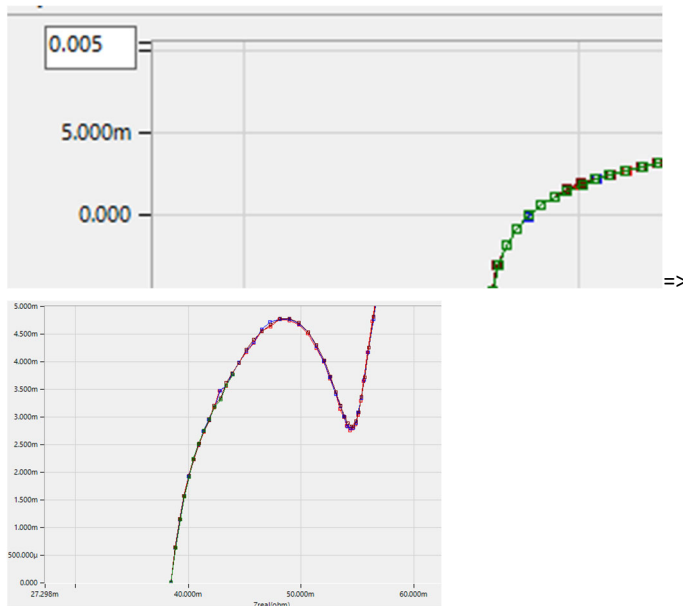
D. Manual axis Scaling

You can change each axis scaling.



You can input max and min value by clicking max/min value on graph axis. (Red value on the above graph)

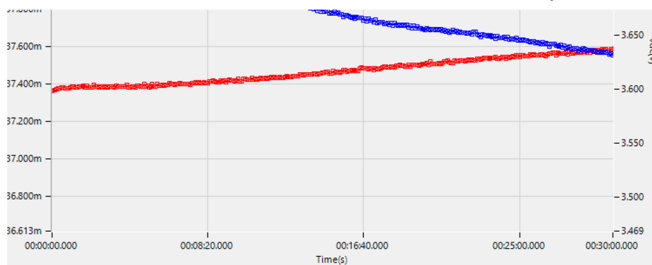
If you want to change max -Zimg as 0.005 value then click 10.427m value and input 0.005. Then -Zimg axis max value will be changed to 5m value



Note: Input value for time must be as **:**:** format.

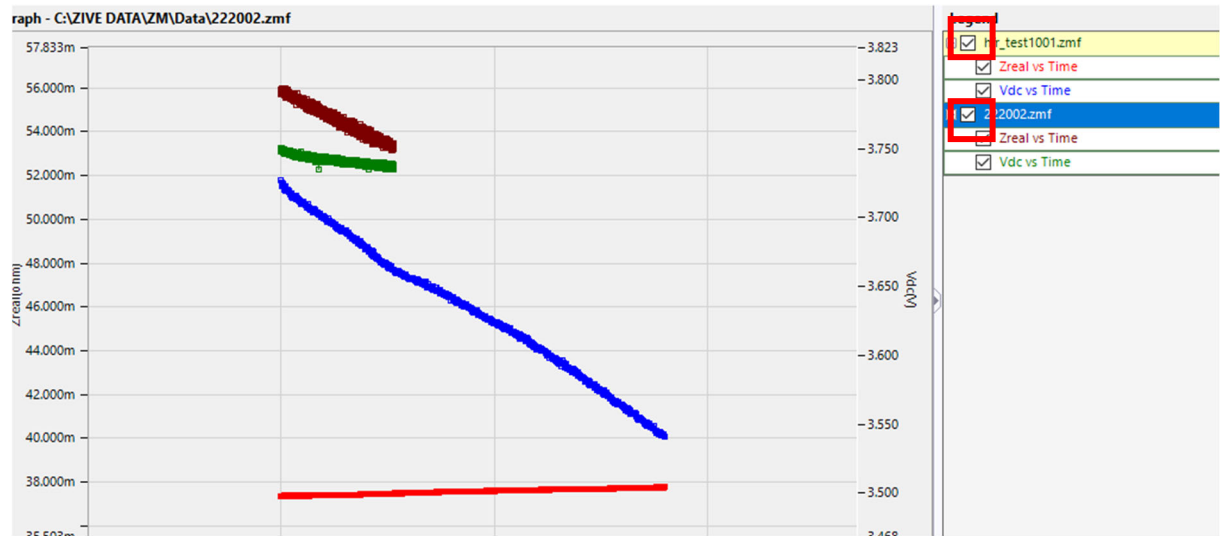


If you want to set 30 minutes



E. Legend selection

1) Data set on/off display



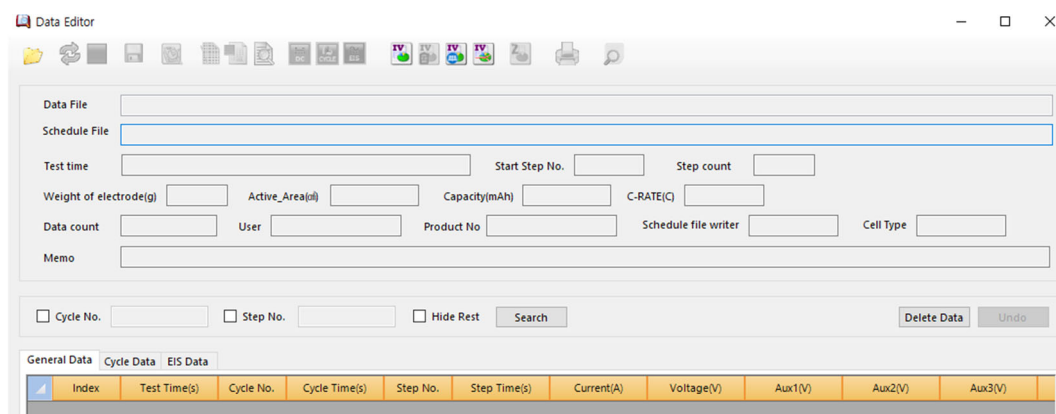
You can select visible on/off selection at check box nearby file name.

Chapter 10. Data editor

1. Data editor

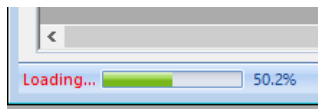
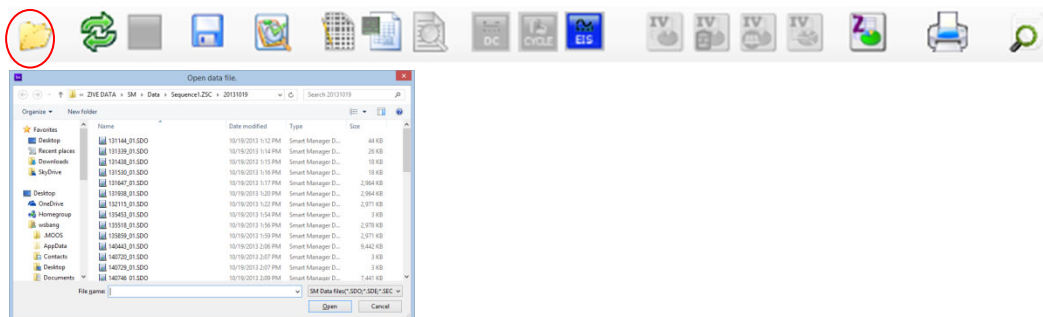
This function can be used simply as a data list or data editor while converting data files provided by this software, and you can use it from there since it is also supported by the original software.

The following menu will display if click this  icon on tool bar.



A. Open data file.

You can select data file by clicking the following icon on tool bar.



BATTERY IMPEDANCE ANALYZER (BZA series) Manual

Data Editor

Data File: C:\ZIVE DATA\ZM\Data\sf6005_m500_40mA_500cable002.zmf Run IV/Man Tafel Analysis

Schedule File: C:\ZIVE DATA\ZM\Sch\technique_EIS3_sf6005_300mA.eis

Test time: 2022/07/20 13:51:05 ~ 2022/07/20 13:59:48 Start Step No. Step count

Weight of electrode(g) Active_Area(cm²) Capacity(mAh) 1000 C-RATE(C)

Data count: 107 User Product No Schedule file writer Battery ID

Memo

☐ Cycle No. ☐ Step No. ☐ Hide Rest Search Delete Data Undo

General Data Cycle Data EIS Data

Index	Test Time(s)	File No.	File Time(s)	Cycle No.	Cycle Time(s)	Step No.	Step Time(s)	Frequency(Hz)	Zre(ohm)	Zim(ohm)	Zp
1	00:00:02.655	1	00:00:02.655	1	00:00:02.655	1	00:00:02.655	10000.000	59.16051e-3	8.22405e-3	59.16051e-3
2	00:00:03.169	1	00:00:03.169	1	00:00:03.169	1	00:00:03.169	7943.280	54.4739e-3	7.5236e-3	54.4739e-3
3	00:00:03.683	1	00:00:03.683	1	00:00:03.683	1	00:00:03.683	7079.460	53.02217e-3	6.85166e-3	53.02217e-3
4	00:00:04.197	1	00:00:04.197	1	00:00:04.197	1	00:00:04.197	6309.570	51.65327e-3	6.21947e-3	51.65327e-3
5	00:00:04.711	1	00:00:04.711	1	00:00:04.711	1	00:00:04.711	5623.410	50.57104e-3	5.61735e-3	50.57104e-3
6	00:00:05.225	1	00:00:05.225	1	00:00:05.225	1	00:00:05.225	5011.870	49.70928e-3	5.02061e-3	49.70928e-3
7	00:00:05.738	1	00:00:05.738	1	00:00:05.738	1	00:00:05.738	4466.840	48.96896e-3	4.43479e-3	48.96896e-3
8	00:00:06.252	1	00:00:06.252	1	00:00:06.252	1	00:00:06.252	3981.070	48.42773e-3	3.84963e-3	48.42773e-3
9	00:00:06.765	1	00:00:06.765	1	00:00:06.765	1	00:00:06.765	3548.130	48.01295e-3	3.30439e-3	48.01295e-3
10	00:00:07.279	1	00:00:07.279	1	00:00:07.279	1	00:00:07.279	3162.280	47.66948e-3	2.80664e-3	47.66948e-3
11	00:00:07.795	1	00:00:07.795	1	00:00:07.795	1	00:00:07.795	2818.380	47.44697e-3	2.29209e-3	47.44697e-3
12	00:00:08.309	1	00:00:08.309	1	00:00:08.309	1	00:00:08.309	2511.890	47.28616e-3	1.82371e-3	47.28616e-3
13	00:00:08.823	1	00:00:08.823	1	00:00:08.823	1	00:00:08.823	2238.720	47.17891e-3	1.40327e-3	47.17891e-3
14	00:00:09.338	1	00:00:09.338	1	00:00:09.338	1	00:00:09.338	1995.260	47.10251e-3	1.03339e-3	47.10251e-3
15	00:00:09.854	1	00:00:09.854	1	00:00:09.854	1	00:00:09.854	1778.280	47.09625e-3	641.59323e-6	47.09625e-3
16	00:00:10.369	1	00:00:10.369	1	00:00:10.369	1	00:00:10.369	1584.890	47.10998e-3	297.08172e-6	47.10998e-3
17	00:00:10.886	1	00:00:10.886	1	00:00:10.886	1	00:00:10.886	1412.540	47.15126e-3	-16.27973e-6	47.15126e-3
18	00:00:11.402	1	00:00:11.402	1	00:00:11.402	1	00:00:11.402	1258.930	47.18131e-3	-331.45864e-6	47.18131e-3
19	00:00:11.920	1	00:00:11.920	1	00:00:11.920	1	00:00:11.920	1122.020	47.31154e-3	-613.70612e-6	47.31154e-3
20	00:00:12.440	1	00:00:12.440	1	00:00:12.440	1	00:00:12.440	1000.000	47.42697e-3	-900.64471e-6	47.42697e-3
21	00:00:12.965	1	00:00:12.965	1	00:00:12.965	1	00:00:12.965	891.251	47.56259e-3	-1.13222e-3	47.56259e-3

Loaded

[Test information]

Battery ID: Li_pack5000

Nominal capacity(mAH): 3000

Creator: Tester name_sample

Data File: C:\ZIVE DATA\ZM\Data\lipack test1001

Schedule File: C:\ZIVE DATA\ZM\Sch\technique_EIS1

Test time: 2022/07/19 17:00:02 ~ Start Step No. Step count

Weight of electrode(g) Active_Area Active Area Capacity(mAh) 3000 C-RATE(C)

Data count: 20 User: Tester name_sample Product No Schedule file writer Battery ID: Li_pack5000

Memo: Memo Test 1st row 2nd row

[Information]

☒ Use the parameters of the technique file.

Technique file: C:\ZIVE DATA\ZM\Sch\technique_EIS1.eis

Battery id: Li_pack5000 Nominal capacity(AH): 3000.0

User: Tester name_sample

Memo: Memo Test 1st row 2nd row

[Information]

☐ Use the parameters of the technique file.

Technique file: C:\ZIVE DATA\ZM\Sch\technique_EIS1.eis

Battery id: Li_pack5000 Nominal capacity(AH): 3000.0

User: Tester name_sample

Memo:

1) Data information

Data file related

- ① File name: data file name and path
- ② Duration: experiment start and end time
- ③ Data count: number of data points
- ④ Tester: name of the tester
- ⑤ Memo: related memo

Schedule file related

- ① schedule File: experiment condition file name and path

2) Data item

General Data	Cycle Data	EIS Data
Index	Test Time(s)	

- Index: Index No.
- Test time(s): Elapsed time from experiment started
- File No.: file number for batch file
- File time(s): Elapsed time from new schedule file started on batch file
- Cycle No : Cycle count number
- Cycle Time: Elapsed time from cycle started
- Step No: Schedule file's step number
- Step Time(s): Elapsed time from step started
- Zre(ohm)
- Zim(Ohm)
- Zmag(ohm)
- Zphase
- Vdc
- Idc
- Temperature: Temperature value (TEMP is option)
- Aux1: not available with BZA
- Aux2: not available with BZA
- Aux3: not available with BZA

3) Data deleting

You can delete the value by clicking index number and click "delete data" button.

☐ Cycle No.
☐ Step No.
☐ Hide Rest

General Data
Cycle Data
EIS Data

	Zre(ohm)	Zim(ohm)	Zmag(ohm)	ZPhase(deg)	Vdc(V)	Idc(A)	Eoc(V)	Temp.(°C)	Aux1(V)	Aux2(V)
	37.36409e-3	1.88716e-3	37.41172e-3	2.8914e+0	3.72717e+0	200.0000e-3	3.73372e+0	0.00	0.0000e+0	0.0000e+0
	37.36848e-3	1.88438e-3	37.41596e-3	2.8868e+0	3.72527e+0	200.0000e-3	3.73372e+0	0.00	0.0000e+0	0.0000e+0
	37.37143e-3	1.88344e-3	37.41886e-3	2.88515e+0	3.72317e+0	200.0000e-3	3.73372e+0	0.00	0.0000e+0	0.0000e+0
	37.37869e-3	1.8811e-3	37.42599e-3	2.8810e+0	3.72498e+0	200.0000e-3	3.73372e+0	0.00	0.0000e+0	0.0000e+0
	37.37532e-3	1.88242e-3	37.42269e-3	2.88328e+0	3.72273e+0	200.0000e-3	3.73372e+0	0.00	0.0000e+0	0.0000e+0
	37.38305e-3	1.87618e-3	37.4301e-3	2.87314e+0	3.72185e+0	200.0000e-3	3.73372e+0	0.00	0.0000e+0	0.0000e+0
	37.38353e-3	1.87737e-3	37.43064e-3	2.87494e+0	3.7207e+0	200.0000e-3	3.73372e+0	0.00	0.0000e+0	0.0000e+0
	37.37842e-3	1.8859e-3	37.42596e-3	2.88837e+0	3.72256e+0	200.0000e-3	3.73372e+0	0.00	0.0000e+0	0.0000e+0
	37.37844e-3	1.88475e-3	37.42593e-3	2.88661e+0	3.7199e+0	200.0000e-3	3.73372e+0	0.00	0.0000e+0	0.0000e+0

If you want to undo delete, click “undo” button. You can recover only one recent deleted data.

If you want to delete multiple data, click and drag the region on index column then selected data will change the background color as yellow. You can delete selected data by clicking “delete data” button.

☐ Cycle No.

☐ Step No.

☐ Hide Rest

Delete Data

Undo

General Data

Cycle Data

EIS Data

	Zre(ohm)	Zim(ohm)	Zmag(ohm)	ZPhase(deg)	Vdc(V)	Idc(A)	Eoc(V)	Temp.(°C)	Aux1(V)	Aux2(V)
	37.36409e-3	1.88716e-3	37.41172e-3	2.8914e+0	3.72717e+0	200.0000e-3	3.73372e+0	0.00	0.0000e+0	0.0000e+0
	37.36848e-3	1.88438e-3	37.41596e-3	2.8868e+0	3.72527e+0	200.0000e-3	3.73372e+0	0.00	0.0000e+0	0.0000e+0
	37.37143e-3	1.88344e-3	37.41886e-3	2.88515e+0	3.72317e+0	200.0000e-3	3.73372e+0	0.00	0.0000e+0	0.0000e+0
	37.37869e-3	1.8811e-3	37.42599e-3	2.8810e+0	3.72498e+0	200.0000e-3	3.73372e+0	0.00	0.0000e+0	0.0000e+0
	37.37532e-3	1.88242e-3	37.42269e-3	2.88328e+0	3.72273e+0	200.0000e-3	3.73372e+0	0.00	0.0000e+0	0.0000e+0
	37.38305e-3	1.87618e-3	37.4301e-3	2.87314e+0	3.72185e+0	200.0000e-3	3.73372e+0	0.00	0.0000e+0	0.0000e+0
	37.38353e-3	1.87737e-3	37.43064e-3	2.87494e+0	3.7207e+0	200.0000e-3	3.73372e+0	0.00	0.0000e+0	0.0000e+0
	37.37842e-3	1.8859e-3	37.42596e-3	2.88837e+0	3.72256e+0	200.0000e-3	3.73372e+0	0.00	0.0000e+0	0.0000e+0
	37.37844e-3	1.88475e-3	37.42593e-3	2.88661e+0	3.7199e+0	200.0000e-3	3.73372e+0	0.00	0.0000e+0	0.0000e+0
	37.37805e-3	1.88597e-3	37.4256e-3	2.8885e+0	3.72063e+0	200.0000e-3	3.73372e+0	0.00	0.0000e+0	0.0000e+0
	37.3837e-3	1.88003e-3	37.43095e-3	2.87899e+0	3.72017e+0	200.0000e-3	3.73372e+0	0.00	0.0000e+0	0.0000e+0
	37.38553e-3	1.87863e-3	37.4327e-3	2.87671e+0	3.71941e+0	200.0000e-3	3.73372e+0	0.00	0.0000e+0	0.0000e+0

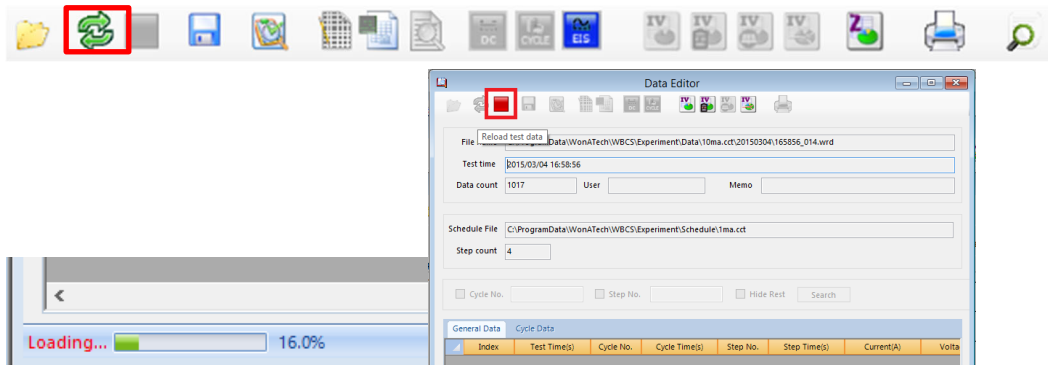
4) Data editing

Index	Test Time(s)	Cycle No.	Cycle Time(s)	Step No.	Step Time(s)	Current(A)	Voltage(V)	Auxiliary(V)	Temperature(°C)
176	000156.970	1	000156.970	2	000000.070	0.0000e+0	1.19797e+0	-152.58789e-6	0.00
177	000157.270	1	000157.270	2	000000.370	0.0000e+0	1.1969e+0	-305.17578e-6	0.00
188	000213.720	1	000213.720	2	000016.820	0.0000e+0	1.18515e+0	-305.17578e-6	0.00
189	000216.440	1	000216.440	2	000019.540	0.0000e+0	1.18408e+0	-457.76367e-6	0.00
190	000219.690	1	000219.690	2	000022.790	0.0000e+0	1.18301e+0	-305.17578e-6	0.00
191	000223.400	1	000223.400	2	000026.500	0.0000e+0	1.18195e+0	-152.58789e-6	0.00
192	000227.060	1	000227.060	2	000030.160	0.0000e+0	1.18088e+0	-305.17578e-6	0.00
193	000231.560	1	000231.560	2	000034.660	0.0000e+0	1.17981e+0	-305.17578e-6	0.00
194	000236.220	1	000236.220	2	000039.320	0.0000e+0	1.17874e+0	-305.17578e-6	0.00
195	000241.300	1	000241.300	2	000044.400	0.0000e+0	1.17767e+0	-305.17578e-6	0.00
196	000247.930	1	000247.930	2	000051.030	0.0000e+0	1.17661e+0	-305.17578e-6	0.00
197	000253.850	1	000253.850	2	000056.950	0.0000e+0	1.17554e+0	-152.58789e-6	0.00
198	000257.320	1	000257.320	3	000000.420	30.55300e-3	1.17172e+0	-305.17578e-6	0.00
199	000257.650	1	000257.650	3	000000.750	-19.88094e-3	1.17065e+0	-152.58789e-6	0.00

You can change the data value by selecting data and input numbers.

5) Reloading data

If you click following icon, original data file will be reloaded.

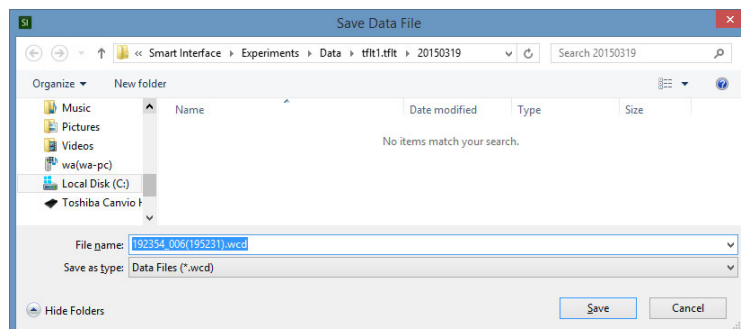


If the data file size is too large and take long time to load the data set, you can stop the loading by pressing stop button .

6) Modified Data file saving



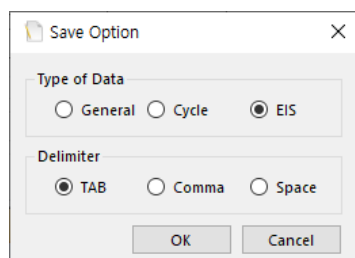
You can save edited data by clicking the above icon. Only loaded data will be saved. File name will be changed as original filename + (Hour-Minute-seconds).



7) ASCII file conversion



This function is to convert WONATECH zm data file format to ASCII format.



If you used searching function, you can convert filtered data set only by selecting “convert Data on report Only”

If you select “Convert selected file”, all data points will be converted.

Also you can select Delimiter (Tab, comma, Line feed). When you select comma, converted file’s extension will be csv. Other’s are txt.

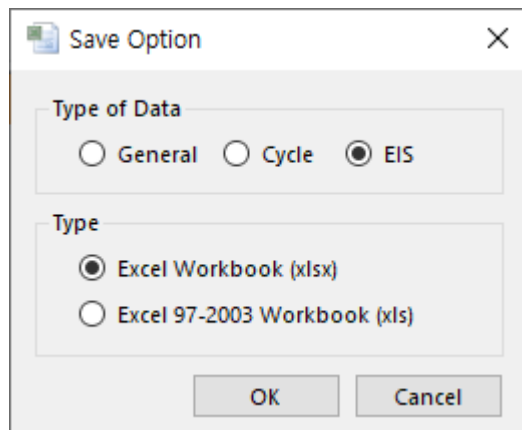
You MUST select EIS for Type of data.

8) Excel file conversion



This function is to convert WONATECH zm data file format to excel format.

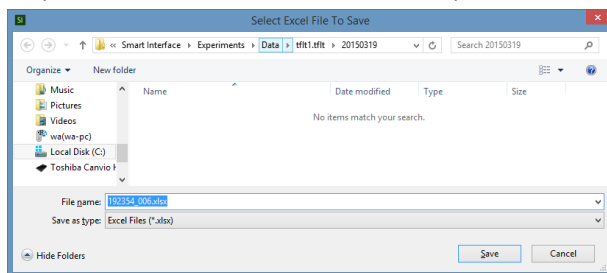
If you click this button, you can select excel version as the follows.



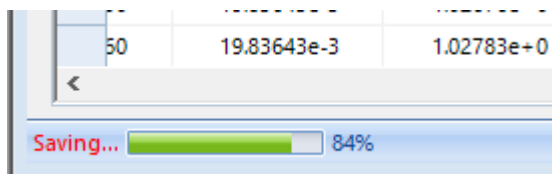
You MUST select EIS for Type of data.

If you used searching function, you can convert filtered data set only by selecting “convert Data on report Only”

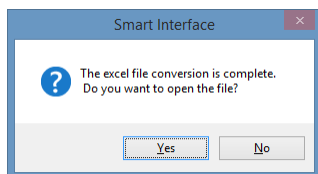
If you select “Convert selected file”, all data points will be converted.



You can select target folder and input file name.



When it finished, select to open Excel program or close the process as the follows.



9) EIS graph menu



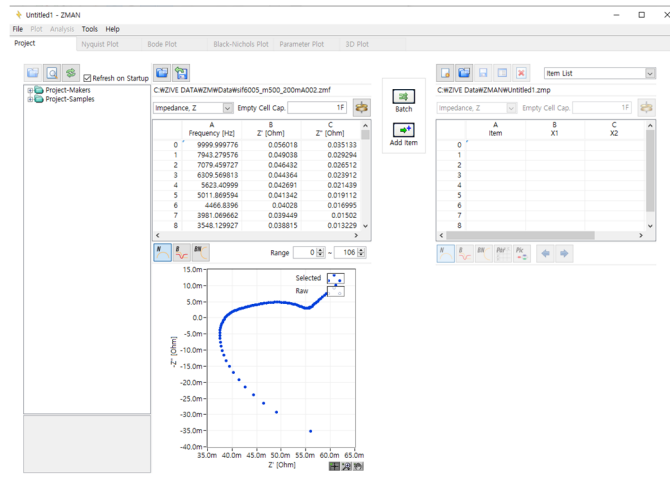
If you click the above icon, you can see graph for the data set.

10) ZMAN link



If you want use ZMAN, a separate independent analysis program, click this icon.

You can analyze the data measured by ZM software without license code if ZMAN software is installed. This link provides the currently uploaded graphs on ZMAN. ZMAN will open only the latest data file you uploaded if you upload multiple overlapping data files. For more information, please refer to separate ZMAN manual.

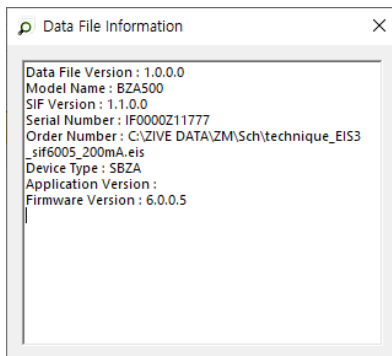


This is an icon for converting WONATECH Smart Interface data into Excel format.

11) BZA information



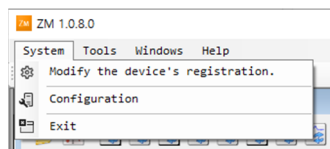
If you click the above icon, display information about the device and data used for data measurement.



Chapter11. Menu & icon in system tray

1. Menu

A. System menu



a. Modify the device's registration

Go to device discovery. Refer to chapter4-3

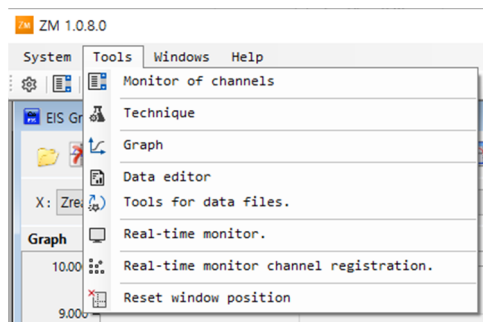
b. Configuration

Refer to chapter4 -5

c. Exit

Terminate ZM software

B. Tools menu



a. Monitor of channels

Refer to chapter5

b. Technique

Refer to chapter7

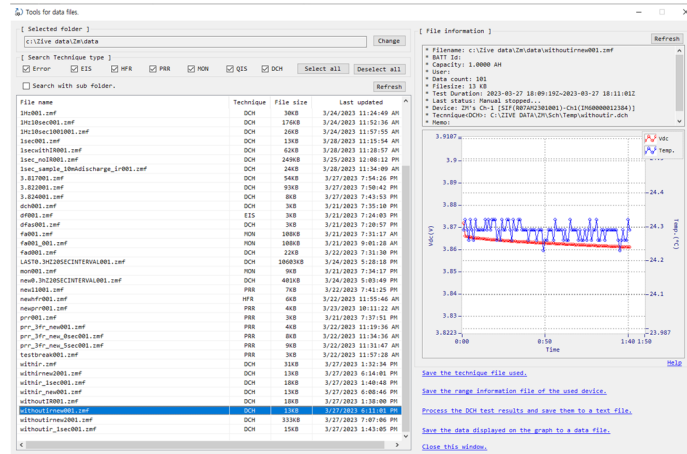
c. Graph

Refer to Chapter9.

d. Data editor

Refer to chapter10

e. Tool for data files

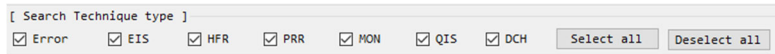


There are several functions in this tool for data files.

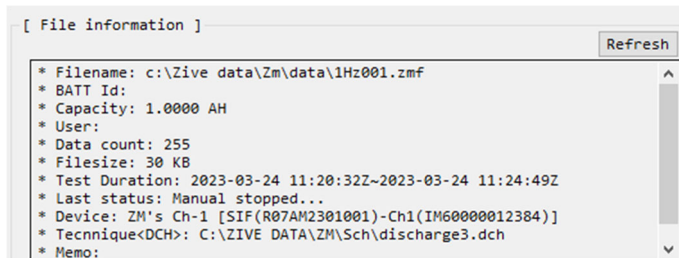
1) Select folder to view the data file



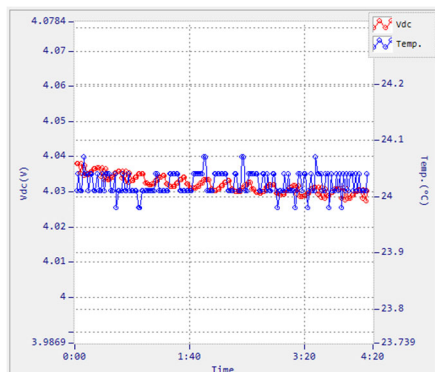
2) sorting data files by techniques.



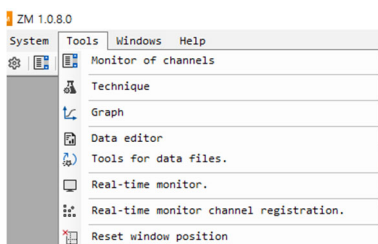
3) summary data files



4) Major Graph format per technique



5) For special technique, software can convert data file with specific data in csv format in tools-tools for data files



The specific technique for this data conversion are;

- HFR [Process the HFR test results and save them to a text file.](#)
Including Cs(F) and Cp(F)
- PRR [Process the PRR test results and save them to a text file.](#)
Including Cs(F) , Cp(F) per frequency and (F3-F2).real(Ω), (F3-F1).real(Ω), (F2-F1).real(Ω)
- DCH [Process the DCH test results and save them to a text file.](#)
Including capacity data.

f. Real-time monitor

User defined channel(s) real-time monitor. Refer to Chapter6.

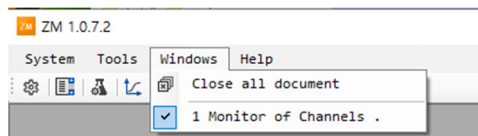
g. Real time monitor channel registration

User defined channel(s) real-time monitor channel registration. Refer to Chapter6

h. Reset window position

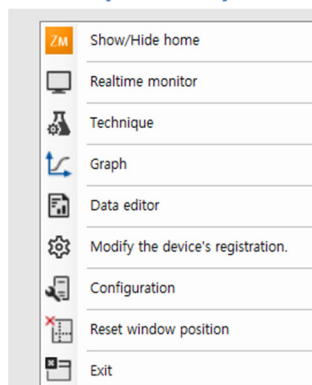
ZM software writes each windows' location and size information. This function will initialize these information.

C. Windows Menu



Close all document: Close all windows which were opened.

2. Icon in system tray



A. Show/Hide home

Toggle function to show or hide ZM icon on tool bar

B. Realtime monitor

User defined channel(s) real-time monitor. Refer to Chapter6.

C. Technique

Refer to chapter7

D. Graph

Refer to chapter 9

E. Data editor

Refer to chapter10

F. Modify the device's registration

Go to device discovery. Refer to chapter4-3

G. Configuration

Refer to chapter4 -5

H. Reset window position

ZM software writes each windows' location and size information. This function will initialize these information

I. Exit

Terminate ZM software

3. Main Tool bar



A. Modify the device's registration

Go to device discovery. Refer to chapter4-3

B. Monitor of channels

Refer to chapter5

C. Technique

Refer to chapter7

D. Graph

Refer to chapter 9

E. Data editor

Refer to chapter10

F. Realtime monitor

User defined channel(s) real-time monitor. Refer to Chapter6.

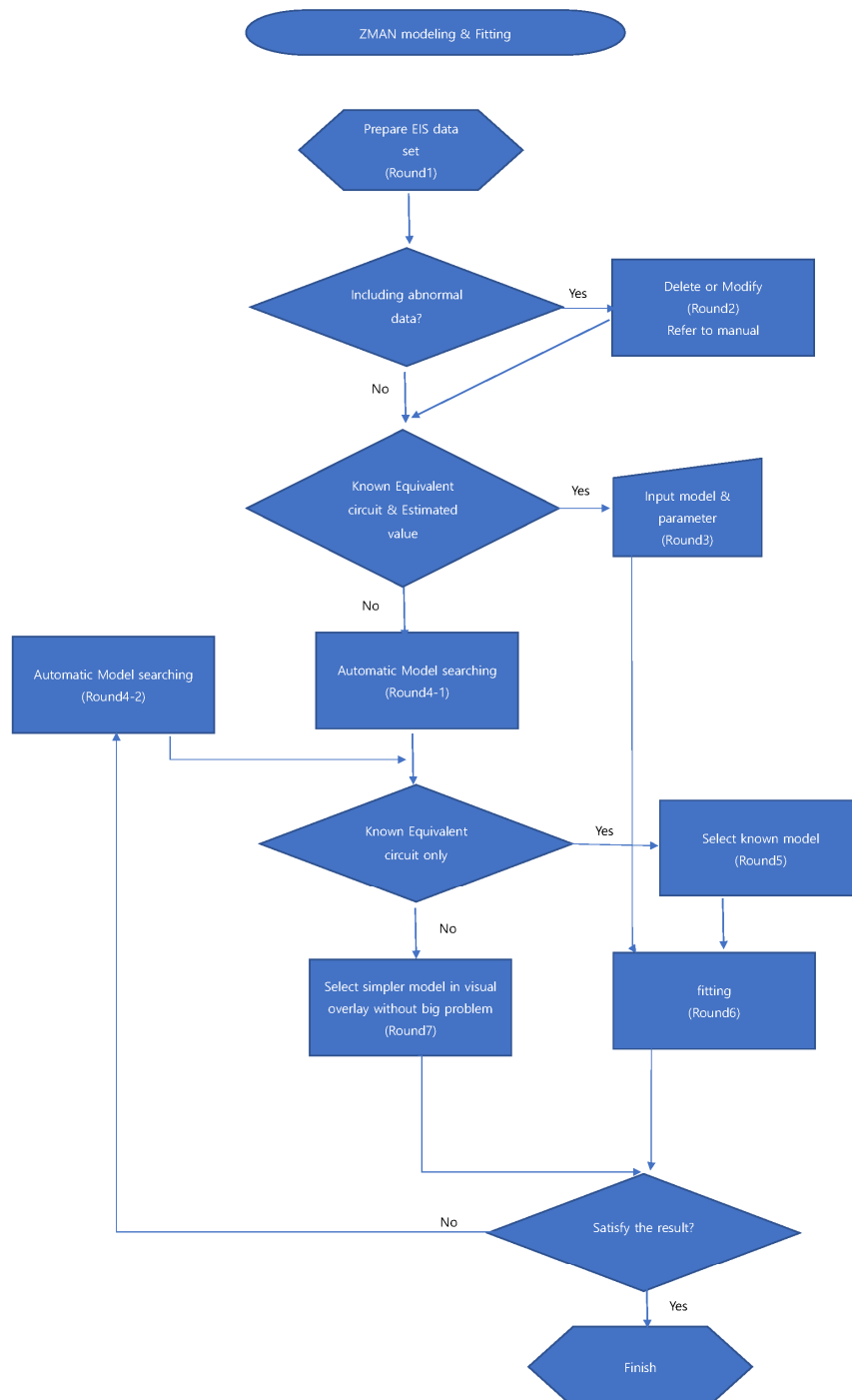
G. Real time monitor channel registration

User defined channel(s) real-time monitor channel registration. Refer to Chapter6

Chapter12. Model searching using ZMAN

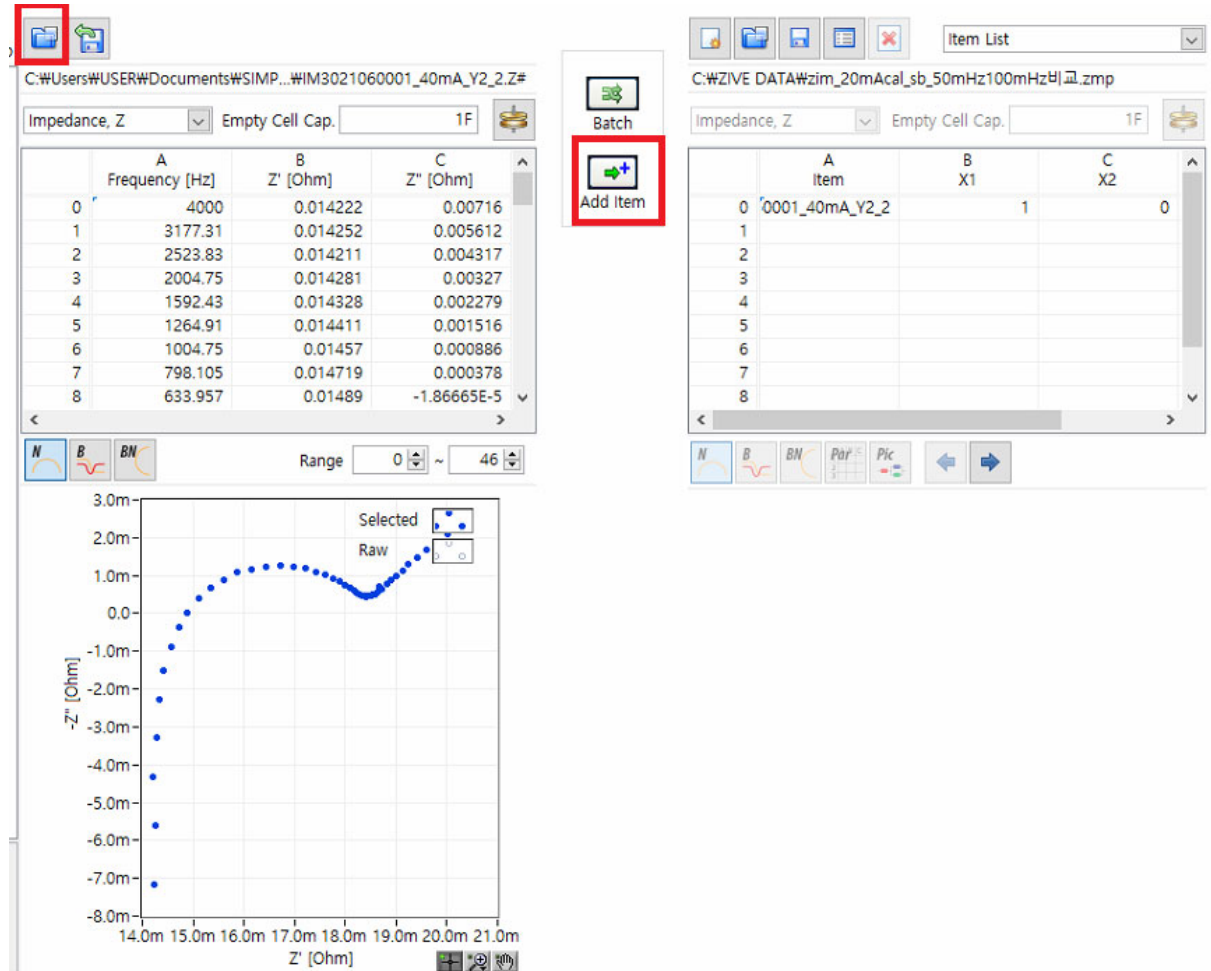
Please refer to the separated ZMAN manual.

This is simple guide how to find proper model using ZMAN..



1. EIS data file loading

Load the BZA data file (*.zmp)



Transfer the data set to analysis section by click "Add item" icon

2. Model searching

There are methods to automatically find the model in 'analysis-modeling' and methods to manually build the equivalent circuit to find an appropriate equivalent circuit. The following is an instruction for the automatic method to find the model. analysis-modeling.



Click the model search button.

(1) Model searching based on default value

Select Target Categories

Target Categories

BASIC (12)
SIMPLE (5)
BATTERY & SUPERCAPACITOR (84)
BIO (5)
CONCRETE (9)

☐ Remove Duplicates

Parameters

Free but must stay positive

Initial Guessing

Method: None

Nonlinear Fitting

Data Sequence: Z: Real & Imag

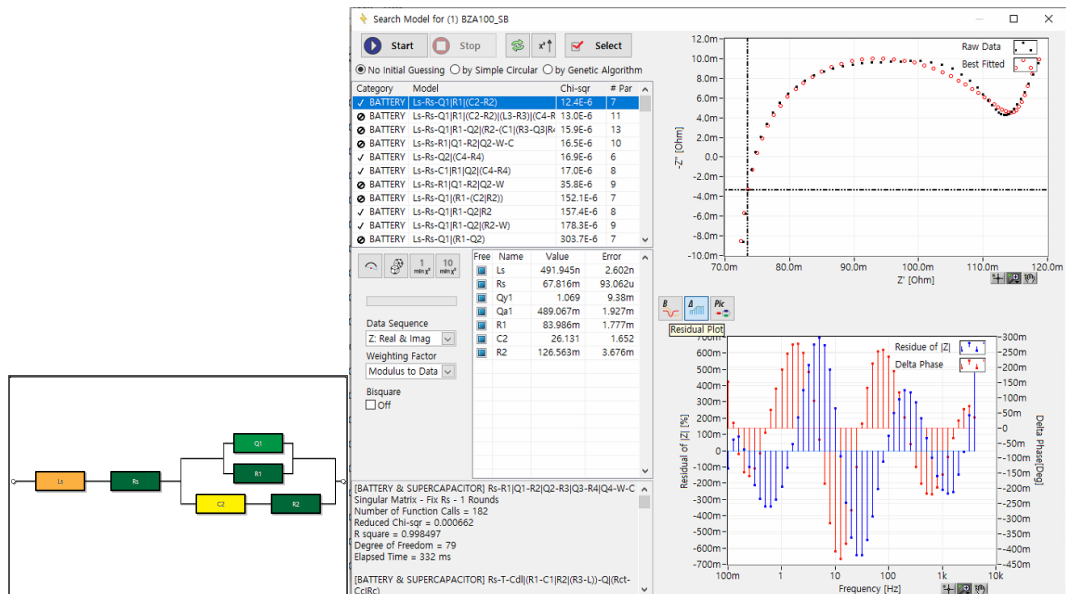
Weighting Factor: Modulus to Data

Bisquare: ☐ Off

Do you want to start searching process?

Yes No

Automatic searching result



(2) Initial guessing using simple circular

Select Target Categories

Target Categories

BASIC (12)
SIMPLE (5)
BATTERY & SUPERCAPACITOR (84)
BIO (5)
CONCRETE (9)

☐ Remove Duplicates

Parameters

Free but must stay positive

Initial Guessing

Method: Simple Circular

Nonlinear Fitting

Data Sequence: Z: Real & Imag

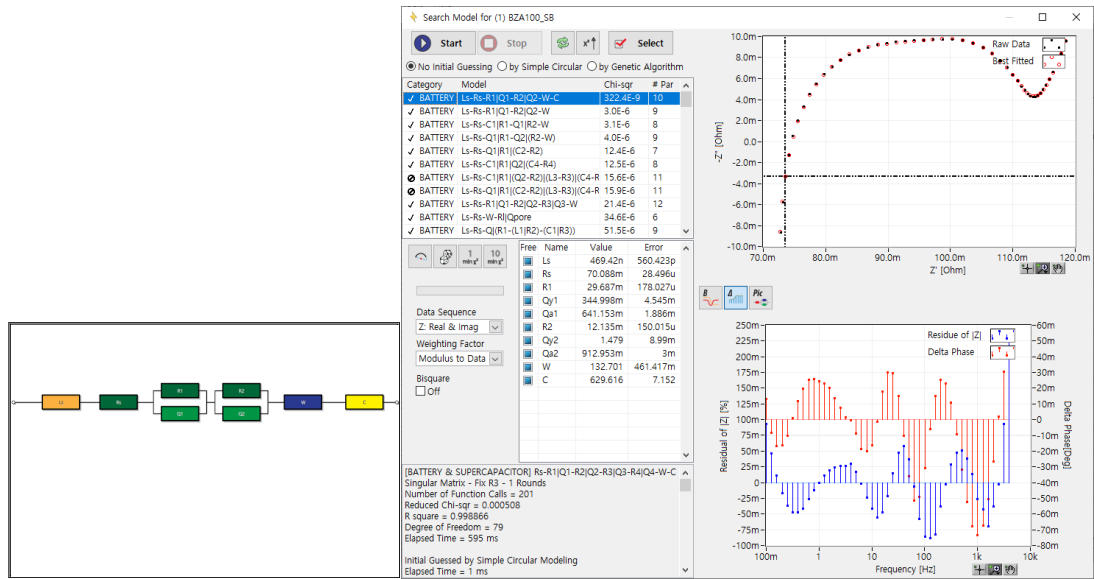
Weighting Factor: Modulus to Data

Bisquare: ☐ Off

Do you want to start searching process?

Yes No

Automatic searching result



(3) Initial guessing using simple circular with proportional weighting

Select Target Categories

Target Categories

- BASIC (12)
- SIMPLE (5)
- BATTERY & SUPERCAPACITOR (84)**
- BIO (5)
- CONCRETE (9)

☐ Remove Duplicates

Parameters

Free but must stay positive

Initial Guessing

Method: Simple Circular

Nonlinear Fitting

Data Sequence: Z: Real & Imag

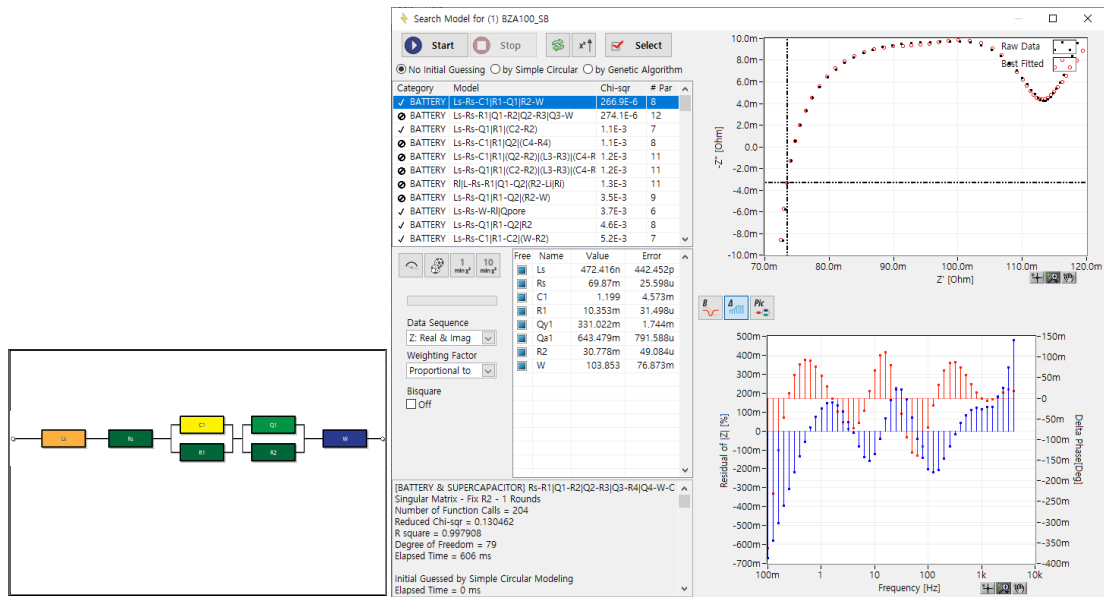
Weighting Factor: Proportional to Data

Bisquare: ☐ Off

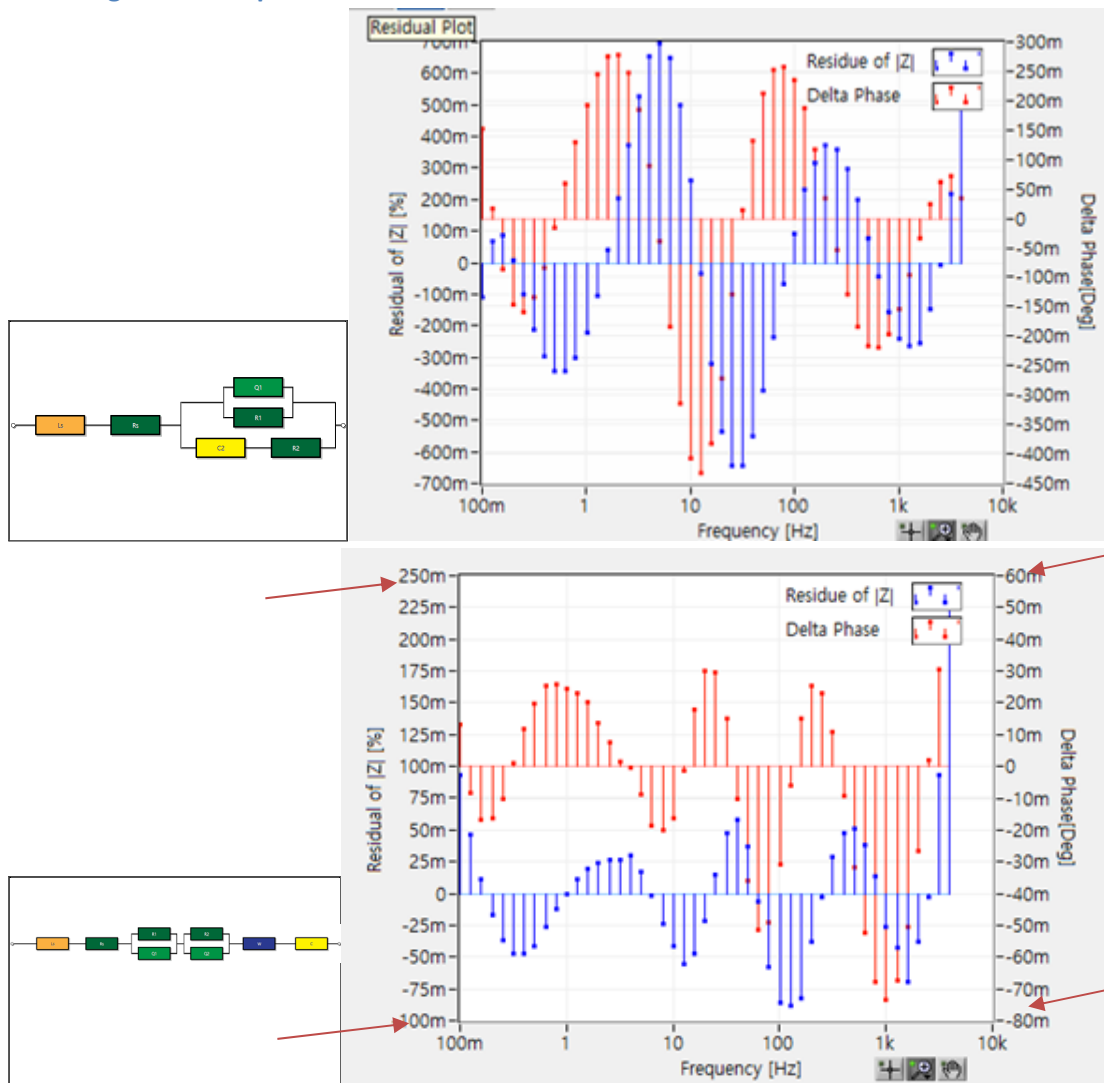
Do you want to start searching process?

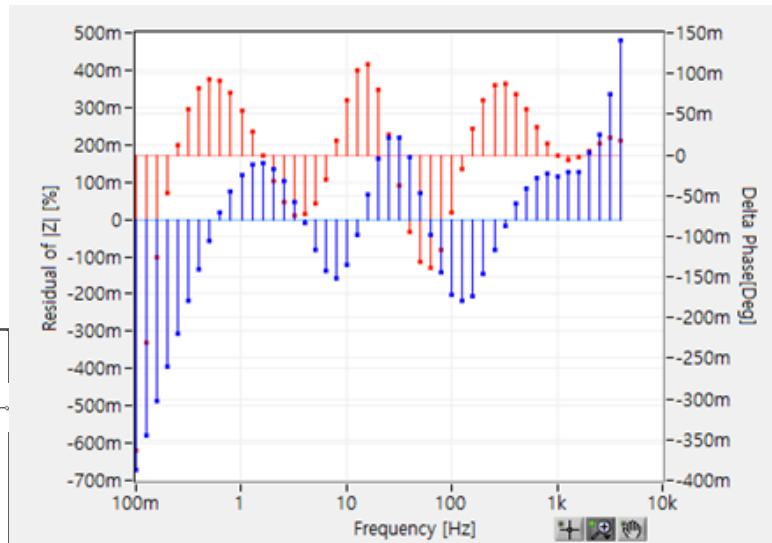
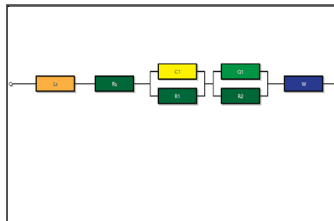
Yes No

Automatic searching result



3. Searching result comparison





Initial guessing using simple circular is good choice for this model searching

Chapter 13. Portable Option

If you purchase portable option for BZA60, You can use them in field purpose

1. Configuration

A. Portable Bag

- Water proof, Dust proof IP67 /RoHS certification
- Temperature range: -40 ~80C degree
- Fiber plastic material

B. Wireless Lan kit

You need to prepare battery pack by yourself

2. Connection

A. Connection with battery pack

- 1) Connect battery pack(User should prepare Battery pack) and BZA



- 2) Connect Battery pack and wireless lan kit; Connect USB cable of wireless Lan kit to Battery pack's USB connector
- 3) Connect Lan cable from wireless Lan kit into BZA lan port



- 4) If you want internet line, connect internet connected lan cable into wireless Lan kit's lan connector.



B. Wireless Lan Kit connection

Lan kit power connection
<-> Battery pack



BZA Lan port
connection

Internet Lan
cable(Optional)

3. IP address using wireless LAN connection

If you use wireless LAN kit, you can find IP address using zm.exe

Chapter 14. Calibration

If you have AC calibration resistors set which WonATech supplied and you had a calibration training from WonATech, Please refer to separated calibration manual.

If not, Please do NOT try to calibrate by yourself. If you need calibration please contact with your local distributor or WonATech.

Wrong calibration might make the instrument's malfunction.

Chapter 15. Specification

Description	Value
Impedance	
Impedance ranges	500uΩ-50Ω (BZA1000/BZA60/BZA500M/BZA60M)
Frequency range	0.05Hz-4kHz (BZA1000), 0.05Hz-10kHz (BZA60/BZA60M/BZA500M)
Current ranges	2A, 400mA, 200mA, 40mA, 20mA, 4mA, 2mA, 400uA (BZA1000/BZA60/BZA500M/BZA60M)
Density	1-65535 per decade
Max data points	100,000
Measurement time for spectrum measurements	9sec (1kH-1Hz, 4points/decade) with screening mode
Amplitude of measurement current	1A, 200mA, 100mA, 20mA, 10mA, 2mA, 1mA, 200uA
VAC	
Range	+/-0.256V
Resolution	30.5nV
IAC	
Range	+/-2.56A
Resolution	305nA
Sine wave	Bias & amplitude: half of current range
Battery Voltage Reading	
Range	0V-100V, 0V-1000V: BZA1000 0V-50V, 0V-500V: BZA500M 0V-6V, 0V-60V: BZA60/BZA60M
Resolution	70uV (1000V range) 7uV (100V range): BZA1000 35uV (500V range) 3.5uV (50V range): BZA500M 8uV (60V range) 0.8uV (6V range): BZA60/BZA60M
Input resistance	1 MΩ (BZA1000/BZA500M)
Temperature measurement	
Type	PT100
Range	-200°C - +850°C
Resolution	0.03°C
Accuracy	±0.5°C
External sensor	100Ω at 0°C
Type	Surface measurement
Communication	TCP/IP
LAN cable length	<u>1meter</u>
Data file	
EIS Data format	Binary (ZMAN compatible), ASCII format
Power supply	
Voltage	5V
Multichannel Model channel No	4
Configurable channel No per PC	32
EIS data analysis	ZMAN software
	1 meter banana terminals with alligator clips Current flowing: 2, voltage sensing: 2

